

GeV-Scale Neutrinos: Modeling, Discovery and Characterization at Forward Physics Experiments

Daniel La Rocco

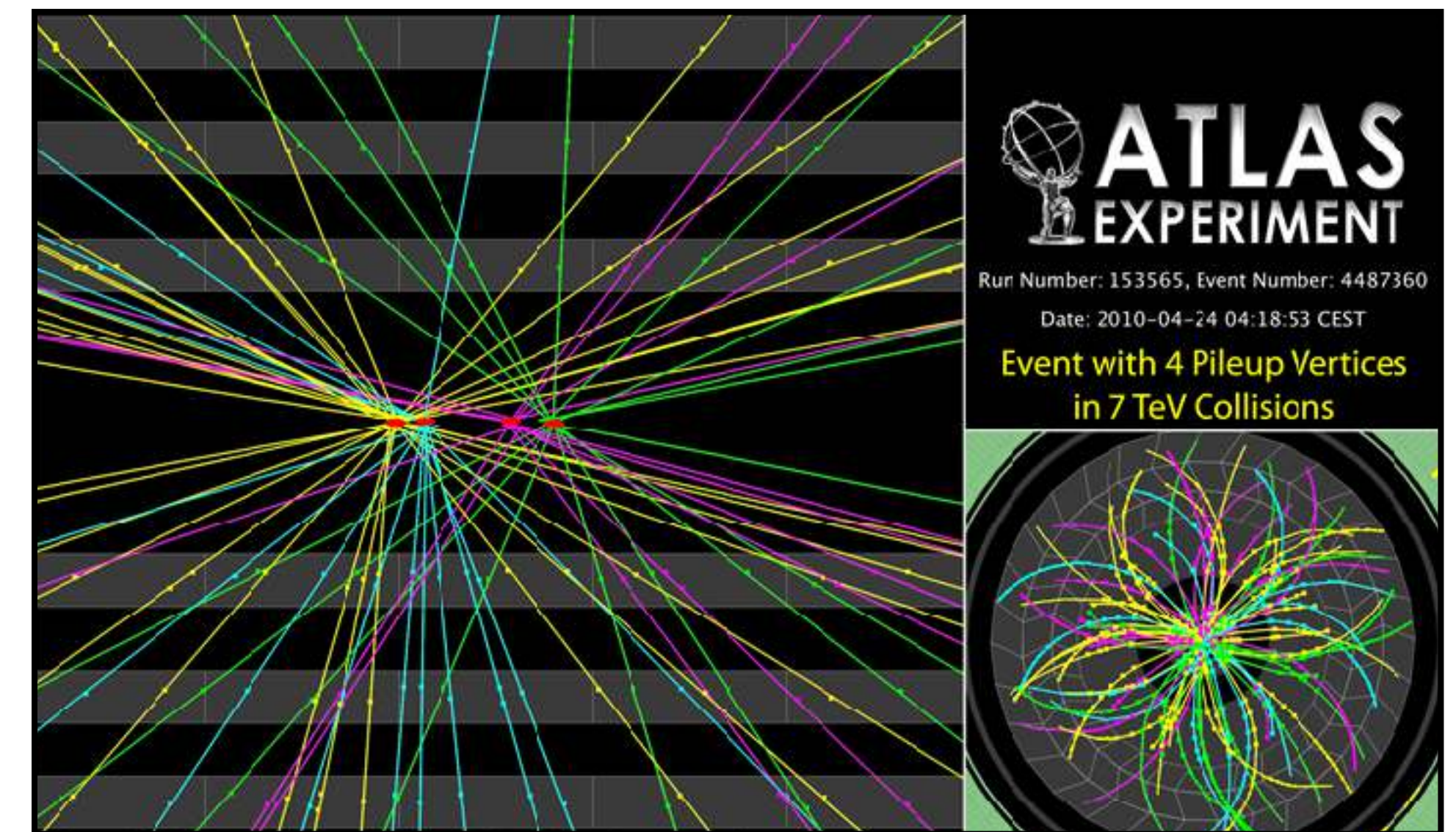
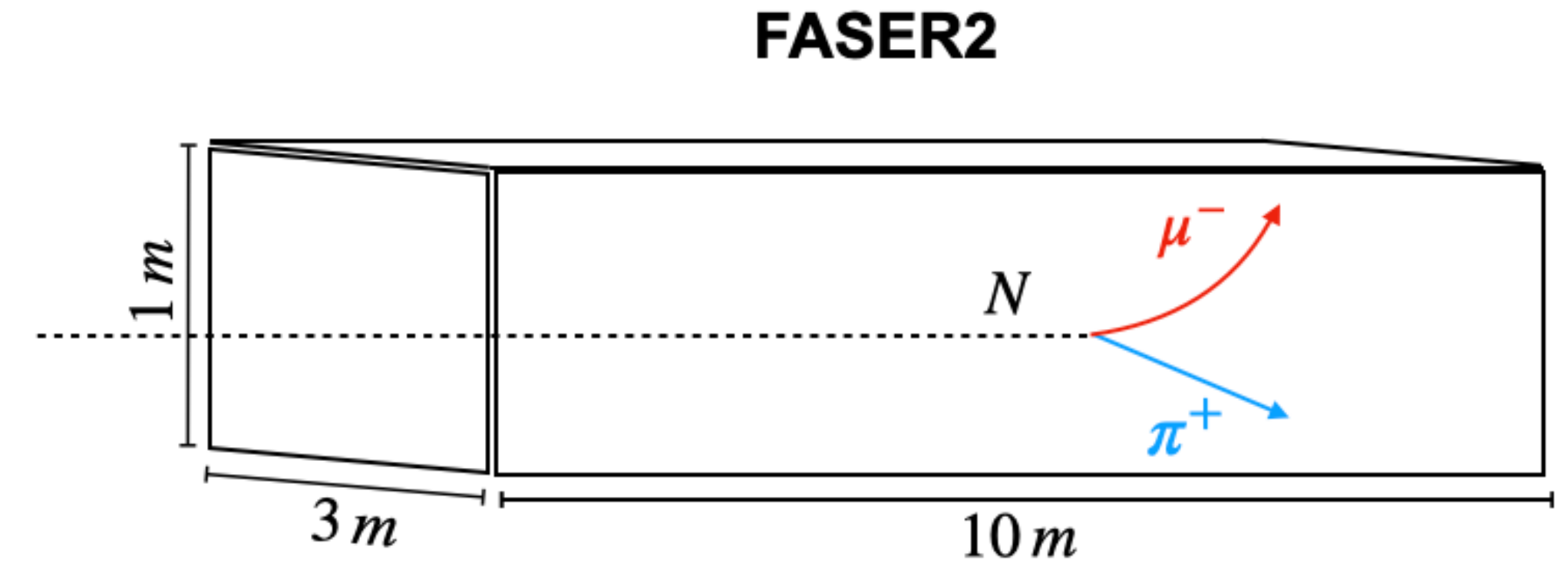
Advisor: Jonathan L. Feng

Committee: Shirley Li, Timothy M.P. Tait, Daniel Whiteson

[2405.07330](#), [2510.16107](#)

Outline

1. The Standard model and the Neutrino Puzzle
2. What are GeV-scale heavy neutral leptons (HNLs) and why are they important?
- 3. Modeling HNLs:** HNLCalc and the FORESEE HNL module
- 4. Discovering HNLs:** Sensitivity of the FASER/FASER2 experiments to HNLs
- 5. Characterizing HNLs:** Parameter estimation, Majorana-vs-Dirac discrimination, FASER2 as a trigger for ATLAS



ATLAS Collaboration

The Standard Model

- The **Standard Model (SM)** is our current theoretical framework to describe the physics of matter and the forces that bind it
- **Matter (fermions):** organized into three "generations" of increasingly heavy particles
 - Quarks: bound by the strong force into protons, neutrons, and mesons
 - Leptons: do not feel the strong force; only the electric & weak forces
- **Forces (bosons):** the photon (electromagnetism), the W and Z bosons (weak nuclear force), gluons (strong nuclear force)
- The **Higgs boson**, the keystone piece discovered at the LHC in 2012, **is responsible for giving every other fundamental particle its mass** (almost)
- **Neutrinos** are electrically neutral, interacting only via the weak force, and **are predicted to be massless by the SM.**

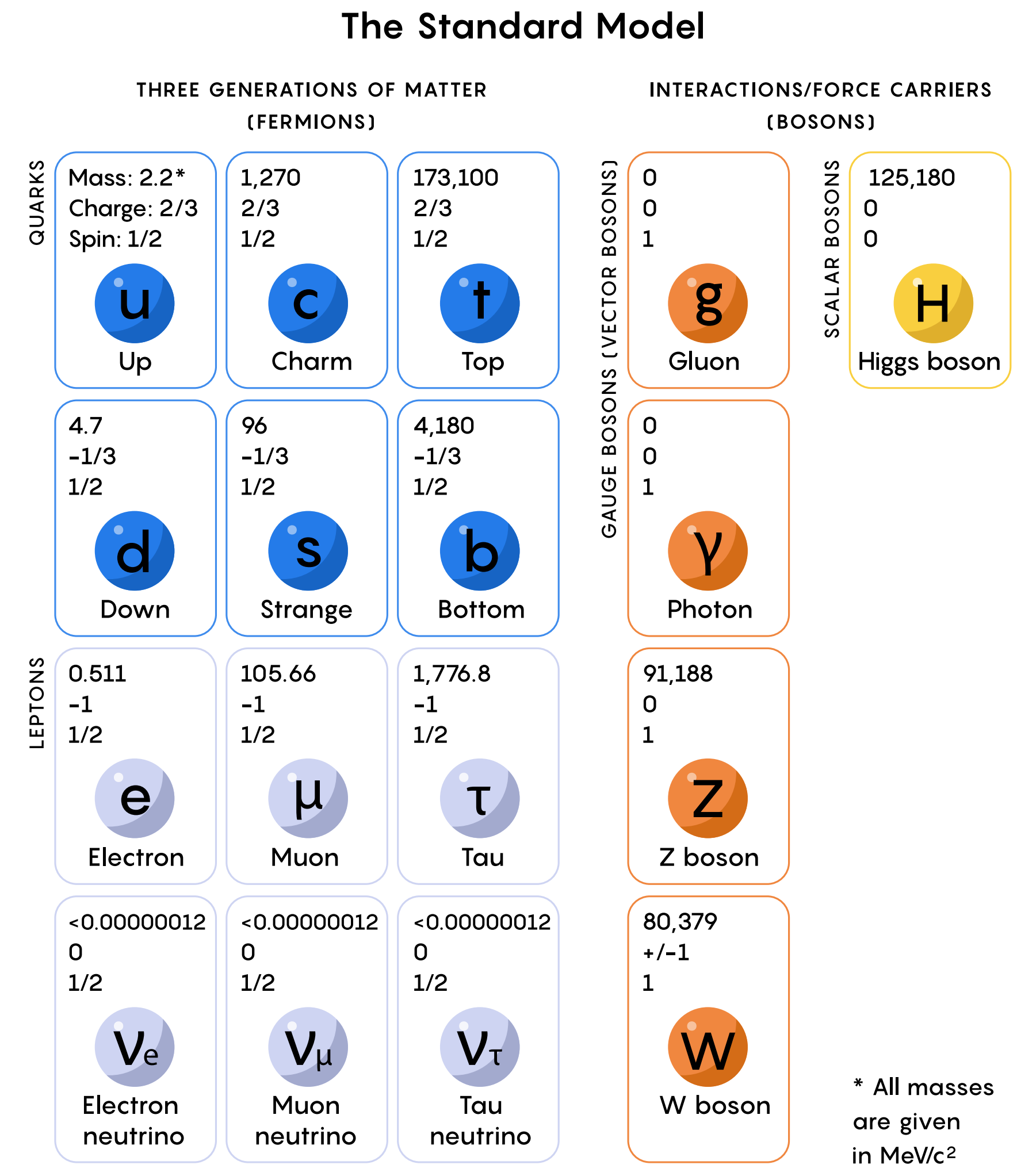


Figure adapted from [Quanta Magazine](#).

Chirality, Mass, and the Neutrino Puzzle

- Each fermion can come in two "chiralities": a left-handed component f_L and a right-handed component f_R
- Due to the symmetries of the SM, the Higgs boson H^0 couples these two fields together to form the Yukawa interactions.
- In the early universe every fundamental particle was massless.
- As the universe expanded and cooled, the Higgs boson obtains a uniform, **non-zero background density** $\langle H^0 \rangle \rightarrow \frac{v}{\sqrt{2}}$ during a phase transition called **electroweak symmetry breaking** (EWSB). Interactions with this background Higgs field are **what gives particles mass**.
- Neutrinos, however, are the **only** fermions in the SM which have no right-handed component ν_R included, since they have not been observed experimentally. Therefore, **neutrinos are predicted to be massless in the SM** ($m_\nu = 0$).

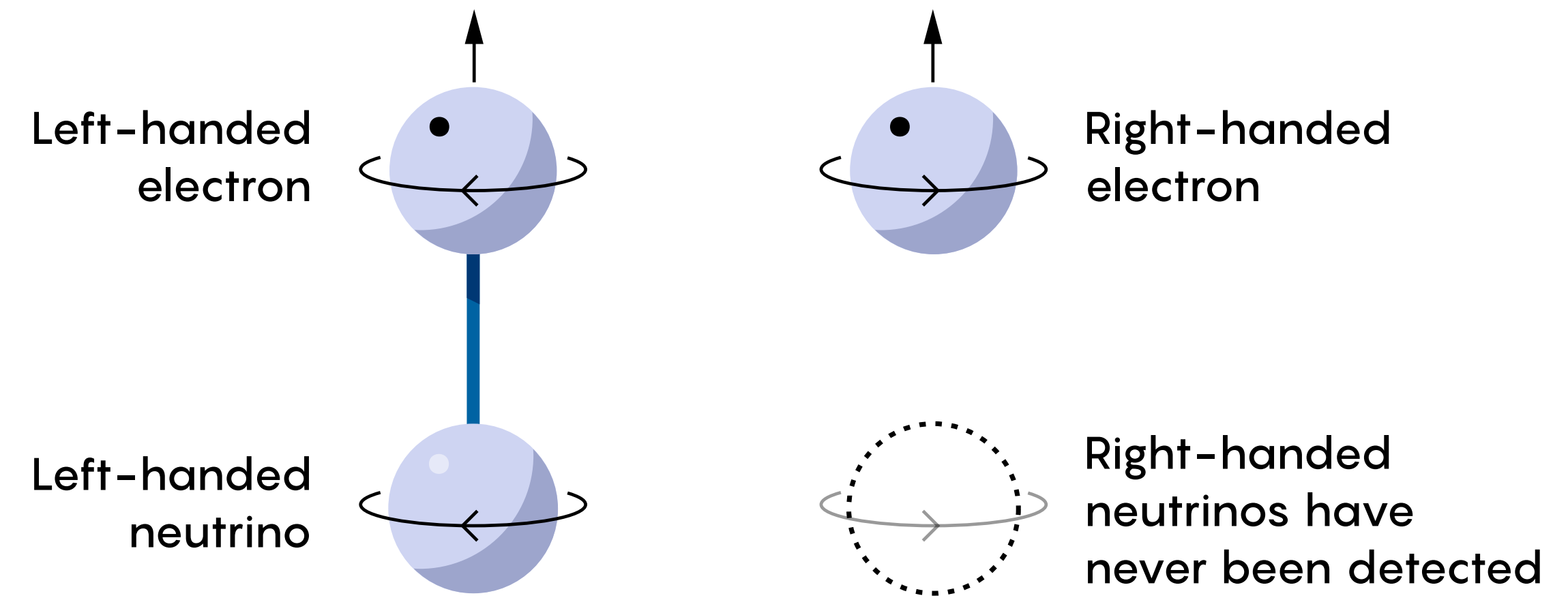


Figure adapted from [Quanta Magazine](#).

$$\mathcal{L} \supset -y_e H^0 (\bar{e}_L e_R + e_R e_L)$$

$$\downarrow \text{EWSB}$$

$$\mathcal{L} \supset -\frac{y_e v}{\sqrt{2}} (\bar{e}_L e_R + e_R e_L) = -m_e (\bar{e}_L e_R + e_R e_L)$$

Evidence for Neutrino Masses

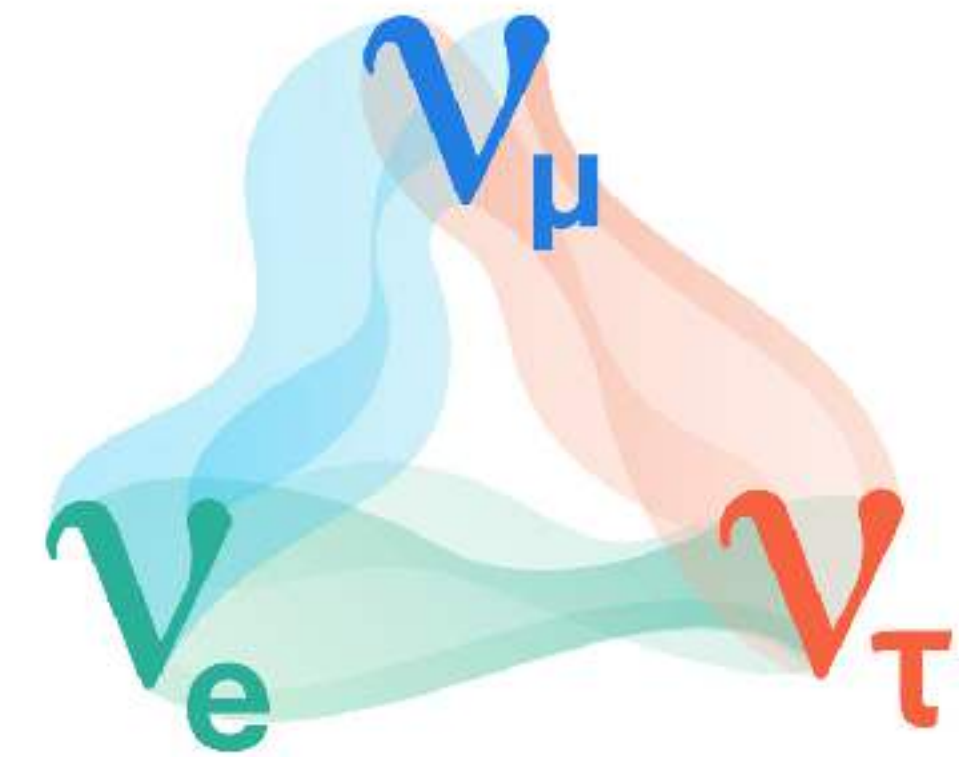
- **Neutrino oscillations** are a phenomenon in which a neutrino's lepton flavor (e , μ , or τ) oscillates as it propagates through space.
- This is because **flavor eigenstates are coherent superpositions of mass eigenstates**, which pick up *different* phases as they propagate a distance L :

$$|\nu_\alpha(L)\rangle = \cos\theta e^{-im_1^2 L/(2E_\nu)} |\nu_1\rangle + \sin\theta e^{-im_2^2 L/(2E_\nu)} |\nu_2\rangle$$

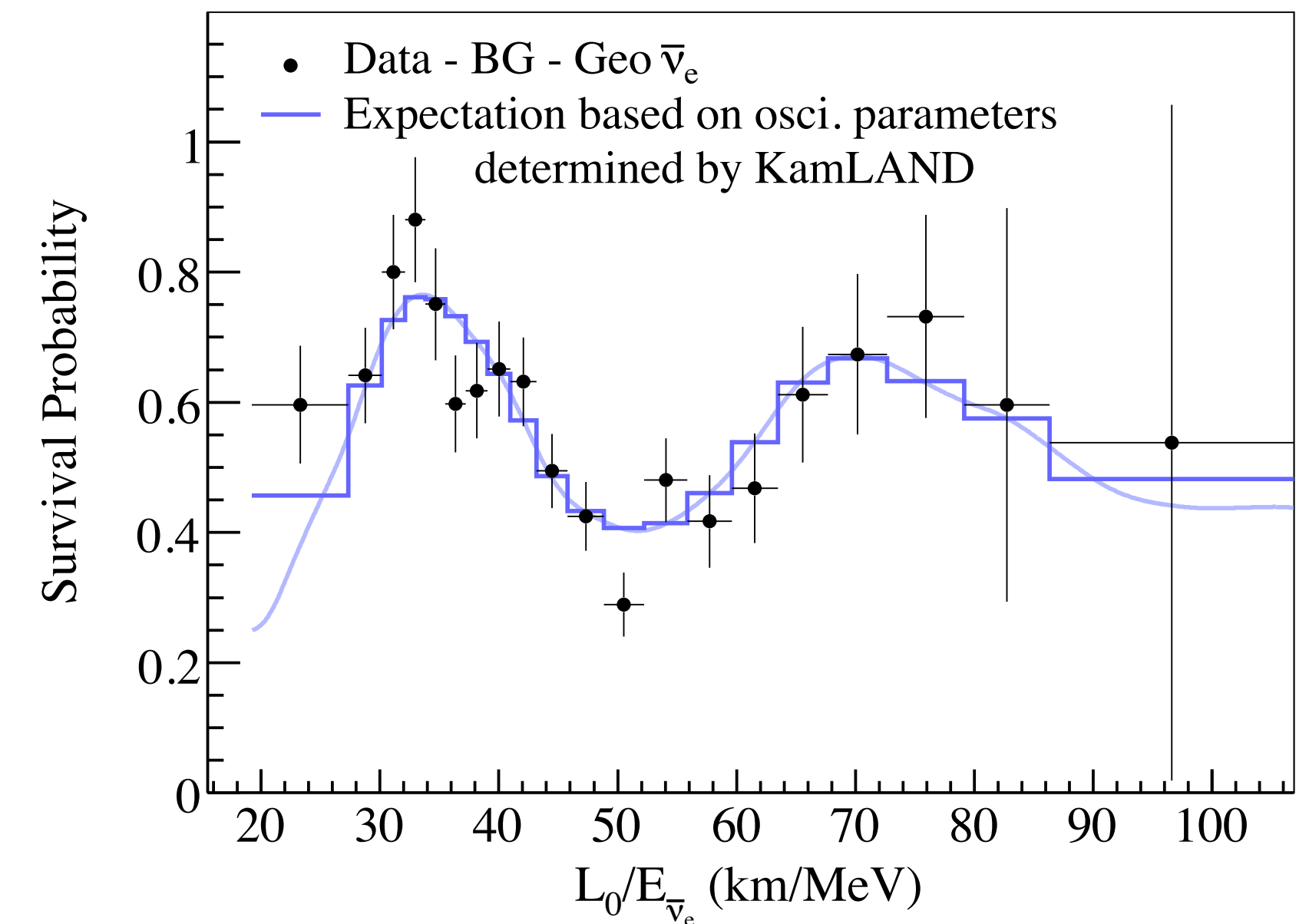
- The **relative phase** $\propto (m_2^2 - m_1^2)L/E_\nu$ is what **drives the flavor oscillation**, which in the two-flavor case is:

$$P(\nu_\alpha \rightarrow \nu_\beta) = \sin^2(2\theta) \sin^2\left(\frac{\Delta m^2 L}{4E_\nu}\right)$$

- This is smoking gun **evidence for physics beyond the Standard Model (BSM)**!



iastate.edu



KamLAND Collaboration [0801.4589](https://arxiv.org/abs/0801.4589)

HNLs, Phenomenologically

- What if we added right-handed neutrinos ν_R to the SM?
- These would be singlets under the SM gauge group $SU(3)_C \times SU(2)_L \times U(1)_Y$, which implies they do not interact directly with the SM ("sterile neutrinos").
- After mixing with active flavors, the mass eigenstates are the **heavy neutral leptons (HNLs)**, denoted N_I

$$\nu_L^\alpha = \sum_{i=1}^3 V_{\alpha i} \nu_i + \sum_I U_{\alpha I} N_I^c$$

- Through mixing, **HNLs acquire weak interactions which are suppressed** by their couplings $U_{\alpha I}$ to the active lepton flavors.

$$\mathcal{L} \supset -\frac{g}{\sqrt{2}} \sum_{\alpha} \bar{\nu}_{\alpha} \gamma^{\mu} P_L \ell_{\alpha} W_{\mu}^{+} - \frac{g}{2 \cos \theta_W} \sum_{\alpha} \bar{\nu}_{\alpha} \gamma^{\mu} P_L \nu_{\alpha} Z_{\mu}$$

↓

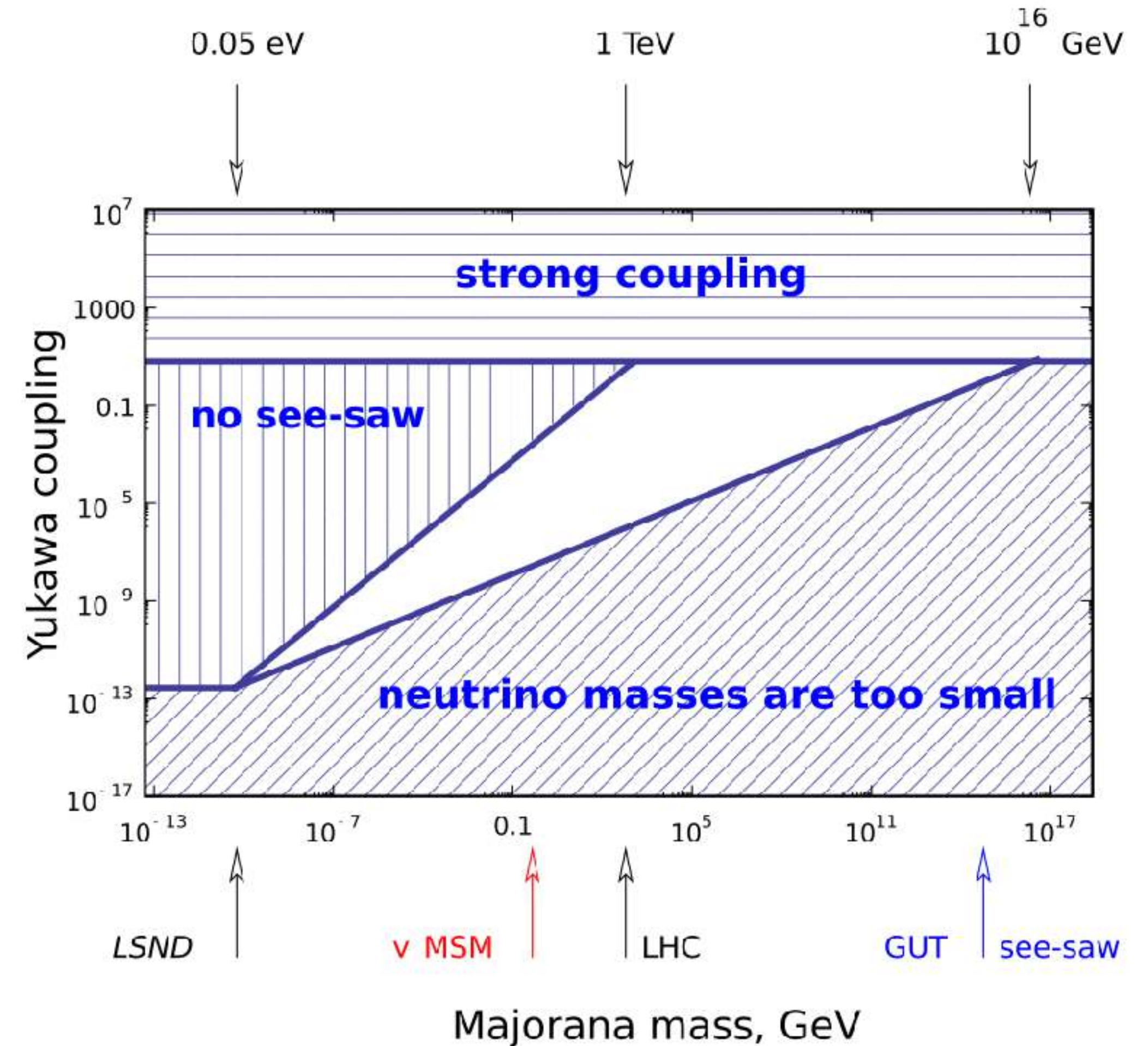
$$\mathcal{L} \supset -\frac{g}{\sqrt{2}} \sum_{\alpha, I} U_{\alpha I}^{*} \bar{N}_I^c \gamma^{\mu} P_L \ell_{\alpha} W_{\mu}^{+} - \frac{g}{2 \cos \theta_W} \sum_{\alpha, I} U_{\alpha I}^{*} \bar{N}_I^c \gamma^{\mu} P_L \nu_{\alpha} Z_{\mu}$$

Every production and decay rate carries one factor of $|U_{\alpha I}|^2$, therefore **the phenomenology is set by the four numbers:**

$$(m_N, |U_{eI}|^2, |U_{\mu I}|^2, |U_{\tau I}|^2)$$

Mass-Scale Landscape

- $\sim \mathbf{eV}$: motivated short-baseline oscillation anomalies; mostly constrained by BBN (ΔN_{eff}) + CMB ($\sum m_\nu$)
- $\sim \mathbf{keV}$: warm dark matter candidates (Dodelson-Widrow, Shi-Fuller); constrained by X-ray searches for $N \rightarrow \nu\gamma$
- $\sim \mathbf{MeV-GeV}$ (**this work**): motivated the ν MSM (explains DM, neutrino masses, and matter/anti-matter asymmetry); constrained by accelerator/collider experiments
- $\gtrsim \mathbf{TeV}$: the original scale for the "vanilla" Type-I seesaw mechanism for generating sub-eV neutrino masses; constrained by charged-lepton flavor violation ($\mu \rightarrow e\gamma$), EDMs, and high-scale thermal leptogenesis



Shaposhnikov 2024

UV Completions: Type-I See-Saw and beyond

- **“Vanilla” Type-I See-saw:** active-neutrino masses are generated from a heavy Majorana scale $M_R \gg M_D$.
- The size of the mixing is therefore tied to the light neutrino mass scale, which assuming $m_\nu \sim 0.05$ eV, gives the relationship:

$$|U|^2 \sim \frac{m_\nu}{m_N} \approx 5 \times 10^{-11} \left(\frac{1 \text{ GeV}}{m_N} \right).$$

- This line lies one to two orders of magnitude **below the sensitivity of any current or proposed accelerator-based search.**

$$\mathcal{L} \supset - \sum_{\alpha I} \underbrace{(y)_{\alpha I} H^{0*} \bar{\nu}_\alpha N'_I}_{\text{Dirac}} - \underbrace{\sum_{IJ} (M_R)_{IJ} \bar{N}'_I{}^c N'_J - \sum_{\alpha\beta} \frac{(C_5)_{\alpha\beta}}{\Lambda} (H^{0*})^2 \bar{\nu}_\alpha \nu_\beta^c + \text{h.c.}}_{\text{Majorana}}$$

$$M_\nu = \begin{pmatrix} 0 & M_D \\ M_D^T & M_R \end{pmatrix} \rightarrow \nu_\alpha = \sum_{i=1}^3 V_{\alpha i} \nu_i + \sum_I U_{\alpha I} N_I^c$$

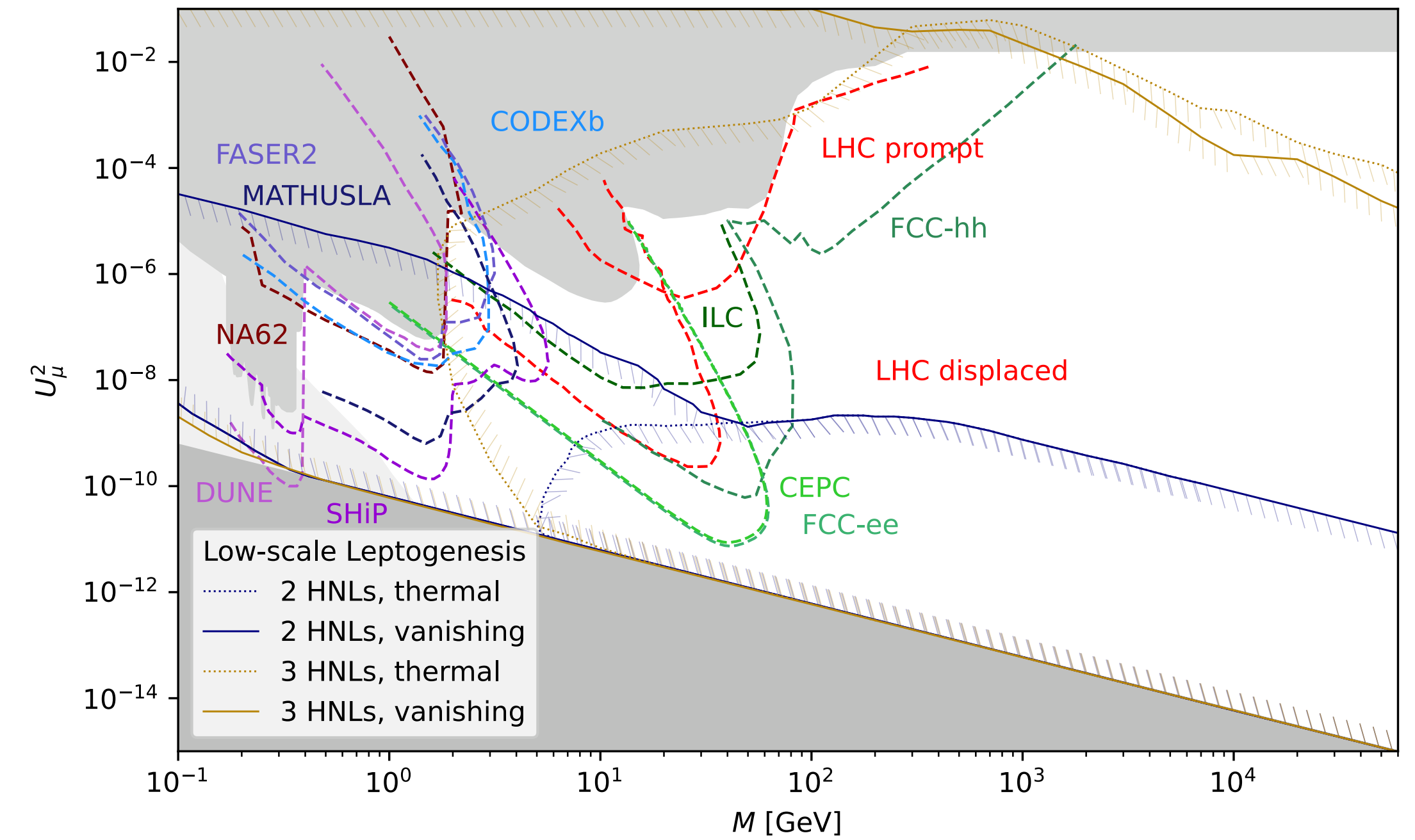
$$m_N \sim m_R, \quad m_\nu \sim \frac{m_D^2}{m_R} \propto \frac{1}{m_N}, \quad |U_\alpha|^2 \sim \frac{m_\nu}{m_N}$$

UV Completions: Type-I See-Saw and beyond

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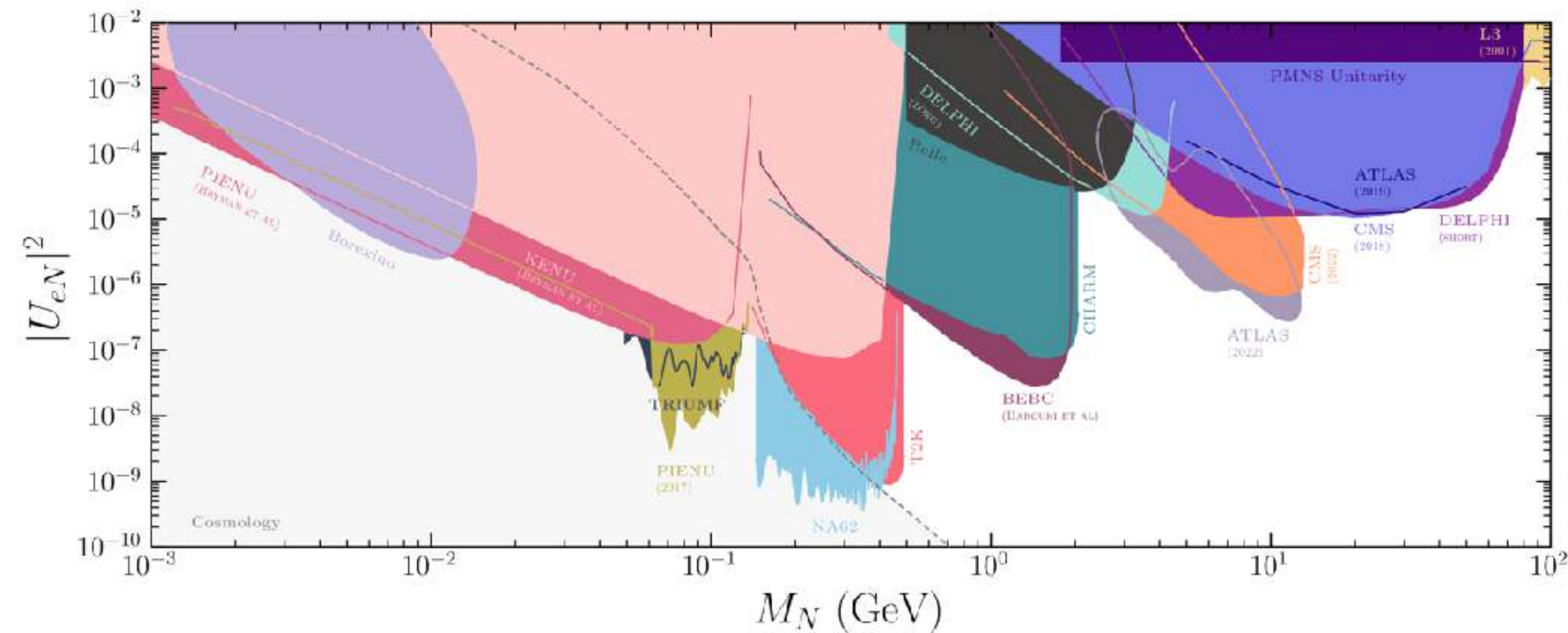
Abdullahi et. al 2203.08039

- **Inverse / Linear / Pseudo-Dirac See-saws:** decouple $|U|^2$ from m_ν via an approximate $U(1)_L$ symmetry e.g. for pseudo-Dirac pairs with mass splitting ΔM_N :

$$|U|^2 \sim \frac{m_\nu}{\Delta M_N} \approx 5 \times 10^{-9} \left(\frac{10 \text{ MeV}}{\Delta M_N} \right).$$

History of HNL Searches

- MeV - GeV scale HNLs have been studied as early as the 1970's, with experimental searches taken up not long after.
- **Peak searches** in $\pi^+/K^+ \rightarrow \ell^+ N$: SIN, KEK, TRIUMF (PIENU), CERN (NA62)
- Peak searches are limited kinematically at $m_N \lesssim m_K - m_\ell \sim 0.4$ GeV and so to reach the GeV window, you need **heavier parents** and **displaced vertex (DV) signatures**
- **Beam-dump searches for DV**: PS191, CHARM, NuTeV, T2K-ND280, MicroBooNE, SHiP (proposed)
- **Collider searches for DV**: DELPHI, ATLAS, CMS, LHCb



Fernández-Martínez et al. 2304.06772

$$\Gamma(M^+ \rightarrow \ell^+ N_I) = |U_{\ell I}|^2 \frac{G_F^2 |V_{qq'}|^2 f_M^2 m_M^3}{8\pi} \times \lambda^{1/2}(1, x_\ell^2, x_N^2) [x_\ell^2 + x_N^2 - (x_\ell^2 - x_N^2)^2]$$

Pseudoscalar decay width into HNLs. ([Shrock 1974](#))

The MeV-GeV Window: LLP Signatures

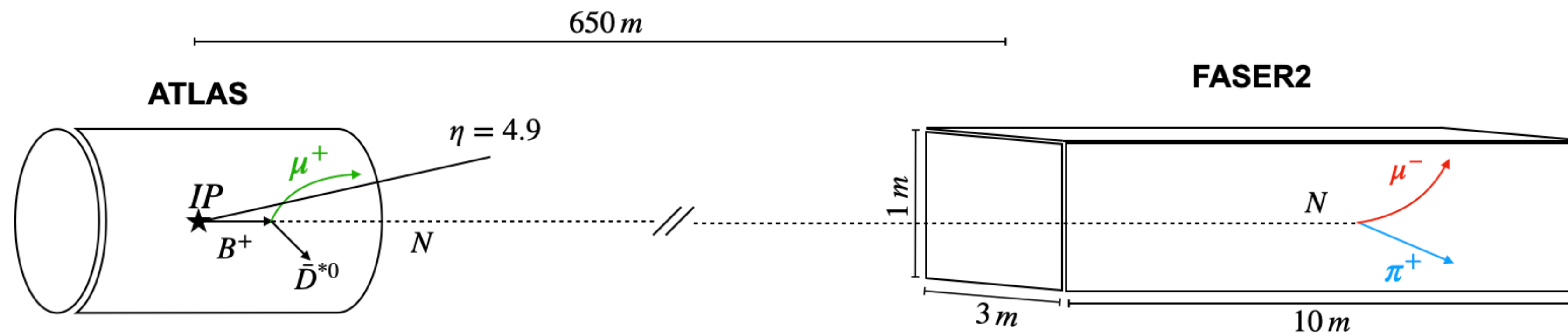
- The **overall active-HNL mixing strength** can be characterized by the term:

$$\epsilon^2 \equiv |U_e|^2 + |U_\mu|^2 + |U_\tau|^2.$$

- The HNL total width is parametrically suppressed by this small mixing:

$$\Gamma_N \propto \epsilon^2 G_F^2 m_N^5 \implies c\tau_N \propto \frac{1}{\epsilon^2 G_F^2 m_N^5}$$

- For MeV-GeV HNLs with TeV-scale energies and mixings $\epsilon^2 \sim 10^{-7} - 10^{-9}$, $\ell_N \approx \frac{E_N}{m_N} c\tau_N$ **spans meters to many kilometers** – the on-axis **far-forward LHC** sites ($L \sim 480 - 650$ m) sit squarely inside this window
- This is the **long-lived-particle (LLP)** regime: HNLs which travel macroscopic distances before decaying.



Modeling & Discovery: HNLCalc and the FORESEE HNL Module

Collaboration: Jonathan L. Feng, **Alec Hewitt**, Felix Kling, and **Daniel La Rocco**

PHYSICAL REVIEW D **110**, 035029 (2024)

Simulating heavy neutral leptons with general couplings at collider
and fixed target experiments

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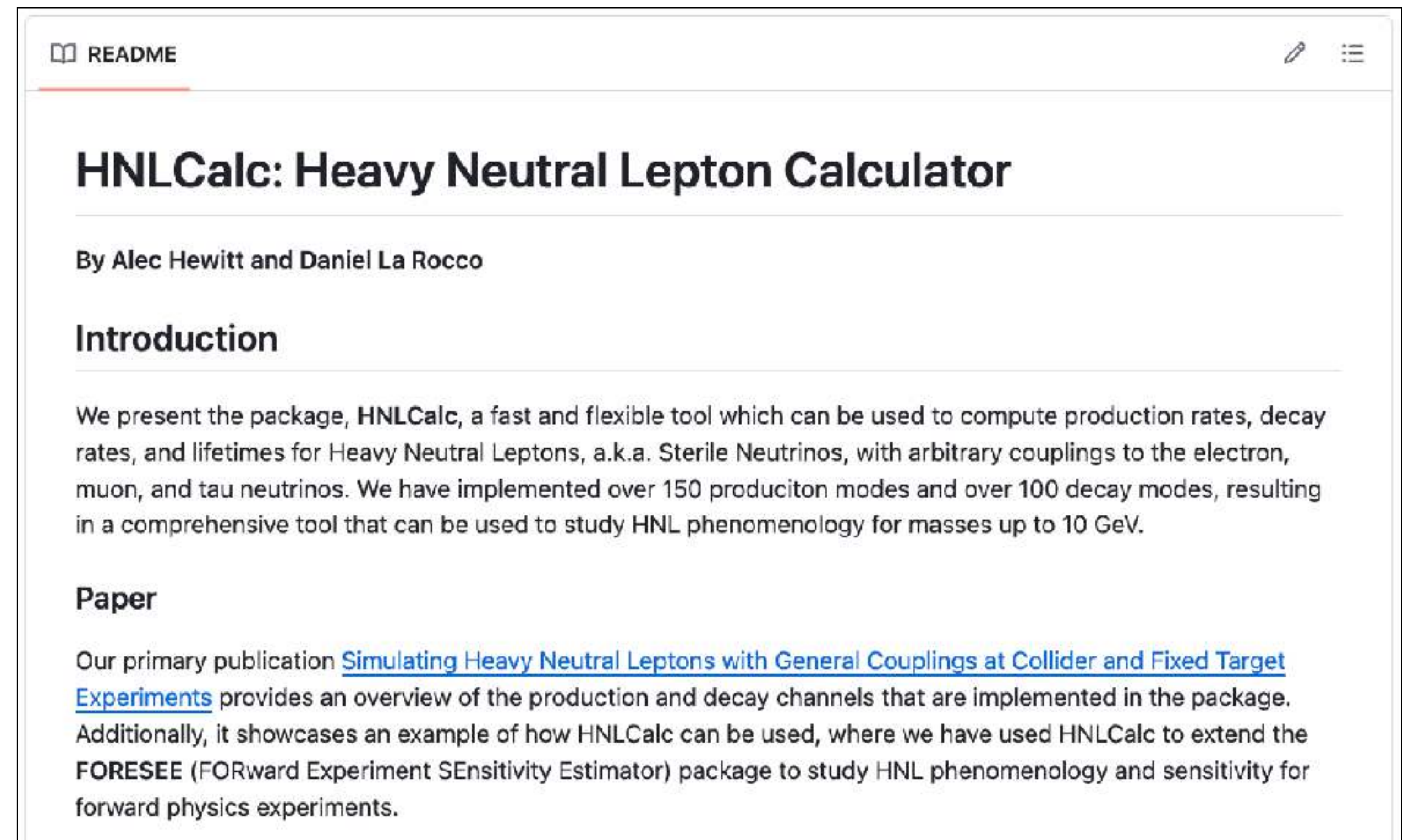


(Received 11 June 2024; accepted 5 July 2024; published 26 August 2024)

[**PRD 110, 035029 \(2024\)**](#)

HNLCalc: A Comprehensive HNL Phenomenology Library

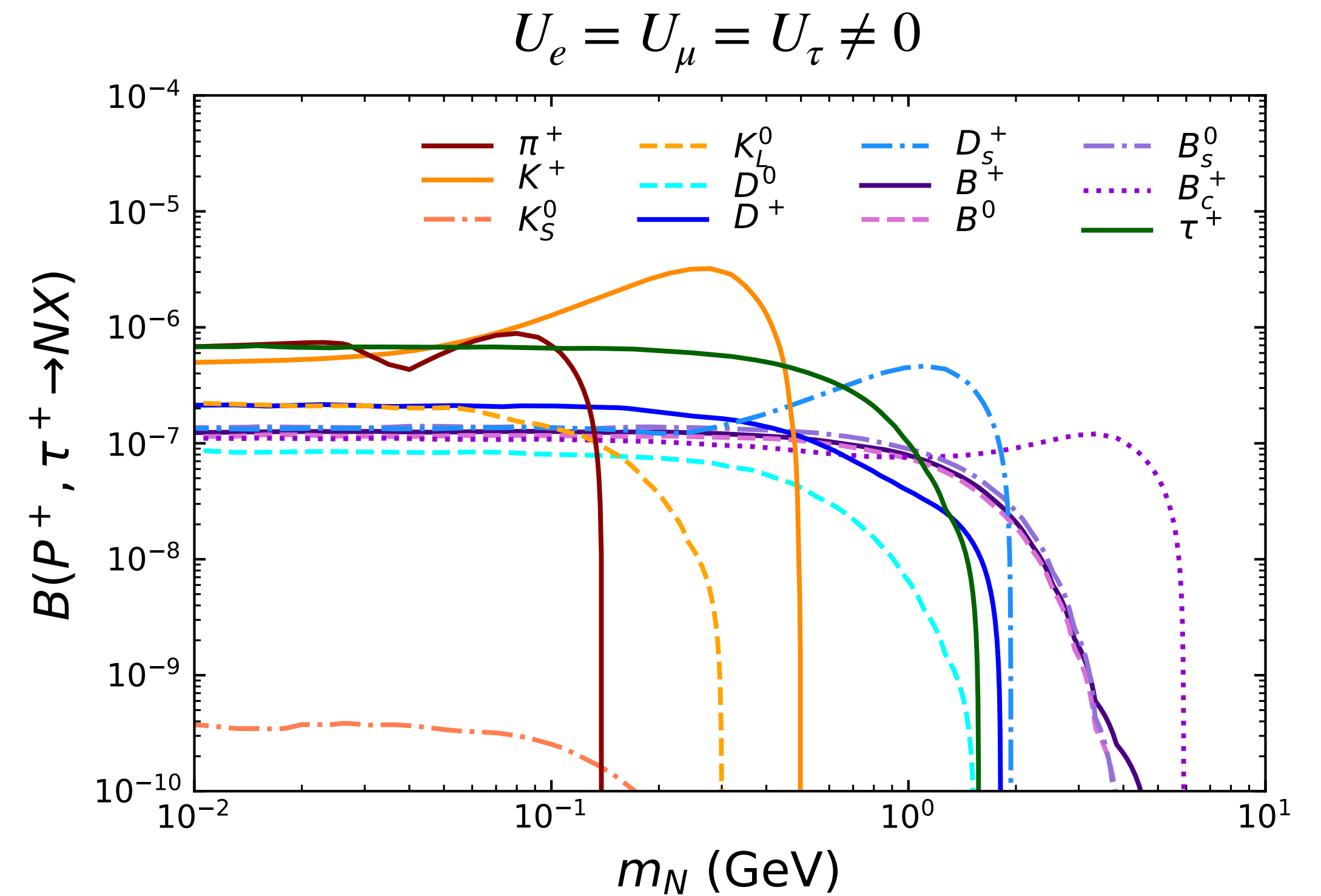
- A fast, flexible, and comprehensive python package for calculating HNL production and decay rates with arbitrary coupling ratios, developed by myself and Alec Hewitt.
- Computes HNL production rates $BR(A \rightarrow N\dots)$, decay branching fractions $BR(N \rightarrow A\dots)$, and lifetimes $c\tau_N$ for a given coupling ratio $|U_e|^2 : |U_\mu|^2 : |U_\tau|^2$.
- Over 150 production modes and 100 decay channels; valid up to $m_N \sim 10$ GeV.
- Produces the relevant outputs for Monte-Carlo (MC) event generators and includes plotting tools for visualization



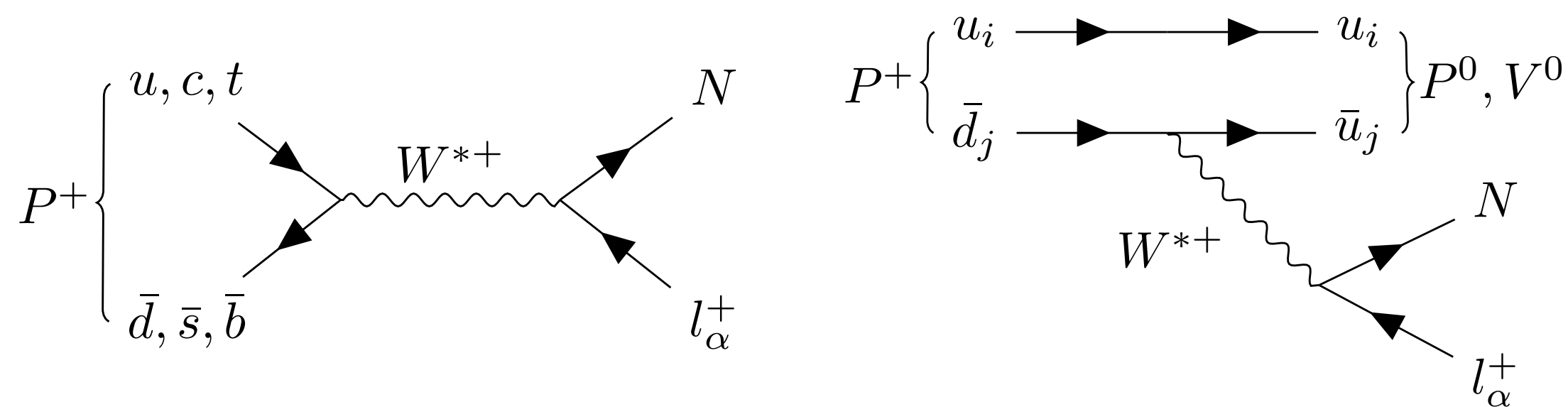
Available on [GitHub](#).

HNL Production

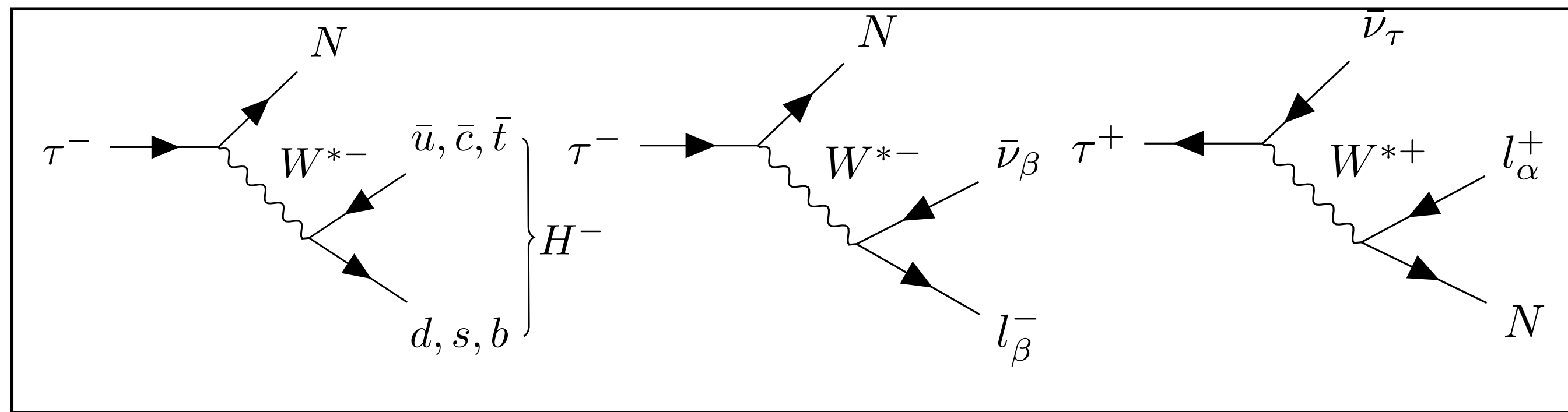
- HNLCalc includes HNL production from the decay of pseudoscalar mesons and tau leptons
- For HNLs in the mass range 100 MeV to 10 GeV, these are the dominant production mechanisms



Pseudoscalar



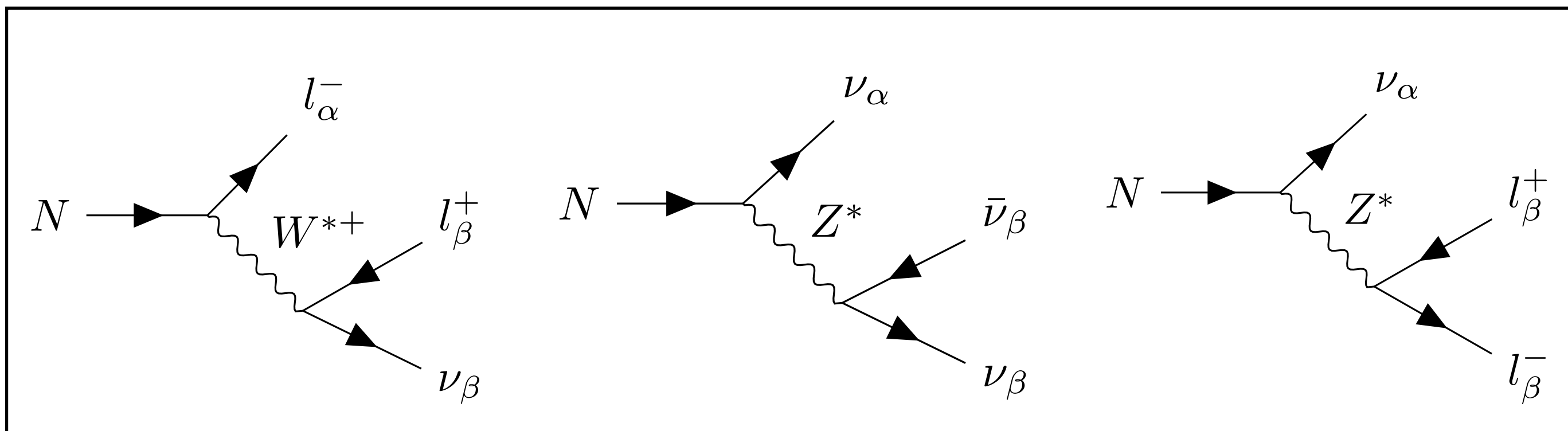
Tau



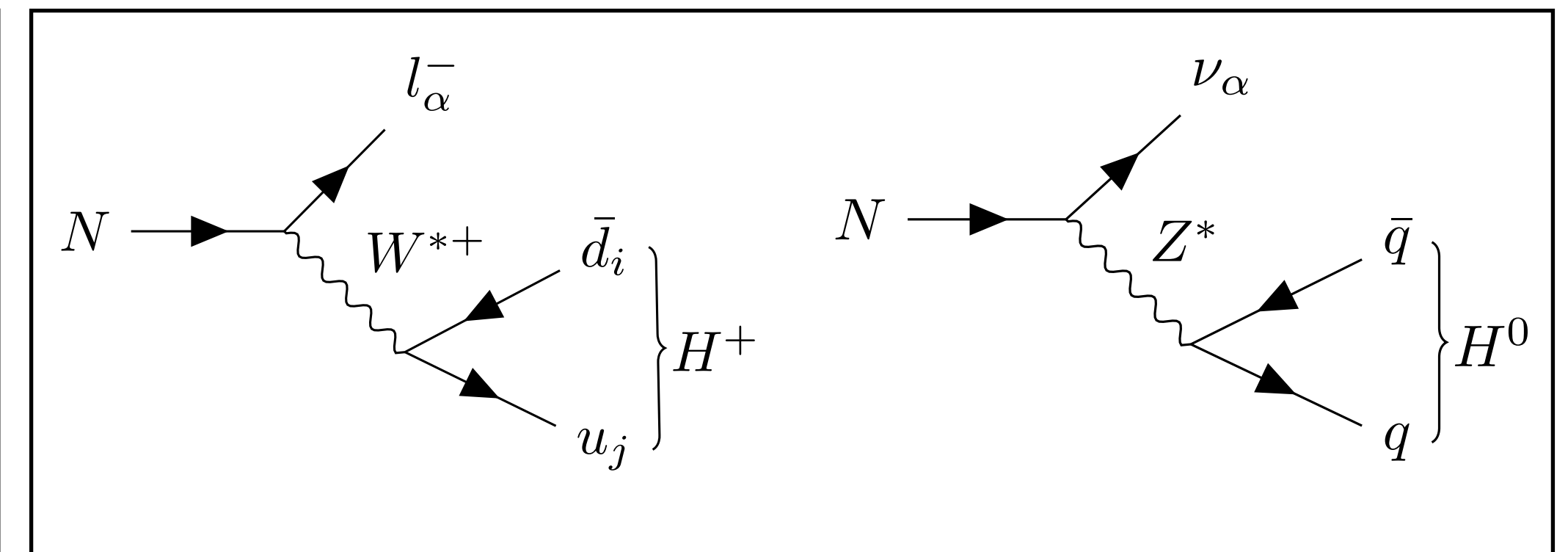
HNL Decay

- HNLs can decay at tree-level into leptonic or hadronic final states through virtual W and Z exchange
- In order to properly simulate detector response, one must accurately represent the final states
 - i.e. $N \rightarrow \nu \rho \rightarrow \nu \pi^+ \pi^-$ vs $N \rightarrow \nu \pi^0 \rightarrow \nu \gamma \gamma$
 - Quark-level final states are insufficient!

Leptonic



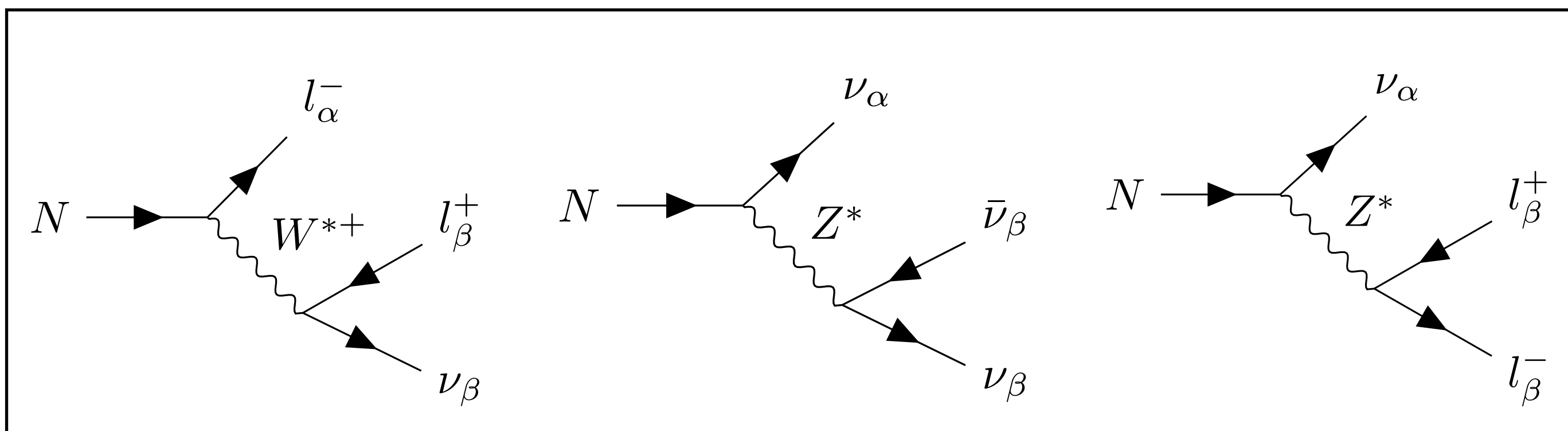
Hadronic



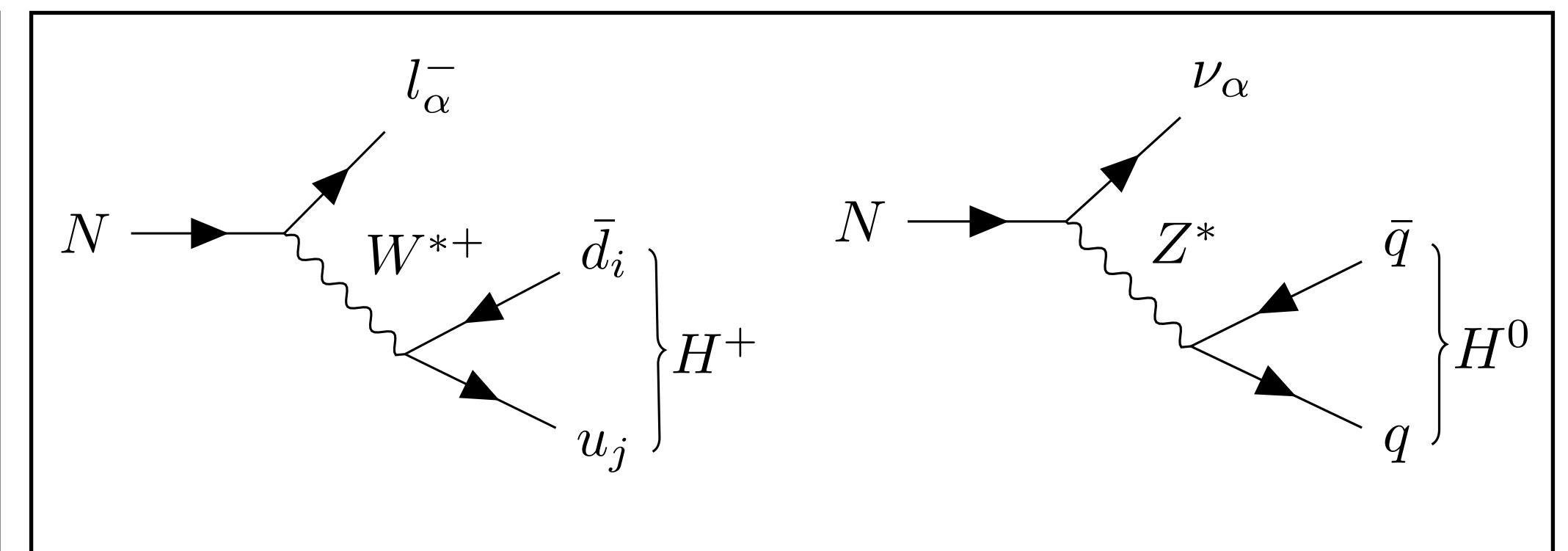
HNL Decay

- However, for $m_N > 1$ GeV, decays into single hadronic states are insufficient for modeling the total HNL decay width, since in this regime multi-meson final states become relevant
- We adopt the approach in Bondarenko et al. (1805.08567) for calculating the total hadronic width:
 - For $m_N < 1$ GeV: Single meson decays
 - For $m_N > 1$ GeV: Quark-level decays + QCD Correction

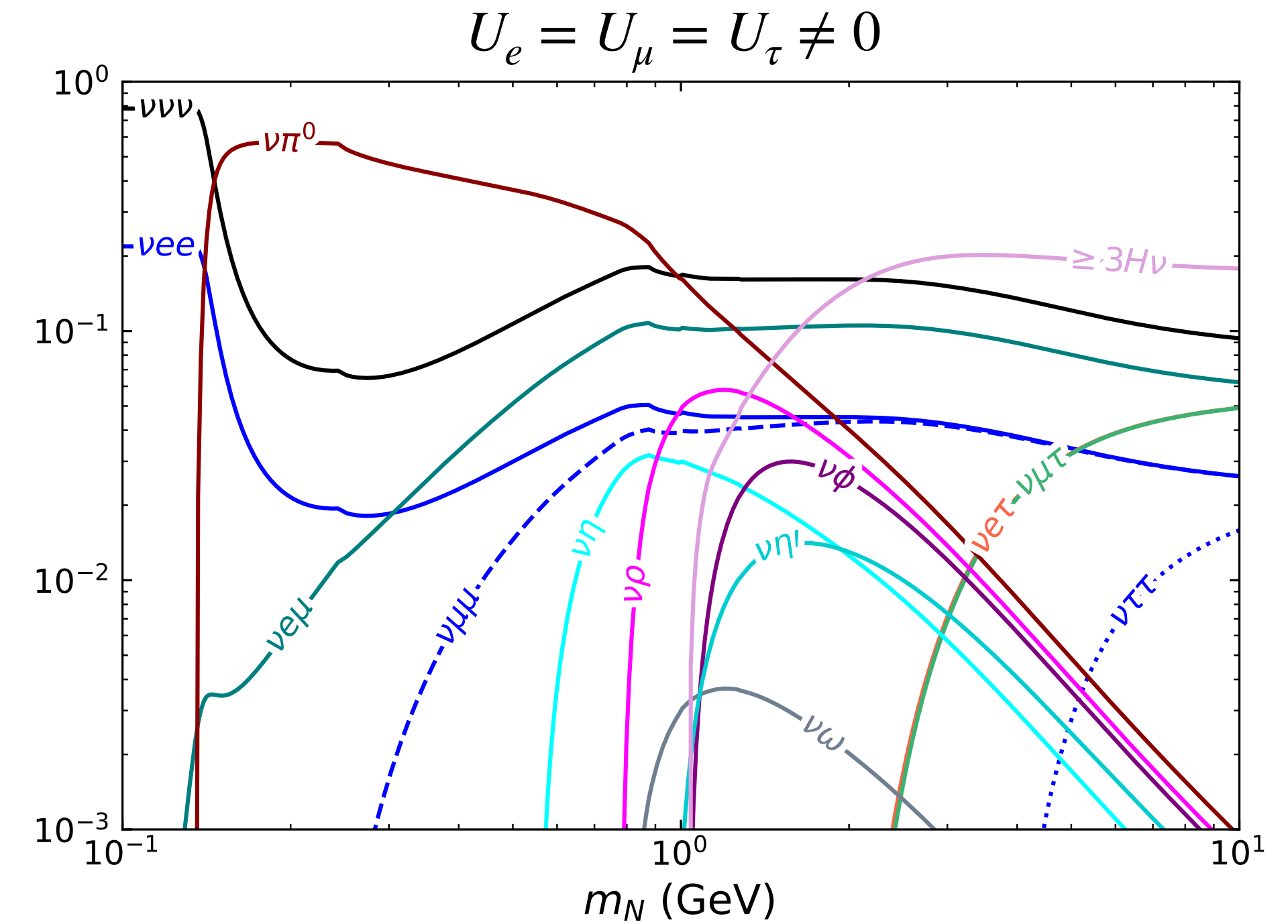
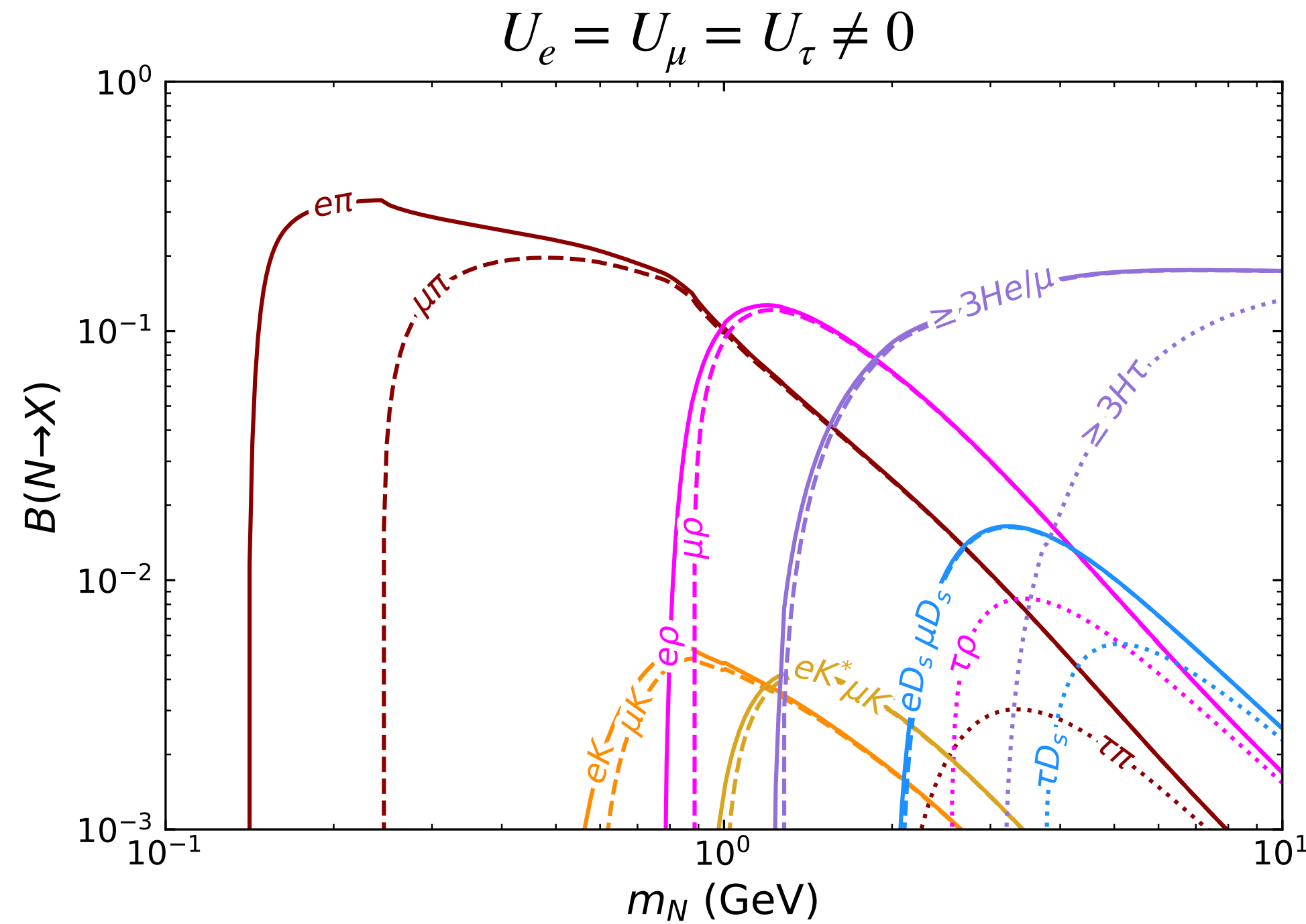
Leptonic



Hadronic



HNL Decay



- HNLCalc includes HNL decays into leptonic and hadronic final states
- HNLs with masses ~ 130 MeV - 1 GeV decay dominantly into $\pi^\pm/\pi^0$
- Above 1 GeV, multi-hadron and leptonic decays become most relevant

HNLCalc Summarized

HNL Decay Modes							
$\nu l^+ l^-$	Fig. 3 (a)	$\nu_l e^+ e^-$	$\nu_l \mu^+ \mu^-$	$\nu_l \tau^+ \tau^-$			
$l^\pm \nu_l l'^\mp$	Fig. 3 (b)	$l^\pm \nu_e e^\mp$	$l^\pm \nu_\mu \mu^\mp$	$l^\pm \nu_\tau \tau^\mp$			
$\nu_l \bar{\nu} \nu$	Fig. 3 (c)	$\nu_l \bar{\nu}_e \nu_e$	$\nu_l \bar{\nu}_\mu \nu_\mu$	$\nu_l \bar{\nu}_\tau \nu_\tau$			
$\nu_l H^0$	Fig. 3 (d)	$\nu_l \pi^0$	$\nu_l \eta$	$\nu_l \eta'$	$\nu_l \rho^0$	$\nu_l \omega$	$\nu_l \phi$
$l^\pm H^\mp$	Fig. 3 (e)	$l^\pm \pi^\mp$	$l^\pm K^\mp$	$l^\pm D^\mp$	$l^\pm D_s^\mp$	$l^\pm \rho^\mp$	$l^\pm K^{*\mp}$
$\nu_l q \bar{q}$	Fig. 3 (d)	$\nu_l u \bar{u}$	$\nu_l d \bar{d}$	$\nu_l s \bar{s}$	$\nu_l c \bar{c}$	$\nu_l b \bar{b}$	
$l^\pm u \bar{d}$	Fig. 3 (e)	$l^- u \bar{d}$	$l^- u \bar{s}$	$l^- u \bar{b}$	$l^+ \bar{u} d$	$l^+ \bar{u} s$	$l^+ \bar{u} b$
		$l^- c \bar{d}$	$l^- c \bar{s}$	$l^- c \bar{b}$	$l^+ \bar{c} d$	$l^+ \bar{c} s$	$l^+ \bar{c} b$

HNL Production Processes							
$P \rightarrow l N$	Fig. 1 (a)	$\pi^- \rightarrow l^+ N$	$K^+ \rightarrow l^+ N$	$D^+ \rightarrow l^+ N$	$D_s^+ \rightarrow l^+ N$	$B^+ \rightarrow l^+ N$	$B_c^+ \rightarrow l^+ N$
$P \rightarrow P' l N$	Fig. 1 (b)	$K^+ \rightarrow \pi^0 l^+ N$	$K_S \rightarrow \pi^+ l^- N$	$K_L \rightarrow \pi^+ l^- N$	$D^0 \rightarrow K^- l^+ N$	$\bar{D}^0 \rightarrow \pi^+ l^- N$	$D^+ \rightarrow \pi^0 l^+ N$
		$D^+ \rightarrow \eta l^+ N$	$D^+ \rightarrow \eta' l^+ N$	$D^+ \rightarrow \bar{K}^0 l^+ N$	$D_s^+ \rightarrow \bar{K}^0 l^+ N$	$D_s^+ \rightarrow \eta l^+ N$	$D_s^- \rightarrow \eta' l^+ N$
		$B^+ \rightarrow \pi^0 l^+ N$	$B^- \rightarrow \eta l^+ N$	$B^+ \rightarrow \eta' l^+ N$	$B^+ \rightarrow \bar{D}^0 l^+ N$	$B^0 \rightarrow \pi^- l^+ N$	$B^0 \rightarrow D^- l^+ N$
		$B_s^0 \rightarrow K^- l^+ N$	$B_s^0 \rightarrow D_s^- l^+ N$	$B_c^+ \rightarrow D^0 l^+ N$	$B_c^+ \rightarrow \eta_c l^- N$	$B_c^+ \rightarrow B^0 l^+ N$	$B_c^- \rightarrow B_s^0 l^+ N$
$P \rightarrow V l N$	Fig. 1 (b)	$D^0 \rightarrow \rho^- l^+ N$	$D^0 \rightarrow K^{*-} l^+ N$	$D^+ \rightarrow \rho^0 l^+ N$	$D^+ \rightarrow \omega l^+ N$	$D^+ \rightarrow \bar{K}^{*0} l^+ N$	$D_s^+ \rightarrow K^{*0} l^+ N$
		$D_s^+ \rightarrow \phi l^+ N$	$B^+ \rightarrow \rho^0 l^+ N$	$B^+ \rightarrow \omega l^+ N$	$B^+ \rightarrow \bar{D}^{*0} l^+ N$	$B^0 \rightarrow \rho^- l^+ N$	$B^0 \rightarrow D^{*-} l^+ N$
		$B_s^0 \rightarrow K^{*-} l^+ N$	$B_s^0 \rightarrow D_s^{*-} l^+ N$	$B_c^+ \rightarrow D^{*0} l^+ N$	$B_c^+ \rightarrow J/\psi l^+ N$	$B_c^+ \rightarrow B^{*0} l^+ N$	$B_c^+ \rightarrow B_s^{*0} l^+ N$
$\tau \rightarrow H N$	Fig. 1 (c)	$\tau^+ \rightarrow \pi^+ N$	$\tau^+ \rightarrow K^+ N$	$\tau^+ \rightarrow \rho^+ N$	$\tau^+ \rightarrow K^{*+} N$		
$\tau^+ \rightarrow l^+ \nu N$	Fig. 1 (d,e)	$\tau^+ \rightarrow l^+ \bar{\nu}_\tau N$	$\tau^+ \rightarrow l^+ \nu_l N$				

Over 150 production modes and 100 decay channels implemented!

Sources

Decay Formulae:

Coloma, Fernandez-Martinez, Gonzalez-Lopez, Hernandez-Garcia & Pavlovic [2007.03701](#)

Bondarenko, Boyarsky, Gorbunov, & Ruchayskiy [1805.08567](#)

Production Formulae:

Gorbunov & Shaposhnikov [0705.1729](#)

Form Factors:

Melikhov & Stech [0001113](#)

Decay Constants:

Faustov & Galkin [1212.3167](#)

Ivanov, Korner, & Santorelli [0007169](#)

Chang, Li X.N., Li X.Q., Su [1805.00718](#)

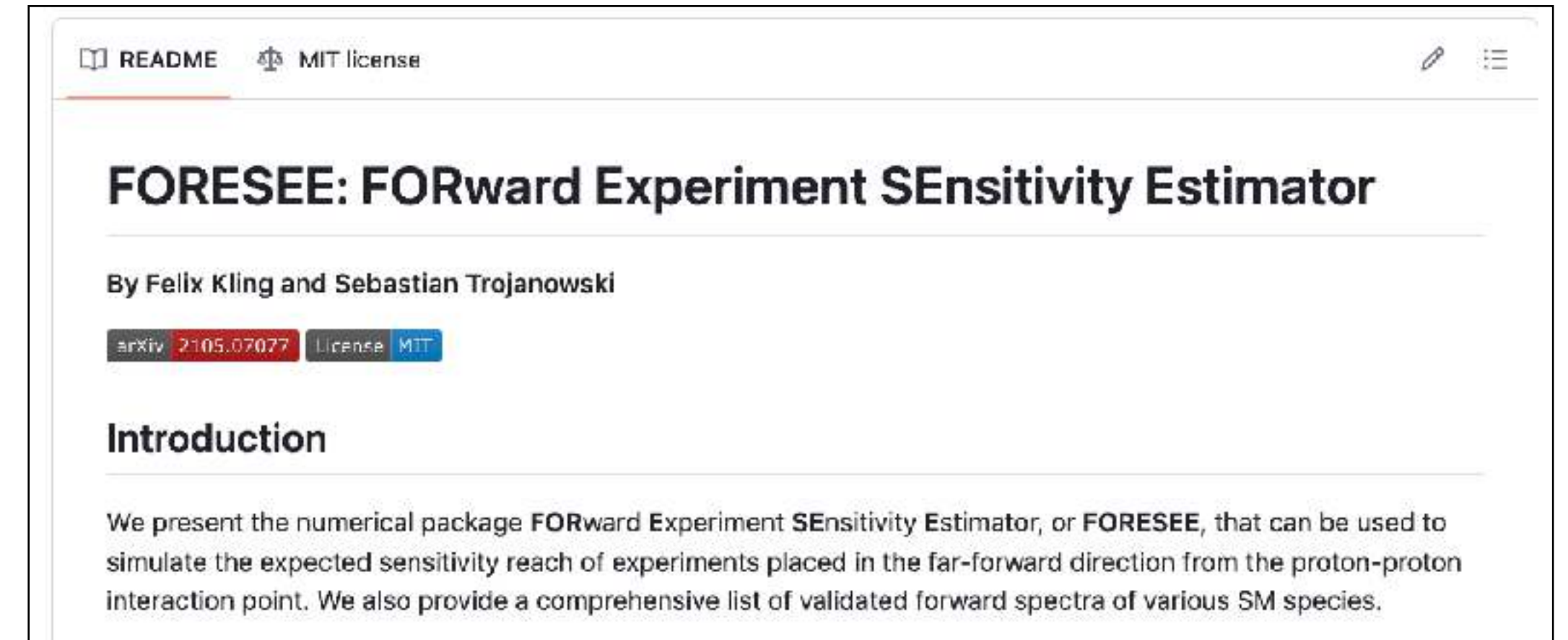
Feldmann [9907491](#)

From HNLCalc to FORESEE: The Full Pipeline

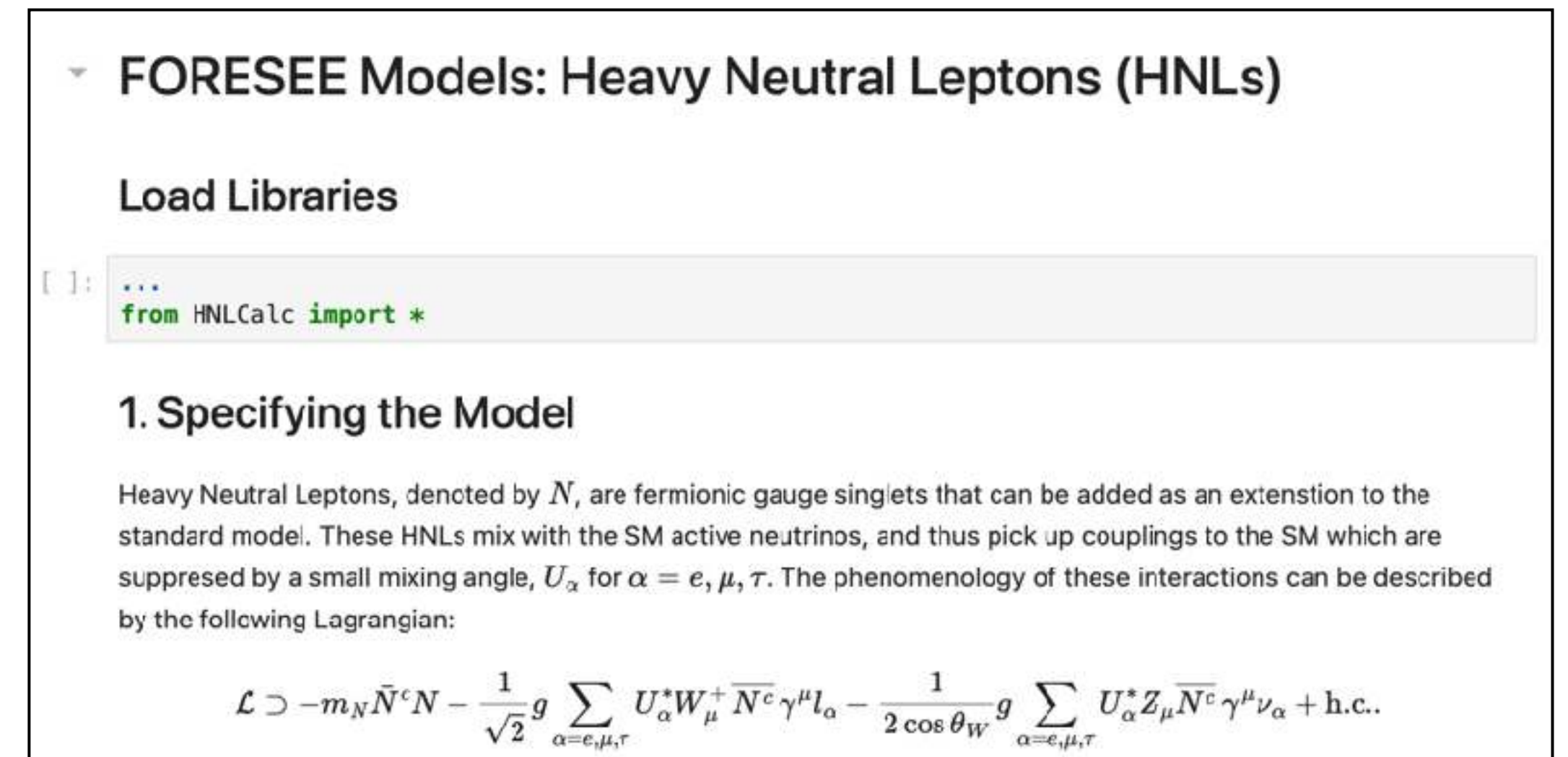
- HNLCalc computes the relevant phenomenological inputs to estimate the sensitivity of LLP experiments to HNLs for a given $(m_N, |U_{eI}|^2, |U_{\mu I}|^2, |U_{\tau I}|^2)$.
- **FORESEE** (Kling & Trojanowski 2105.07077): Monte Carlo event generator / sensitivity estimator for LLPs in the forward direction – existing library covers dark photons, dark Higgs, ALPs
- In our work, we used HNLCalc to **extend the FORESEE model library** to include HNLs, which we use to **compute sensitivity estimates of the FASER and FASER2** experiments to newly proposed HNL benchmarks.

FORESEE Master Formula

$$N = \int dE_N d\theta_N \frac{d^2 N_N}{dE_N d\theta_N} \varepsilon_{geom}(\theta_N) P_{dec}(\ell_N(E_N); L, \Delta) BR(N \rightarrow \text{vis})$$



Available on [GitHub](https://github.com/felixkling/FORESEE)

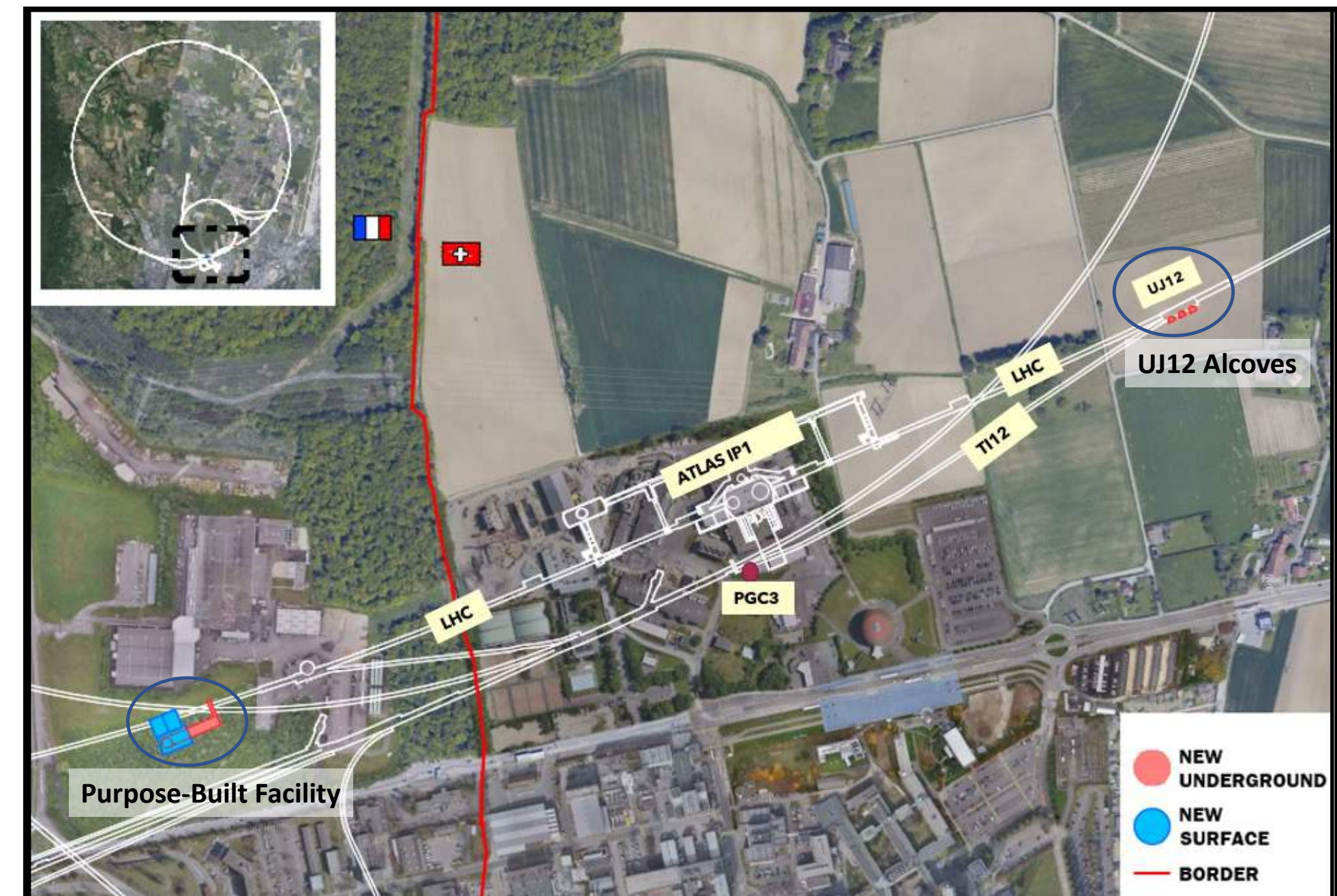


The Forward Physics Program at the LHC

- Two on-axis regions of interest:
 - **UJ12 / T112** (~480 m east of IP1, Swiss side): currently houses the **FASER** and FASERv experiments.
 - **Forward Physics Facility (FPF)** (~650 m west of IP1, French side): proposed underground cavern for the HL-LHC era
- FPF will host **FASER2**, FLArE (LArTPC for neutrinos and BSM scattering), FASERv2, FORMOSA (millicharged particles)
- **Run 3 vs. HL-LHC**: 250 fb^{-1} vs. 3000 fb^{-1} integrated luminosity – 12× more data



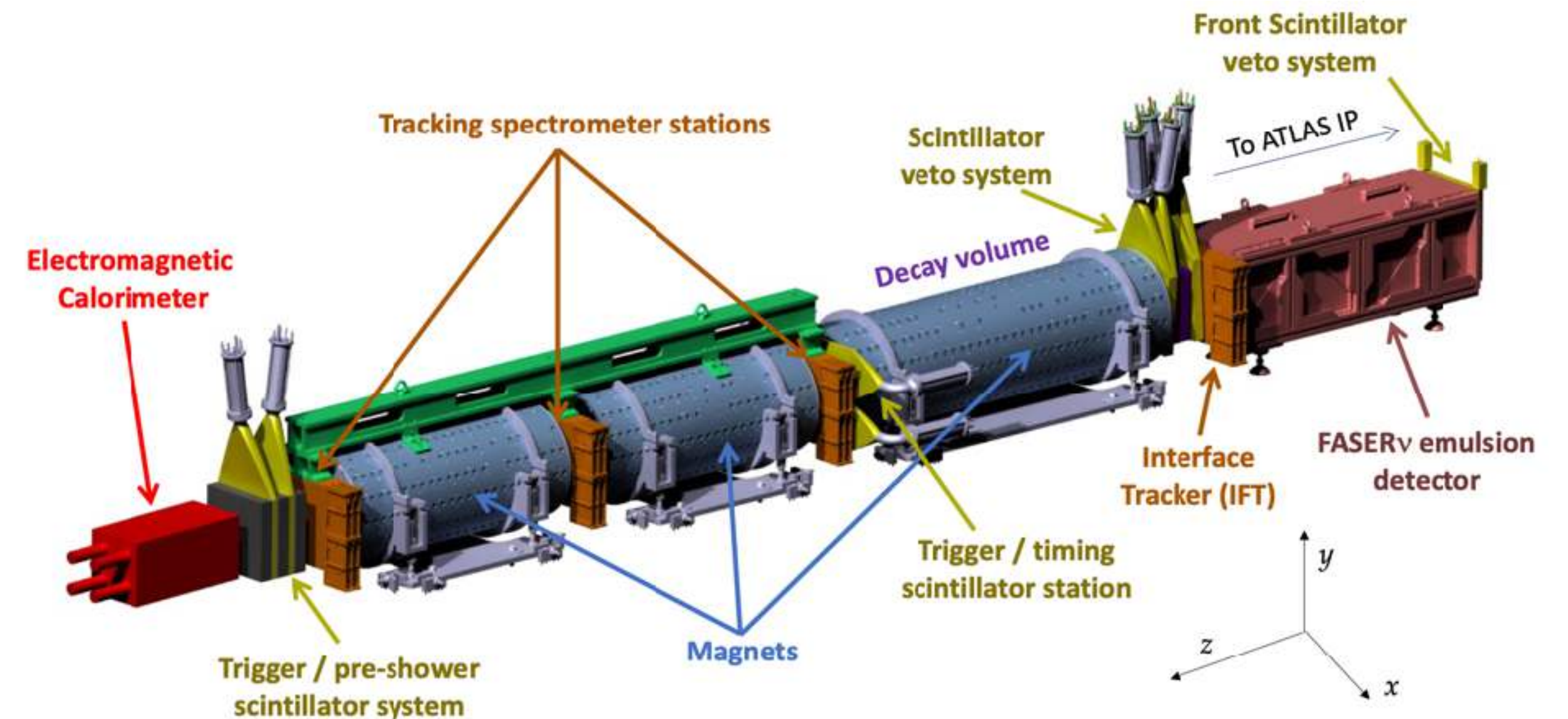
Google Images



Feng et. al 2203.05090

FASER

- The **ForwArd Search ExpeRiment (FASER)** is a currently operating experiment at the LHC designed to detect forward boosted LLPs produced at the ATLAS interaction point
- Located 480 meters down the beam axis, through 100m of rock
- Features a cylindrical decay volume with radius $R = 10$ cm and length $\Delta = 1.5$ m
- Angular coverage of $\theta \lesssim 0.2$ mrad ($\eta \gtrsim 9.2$)

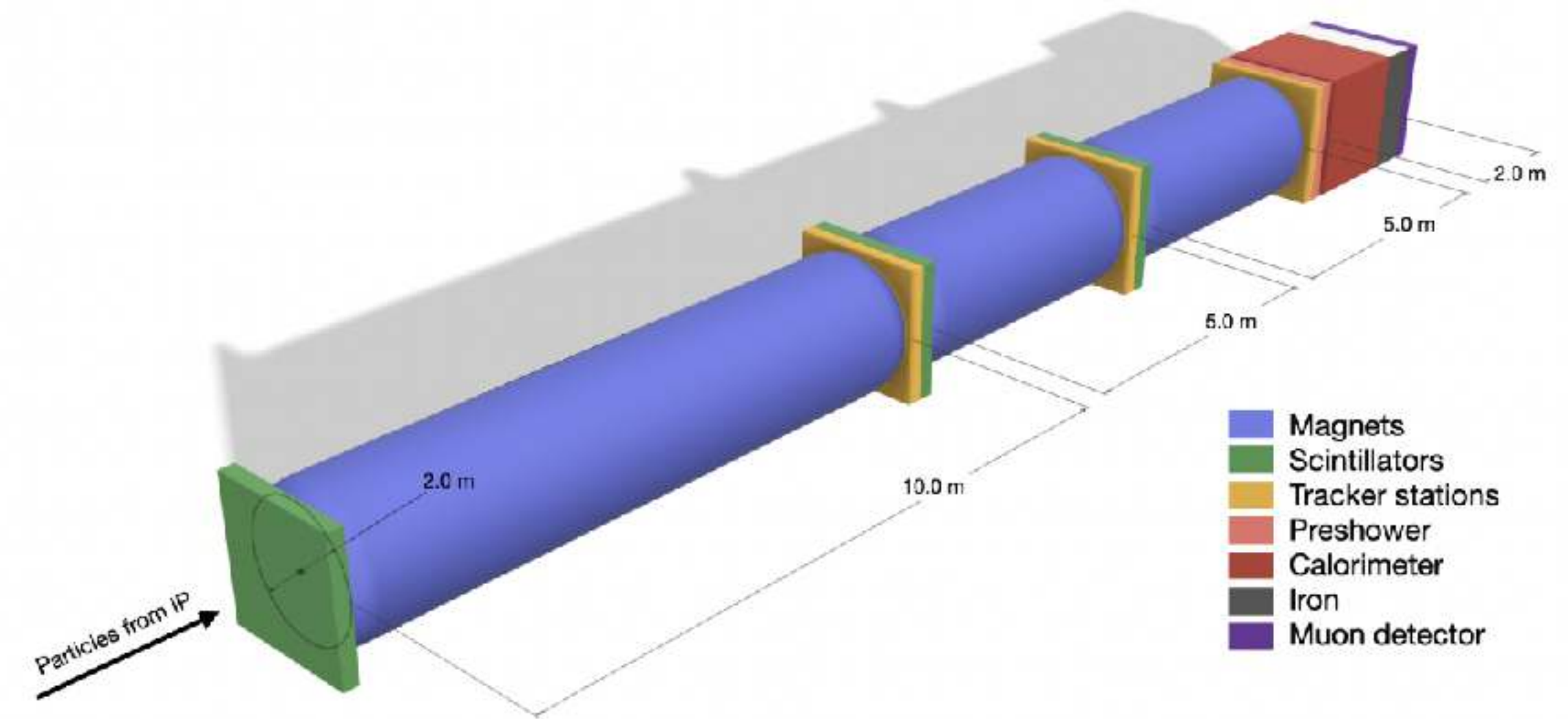


FASER Collaboration 2207.11427

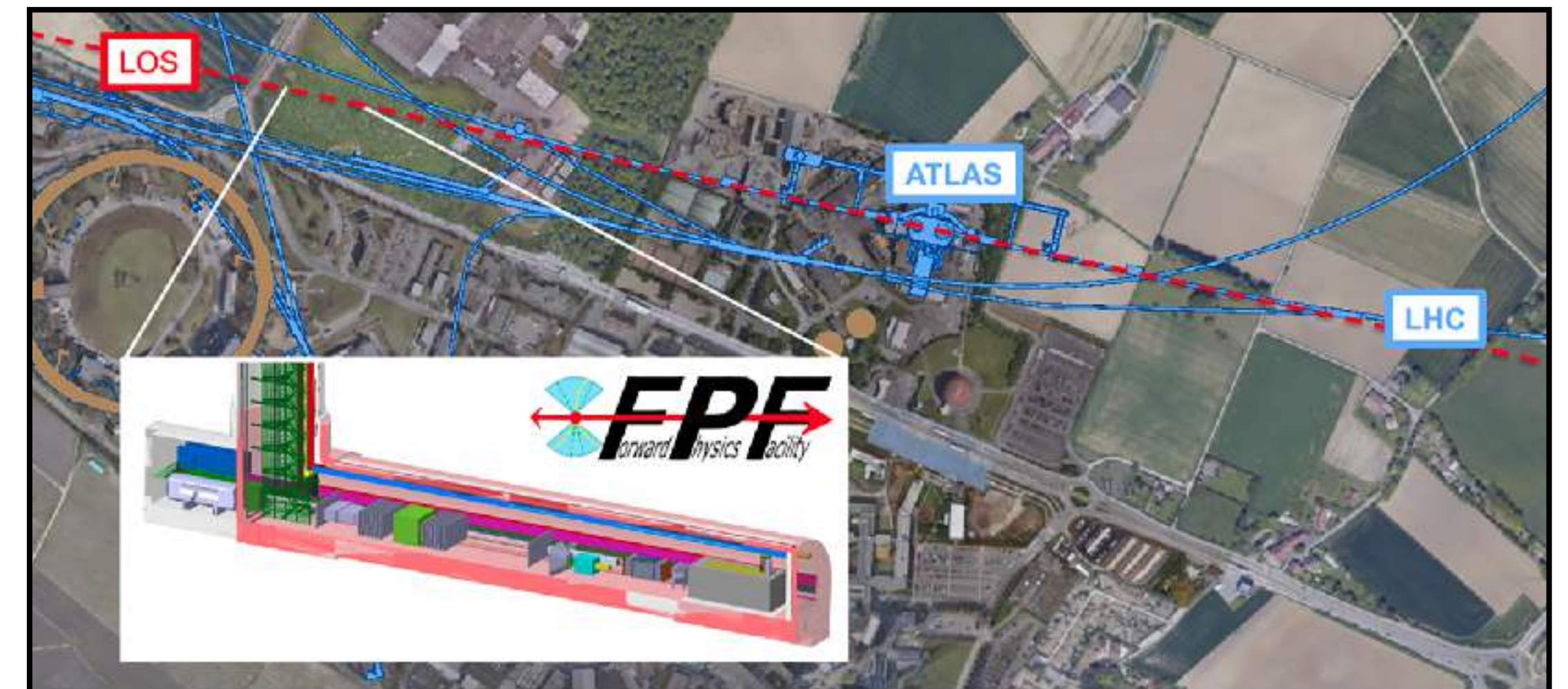


FPF: FASER2

- The Forward Physics Facility (FPF) is a proposed underground facility to house a suite of forward physics experiments during the High-Luminosity LHC era (HL-LHC)
- **FASER2 is a proposed FPF experiment** with an enlarged decay volume compared to FASER, which could extend the physics reach of the FASER program significantly
- Located 650 meters down the beam axis, with a rectangular decay volume with cross-section 3 m × 1 m and length $\Delta = 10$ m.
- Angular coverage of $\theta \lesssim 2.4$ mrad ($\eta \gtrsim 6.7$)



[Feng et al. 2203.05090]



[<https://fpf.web.cern.ch/>]

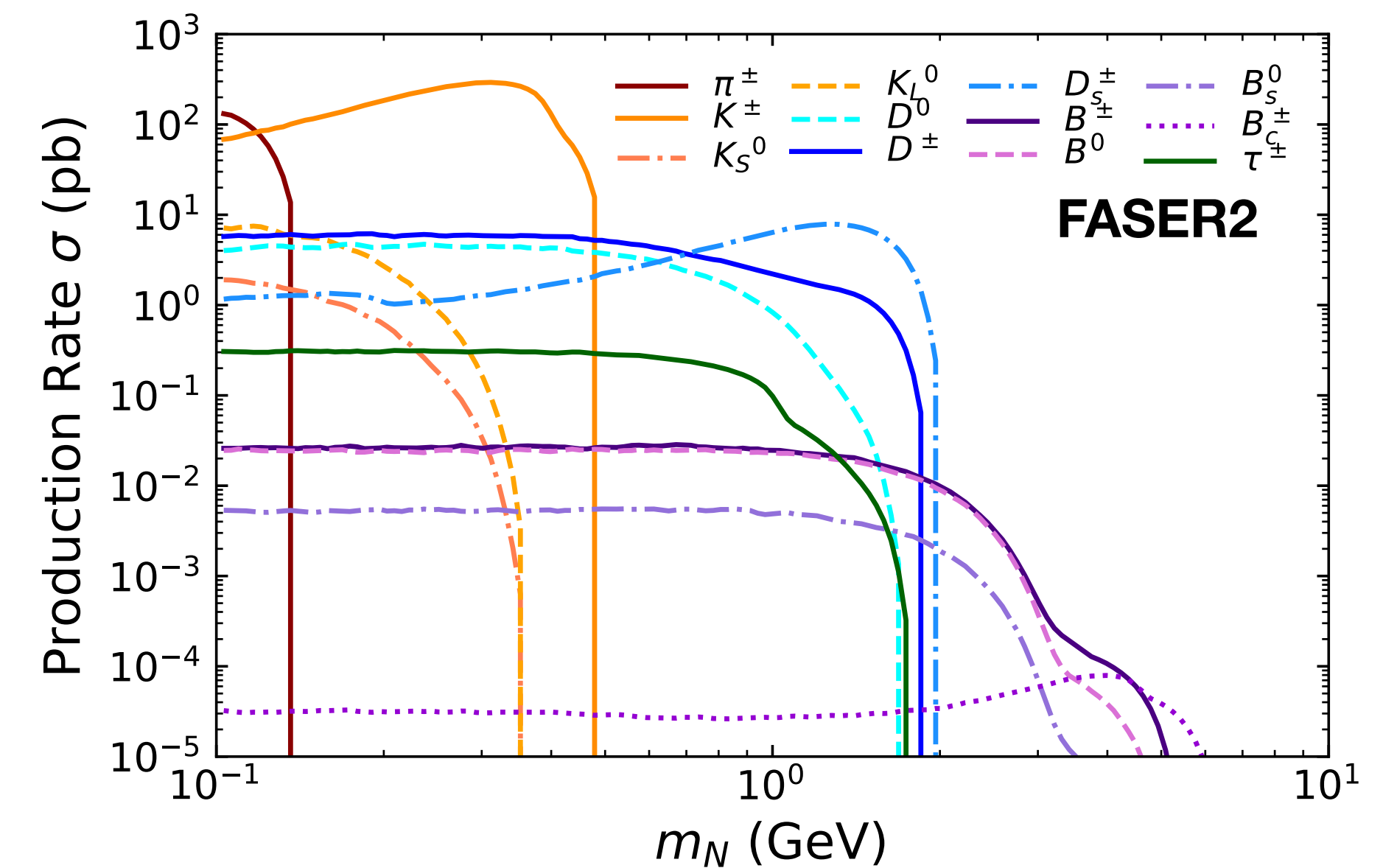
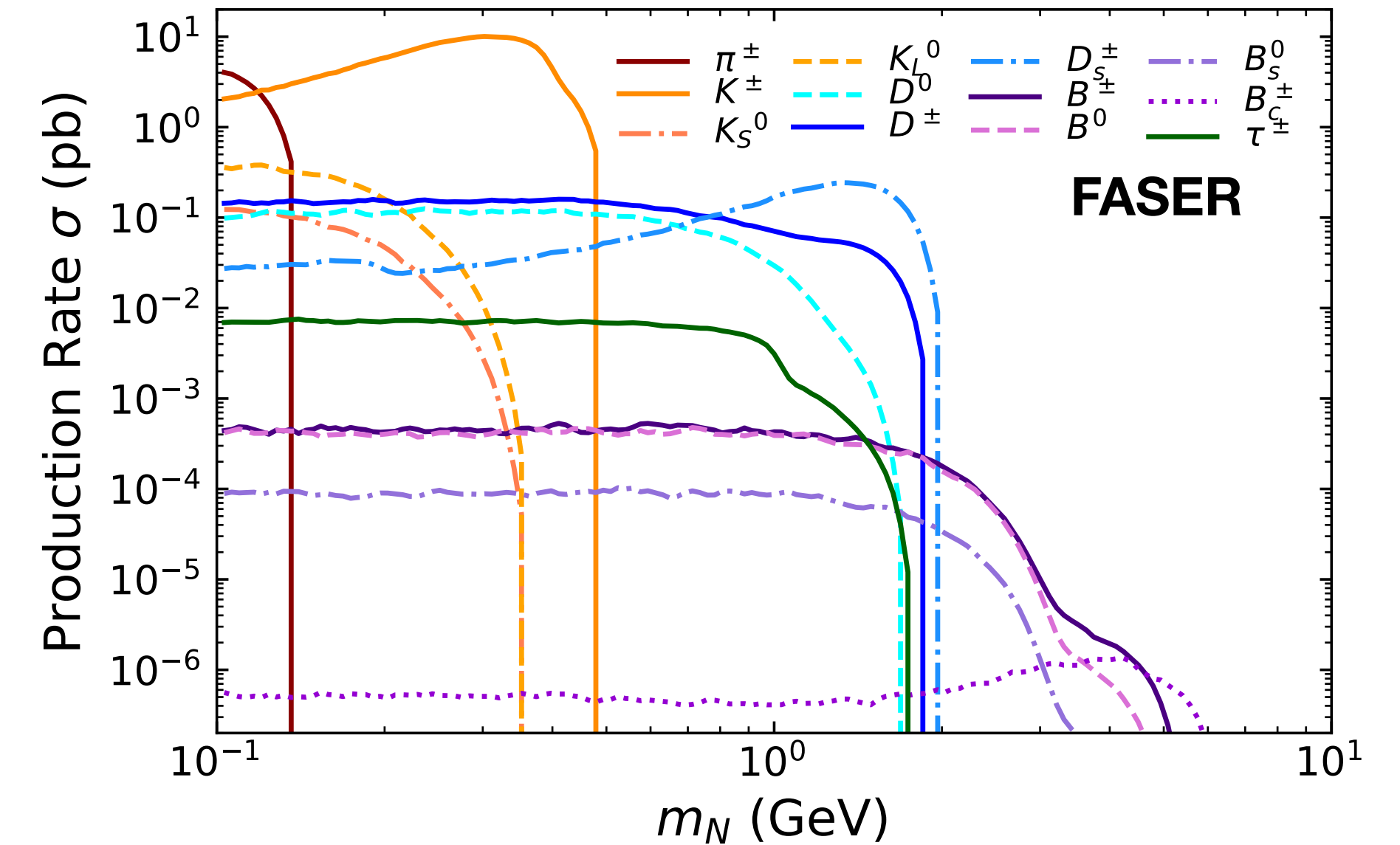


Theoretical Uncertainty: Forward Hadron Production

- FORESEE generates HNL production rates by convolving the HNLCalc branching ratios with the differential flux of forward hadrons which can decay into HNLs:

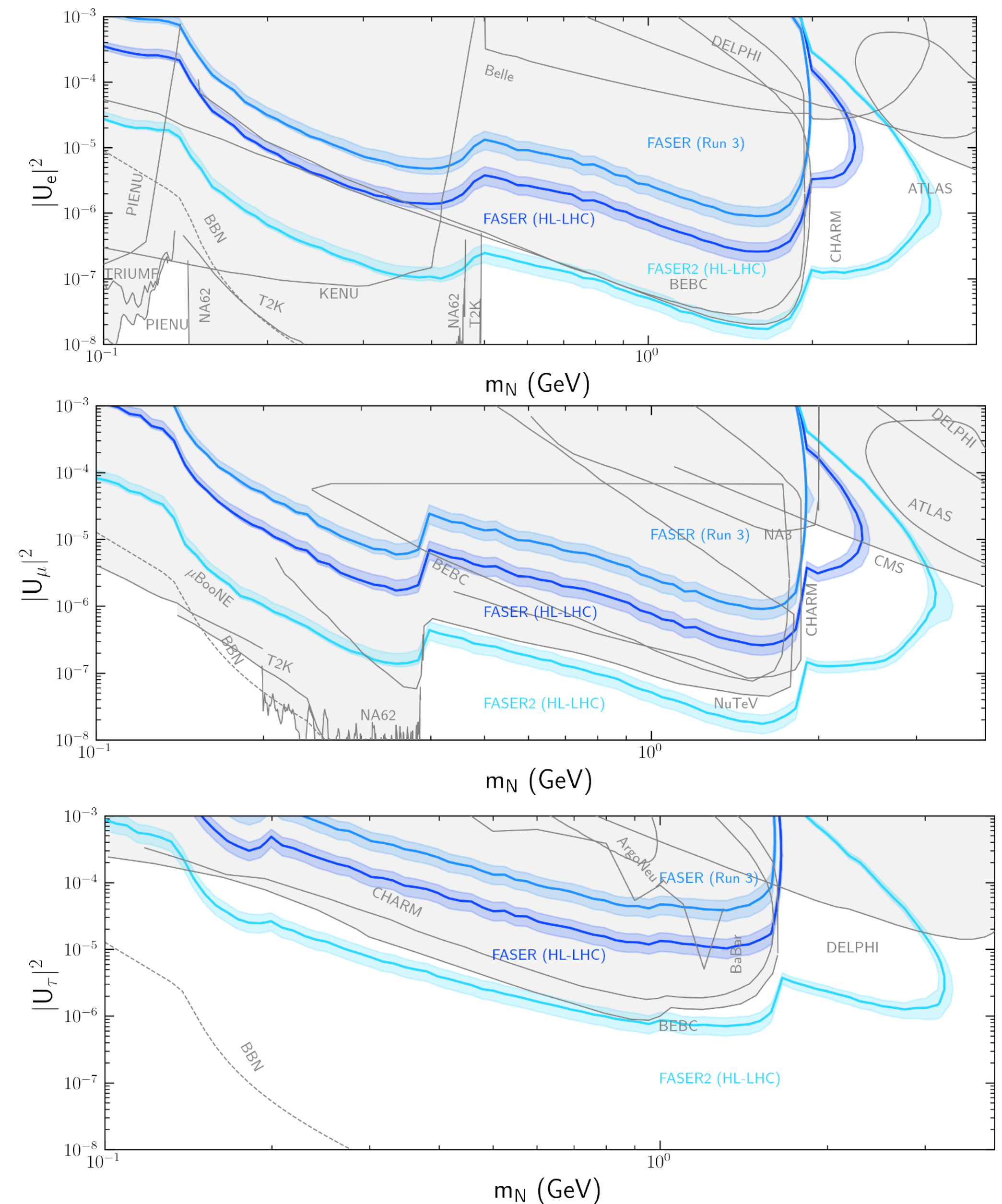
$$\frac{d^2 N_N}{dE_N d\theta_N} = \sum_P \int dE_P d\theta_P \frac{d^2 N_P}{dE_P d\theta_P} \frac{d^2 BR(P \rightarrow N)}{dE_N d\theta_N}(E_P, \theta_P; E_N, \theta_N)$$

- Leading uncertainty on HNL flux: **forward hadron production cross-sections.**
- Light hadrons (π, K): we compare the spread in predictions from EPOS-LHC, QGSJET 2.04, SIBYLL 2.3d
- Heavy hadrons (D, B) and τ : we compare POWHEG+PYTHIA predictions with factorization-scale variations



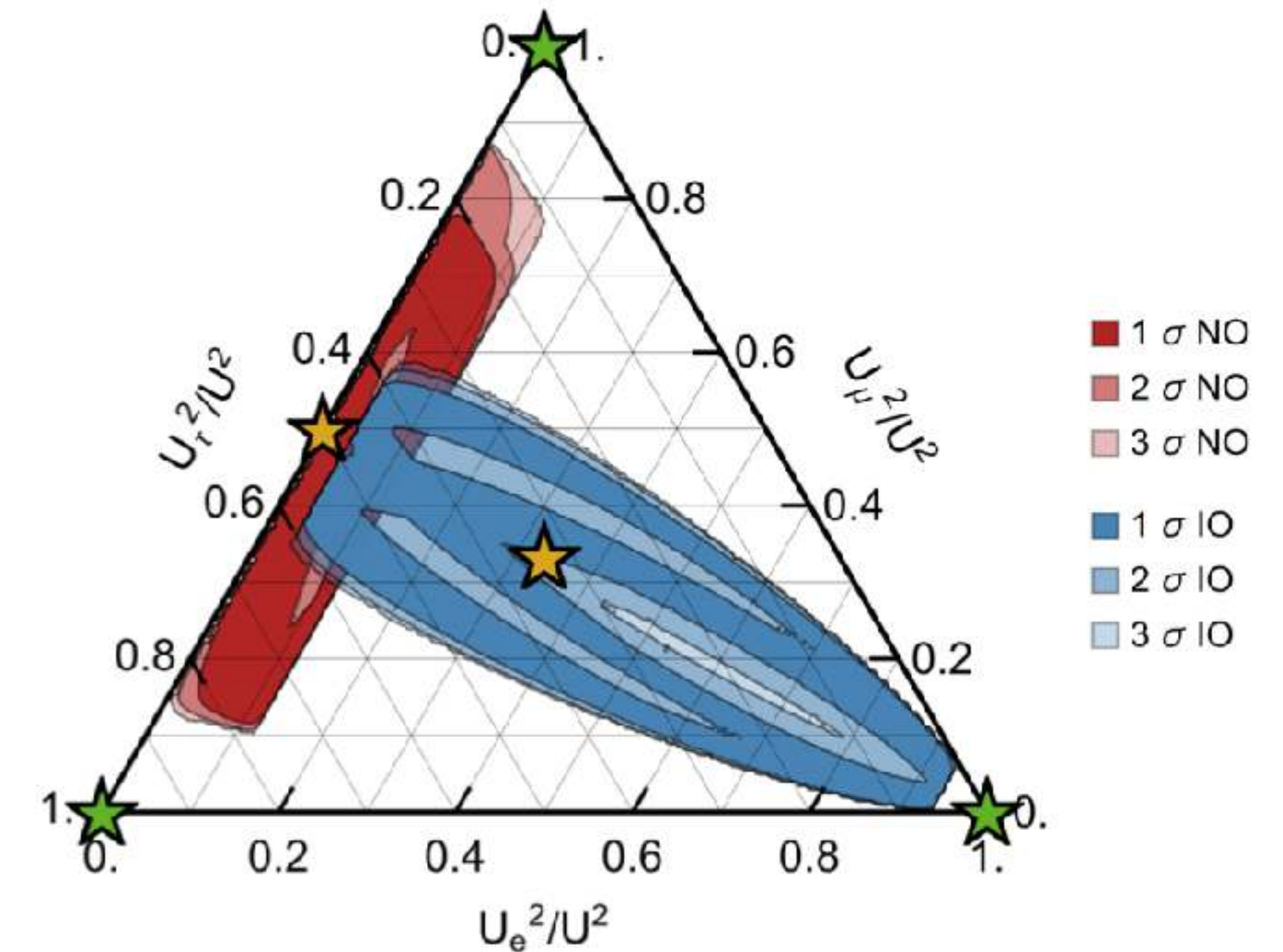
Sensitivity: Single-Coupling Benchmarks (100, 010, 001)

- We present results for FASER during Run 3 ($\mathcal{L} = 250 \text{ fb}^{-1}$), and FASER/FASER2 during HL-LHC ($\mathcal{L} = 3000 \text{ fb}^{-1}$) with shaded bands estimating the forward-hadron uncertainty.
- The reach extends into currently unconstrained parameter space, especially at FASER2 where there is a much larger decay volume.
- Accurately reproduces previous estimates, validating our simulation pipeline. (Kling & Trojanowski 2105.07077)



Mixed-Couplings: Realistic Benchmarks

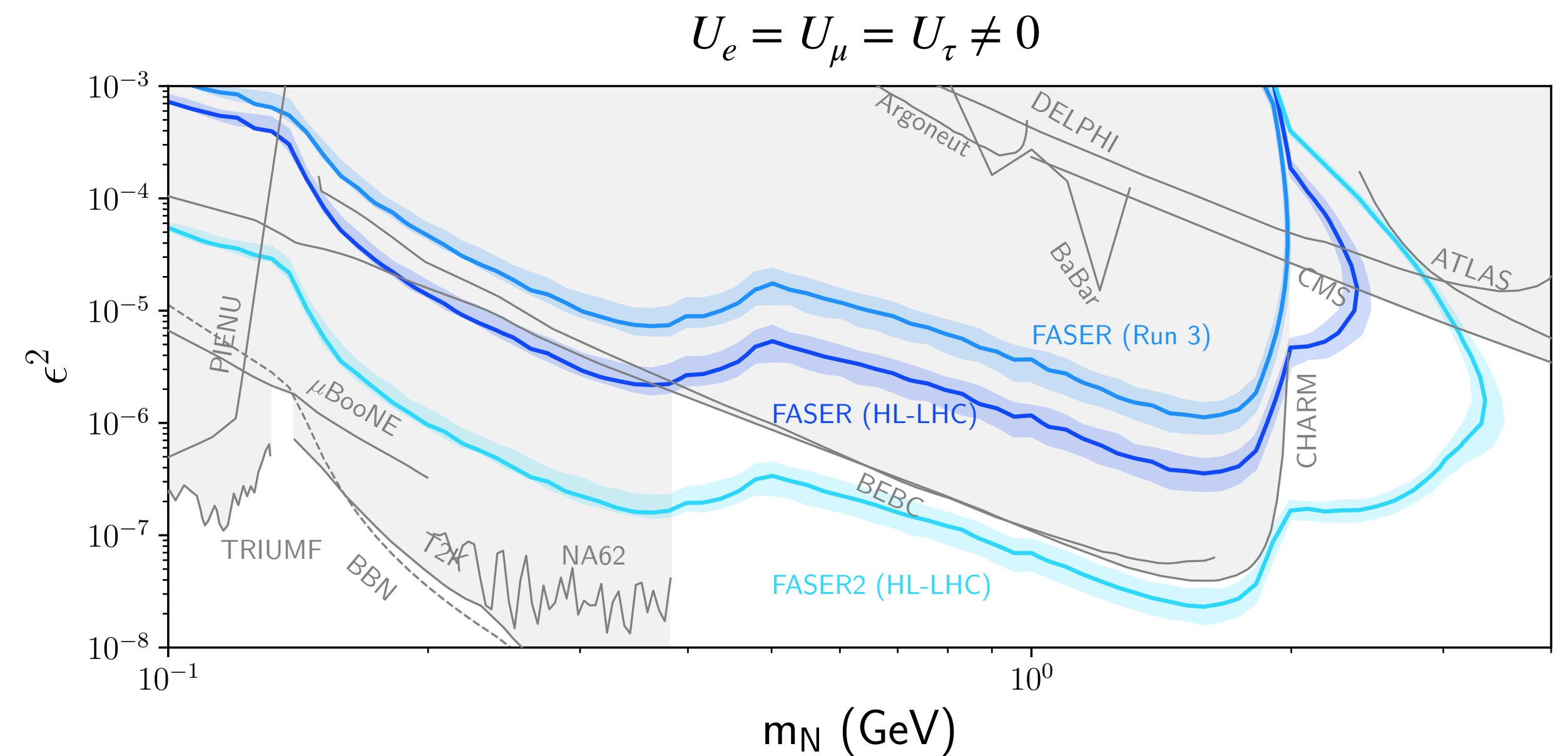
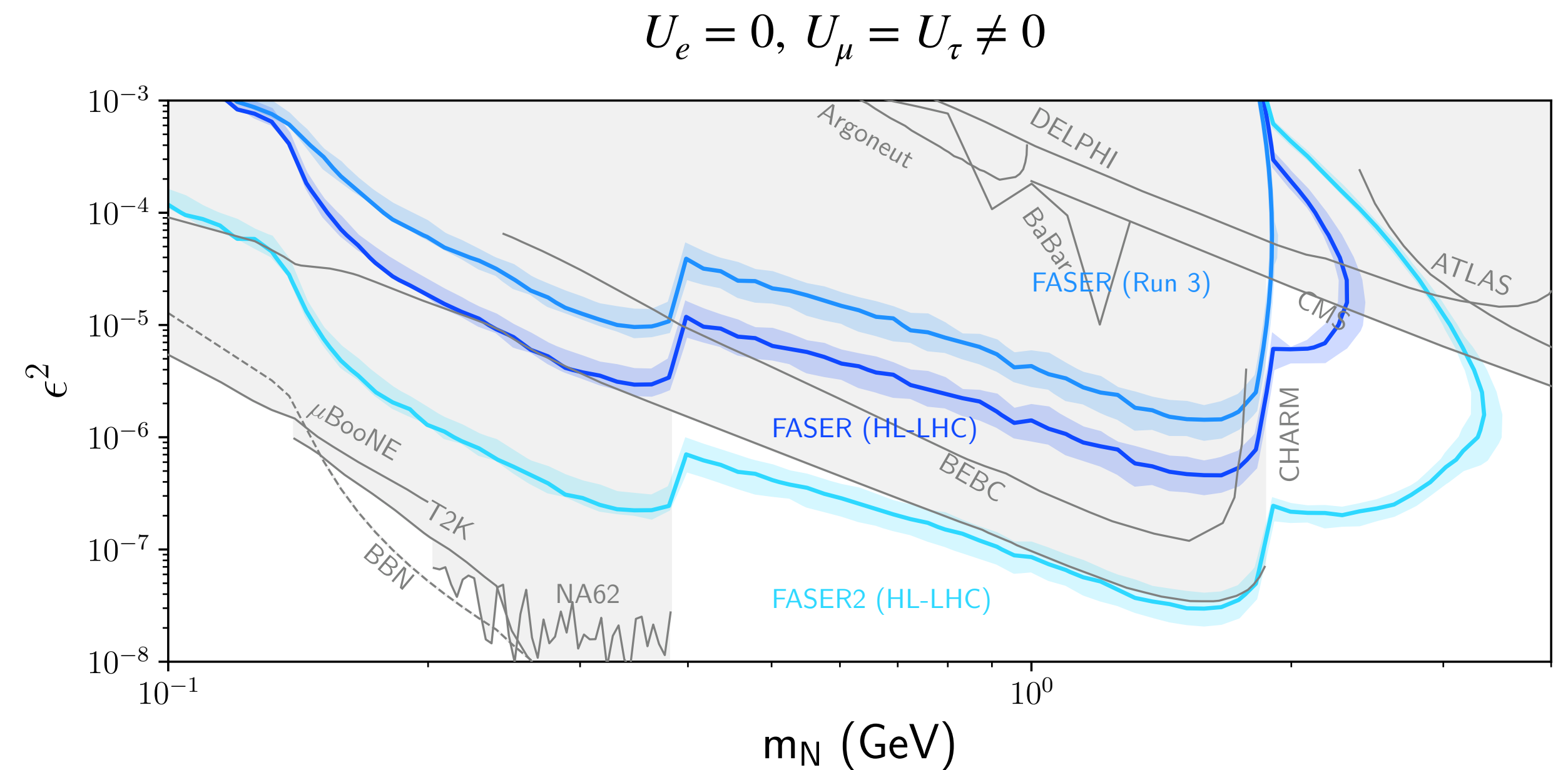
- Singly-coupled HNL models have served as well-defined simple benchmarks for experimental collaborations.
(Physics Beyond Colliders Working Group 1901.09966)
- However, realistic SM extensions – those that actually generate the observed neutrino masses and mixings – typically necessitate multiple non-zero U_α
- Recently, more realistic benchmarks have been proposed (Drewes, Klarić, & López-Pavón 2207.02742):
 - "011": $U_\mu = U_\tau \neq 0, U_e = 0$
 - "111": $U_e = U_\mu = U_\tau \neq 0$



Drewes et. al 2207.02742

Sensitivity: Mixed-Coupling Benchmarks (011, 111)

- We present results for FASER during Run 3 ($\mathcal{L} = 250 \text{ fb}^{-1}$), and FASER/FASER2 during HL-LHC ($\mathcal{L} = 3000 \text{ fb}^{-1}$)
- We find that the HL-LHC upgrade will enhance the reach of FASER/FASER2 into unconstrained parameter space, with FASER2 seeing a significant increase in sensitivity due to its larger decay volume
- **Results are easily obtained for different coupling ratios using the FORESEE Module!**
- Bounds from previous experiments are estimated from singly-coupled studies



Modeling & Discovery Summary (Chapter 3)

- We produced HNLCalc & the FORESEE HNL module: a public, end-to-end pipeline for forward HNL phenomenology with arbitrary coupling ratios.
- Using this pipeline, we estimated the sensitivity of the FASER experiment, and its planned successor FASER2, to more recently proposed mixed-flavor HNL benchmarks.
- These tools are publicly available and can be extended to a large variety of different collider/fixed target experiments (DUNE, FCC, etc.)
- We have now shown **where** to look for HNLs, but now we want to answer the question: *What if we actually detect a signal?*

Model Characterization: Dirac vs. Majorana

Collaboration: Jonathan L. Feng, **Alec Hewitt**, **Daniel La Rocco**, and Daniel Whiteson

PHYSICAL REVIEW D **113**, 035006 (2026)

**Characterizing heavy neutral leptons: Measuring parameters,
discriminating Majorana versus Dirac, and using FASER2
as a trigger for ATLAS**

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Department of Physics and Astronomy, [University of California](#), Irvine, California 92697, USA

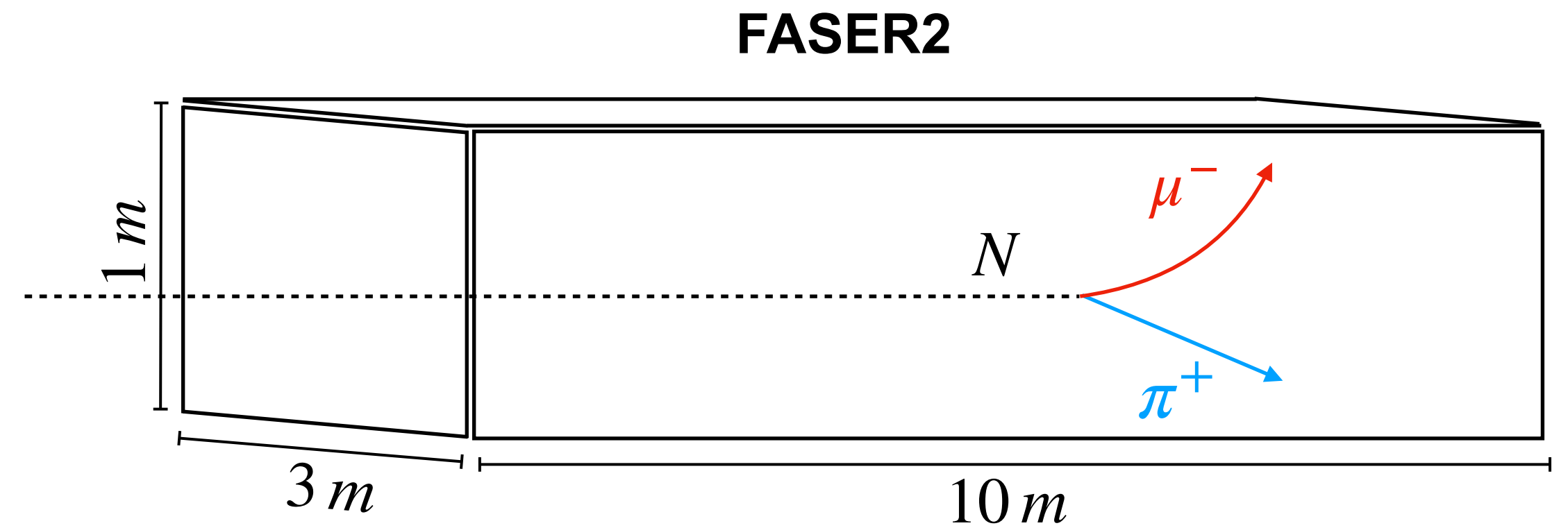


(Received 30 October 2025; accepted 12 January 2026; published 4 February 2026)

[PRD 113, 035006 \(2026\)](#)

Signal Characterization

- In the event an HNL is detected, much work will be done to identify the underlying nature of the observed signal excess.
- In this work, **we assume that a discovery has been made** and answer the following questions:
 - *How well could FASER2 estimate the HNL model parameters?*
 - *Would it be possible to discriminate between Dirac and Majorana HNLs using FASER2?*
 - *Can FASER2 as a trigger for ATLAS improve model discrimination?*
- We answer these questions by performing a likelihood based statistical analysis on fake HNL events simulated in our new FORESEE pipeline.



Benchmark Scenarios

- Simplifications:** We assume that $U_e = U_\tau = 0$ and restrict our focus to the fully reconstructible $N \rightarrow \mu\pi$ decay channel.
- Model 1 (High Statistics):**
 - 1.84 GeV Majorana HNL with $|U_\mu| = 3.6 \times 10^{-3}$
 - predicts ~ 8600 $N \rightarrow \mu\pi$ events at FASER2
 - $D_s \rightarrow \mu N$ dominates production
- Model 2 (Low Statistics):**
 - 2.0 GeV Majorana HNL with $|U_\mu| = 2.0 \times 10^{-3}$
 - predicts ~ 80 $N \rightarrow \mu\pi$ events at FASER2

		Model 1		Model 2		
HNL Model Parameters		$m_N = 1.84 \text{ GeV}, U_\mu = 0.0036, \text{Majorana}$		$m_N = 2.00 \text{ GeV}, U_\mu = 0.002, \text{Majorana}$		
CT_N		2.03 m		4.33 m		
$\langle E_N \rangle_{\text{FASER2}}$		820 GeV		828 GeV		
$\langle E_N \rangle_{\text{FASER2+ATLAS}}$		458 GeV		452 GeV		
			FASER2		FASER2	FASER2+ATLAS
Production Modes	N_{events} at HL-LHC (in 3 ab^{-1})	$D_s \rightarrow \mu N$	119,000	$B \rightarrow D^{*0} \mu N$	447	56.1
		$B \rightarrow D^{*0} \mu N$	1670	$B^0 \rightarrow D^* \mu N$	405	50.7
		$B^0 \rightarrow D^* \mu N$	1570	$B \rightarrow D^0 \mu N$	199	26.4
		$B_s^0 \rightarrow D_s^* \mu N$	333	$B^0 \rightarrow D \mu N$	183	22.4
Decay Modes	Branching Fraction	$\mu\rho$	0.187	$\mu\rho$	0.162	
		$\nu\nu\nu$	0.129	$\nu\nu\nu$	0.128	
		$\nu e\mu$	0.125	$\nu e\mu$	0.125	
		$\nu\mu\mu$	0.072	$\nu\mu\mu$	0.073	
		$\mu\pi$	0.071	$\mu\pi$	0.060	
		$\nu\pi^0$	0.038	$\nu\pi^0$	0.032	
		Other	0.378	Other	0.419	
Total FASER2 $N \rightarrow \mu\pi$ Events			8610		81.8	10.6

Kinematic Observables

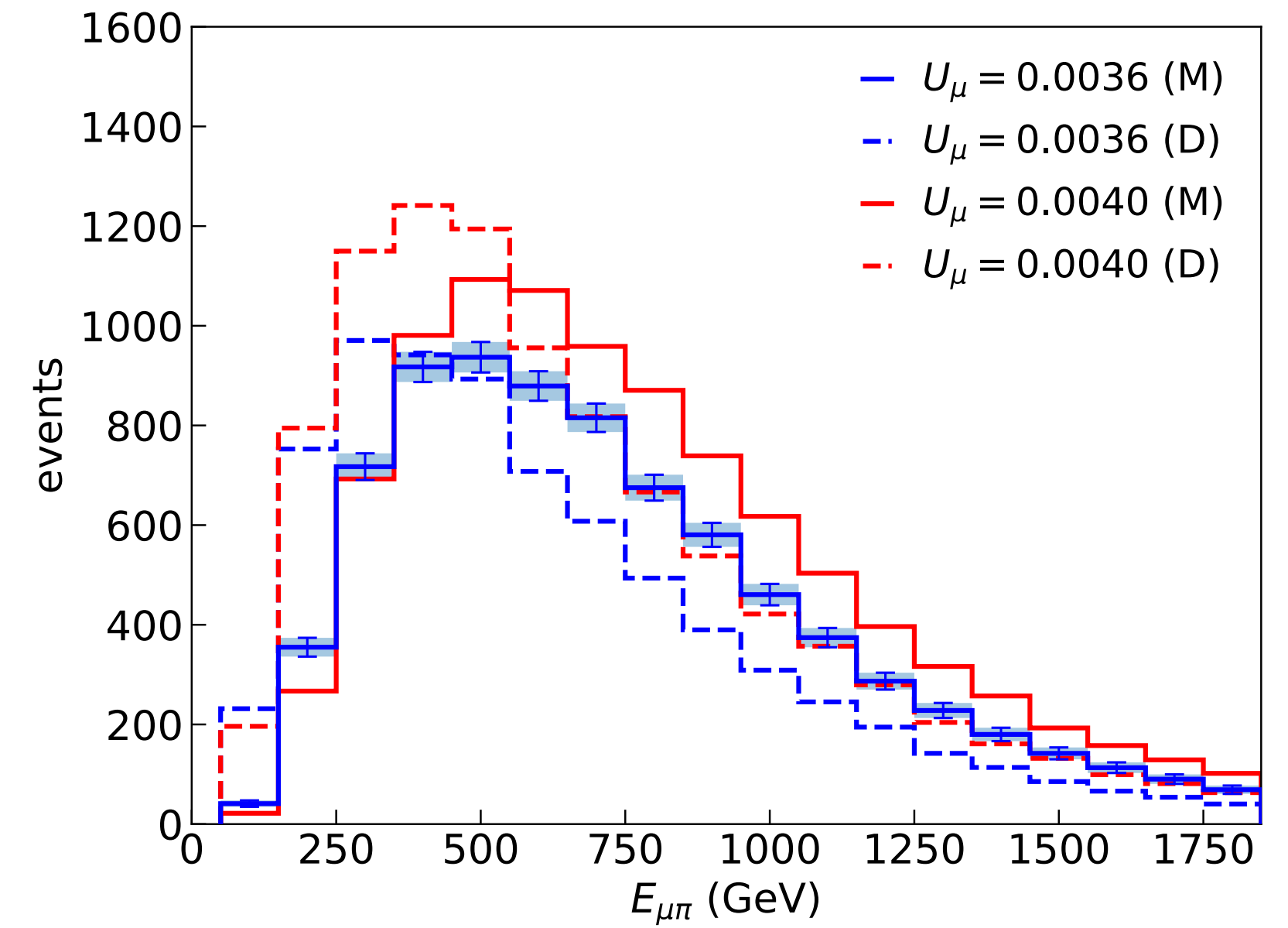
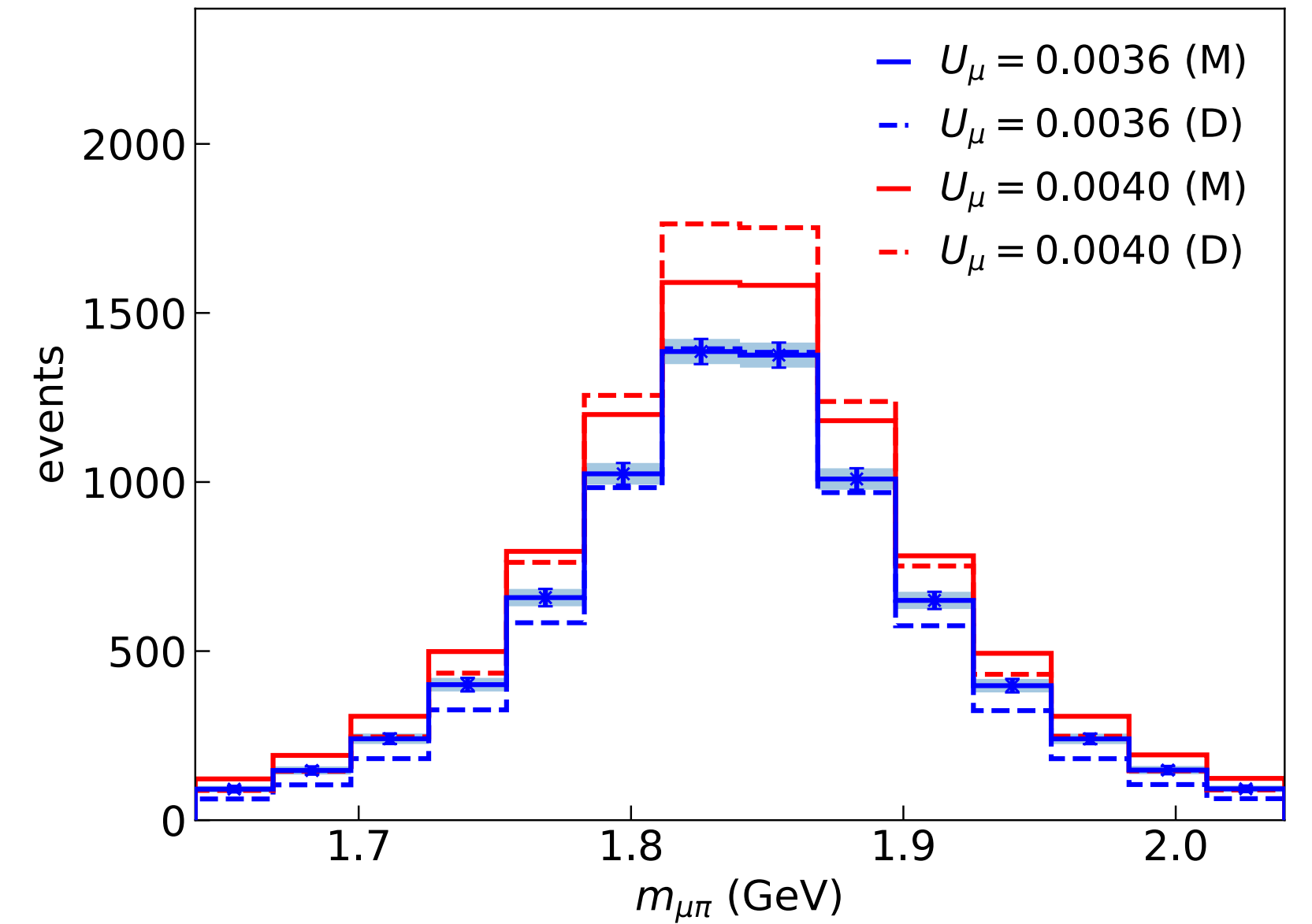
$$m_{\mu\pi}^2 = m_\mu^2 + m_\pi^2 + 2(E_\mu E_\pi - p_\mu p_\pi \cos \theta_{\mu\pi})$$

- Dual-readout hadronic calorimeter
 - Combines scintillation and Cherenkov signals for high precision measurement of the pion energy:

$$\frac{\delta E_\pi}{E_\pi} = \frac{35\%}{\sqrt{E_{\pi^\pm}/\text{GeV}}} \oplus 0.5\%$$

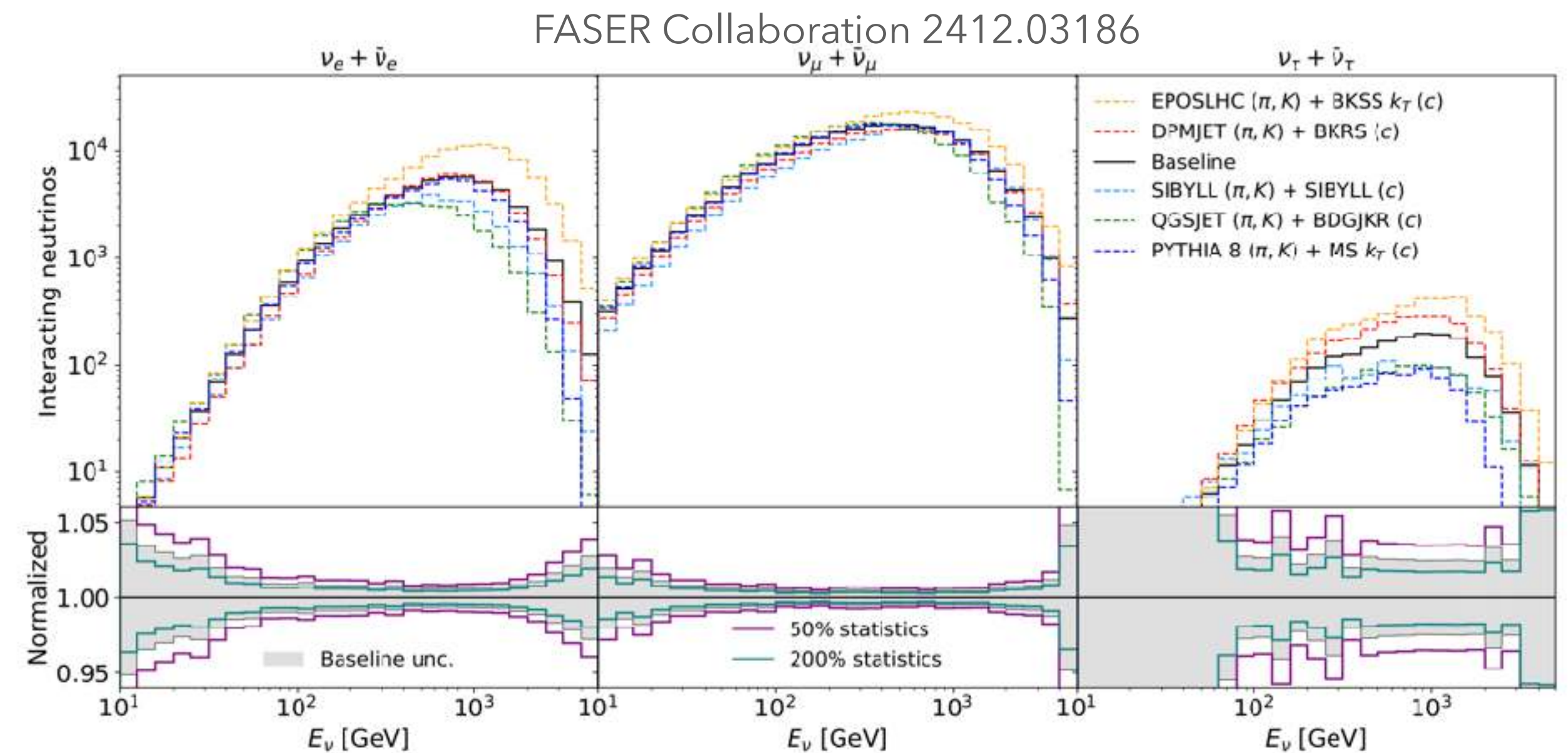
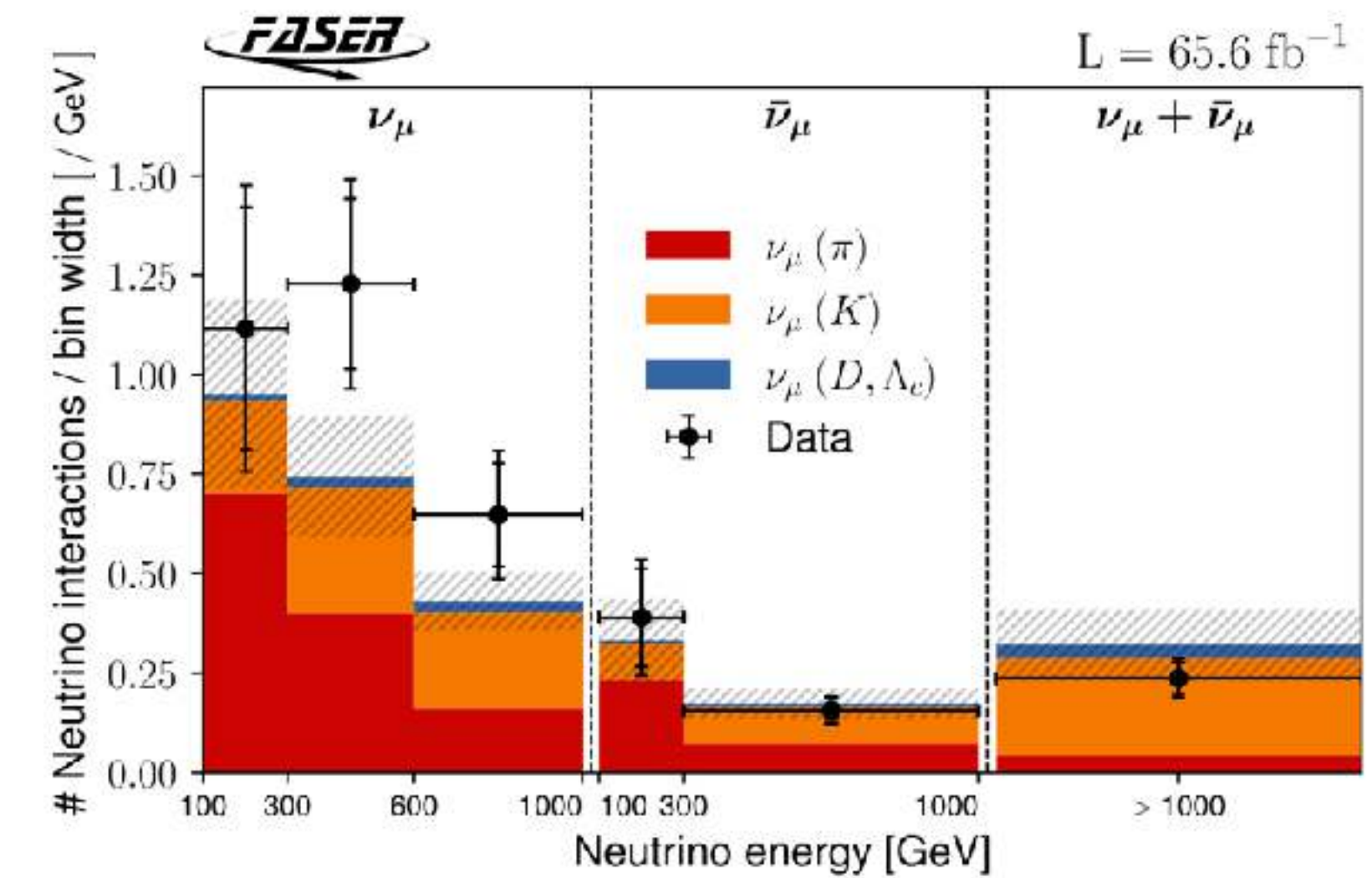
- Scintillating Fiber Tracking Stations
 - Measure muon momentum and angular separation:

$$\frac{\delta p_\mu}{p_\mu} = 3.3 \times 10^{-5} \left[\frac{p_\mu}{\text{GeV}} \right], \quad \delta \theta_{\mu\pi} = 250 \mu\text{rad}$$



Theoretical Uncertainty

- The leading theoretical uncertainty in the HNL flux comes from **forward hadron production**
 - Hadron production in the forward direction at the LHC has not been measured historically
 - MC generators are not optimally tuned for the forward direction
- Fortunately, modeling uncertainties will certainly be reduced as FASER ν continues to measure forward neutrino fluxes
 - “Latest neutrino results from the FASER experiment and their implications for forward hadron production” ([Ohashi et al. 2025](#))



Kling, Makela, and Trojanowski 2309.10417

Statistical Framework

- We perform an asymptotic likelihood-based analysis to obtain the expected constraints on the HNL model parameters using Wilk's Theorem.
- Nuisance parameters are incorporated to represent the theoretical uncertainties in the HNL flux normalization and shape
- We consider the following scenarios:
 - $\sigma_\eta = 60\%$ and $\sigma_\gamma = 10\%$ (Current)
 - $\sigma_\eta = 5\%$ and negligible σ_γ (Future)

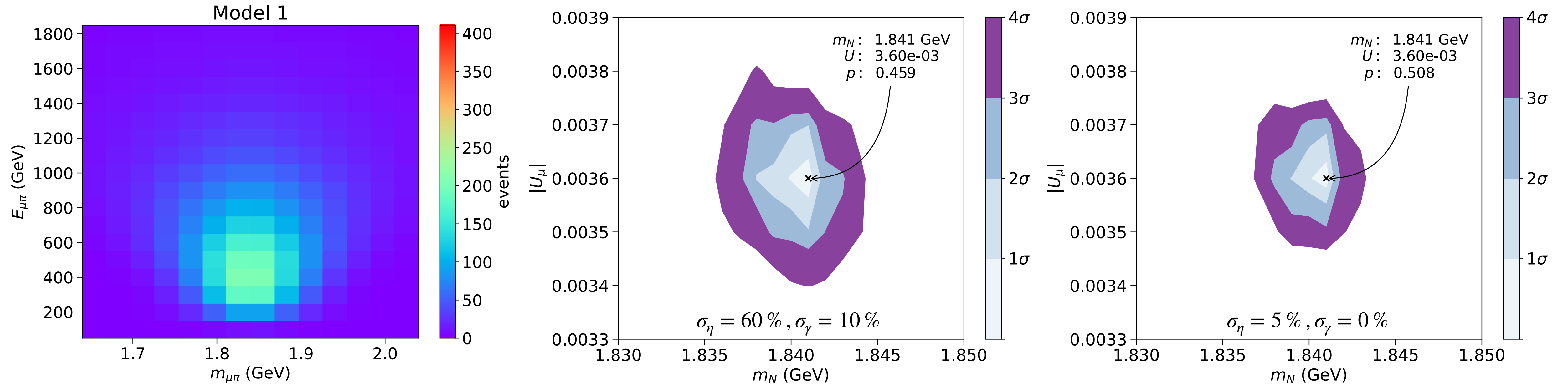
$$L(\vec{d}, m_N; \eta, \vec{\gamma}) = \prod_i f_P(d_i | \eta \gamma_i \mu_i) \underbrace{f_G(1 | \eta, \sigma_\eta)}_{\text{Normalization uncertainty}} \underbrace{f_G(1 | \gamma, \sigma_\gamma \oplus \sigma_i^{MC})}_{\text{Bin-by-bin uncertainty (Shape)}}$$

$$\tilde{t}(\vec{d} | m_N, U_\mu) := -2 \ln \frac{L(\vec{d} | m_N, U_\mu; \hat{\eta}, \hat{\vec{\gamma}})}{L(\vec{d} | \hat{m}_N, \hat{U}_\mu; \hat{\eta}, \hat{\vec{\gamma}})}$$

- d_i : Observed bin-count
- $\mu_i(m_N, U_\mu)$: Mean bin count prediction
- Nuisance Parameters: η, γ_i
- Theoretical Uncertainties:
 - σ_η : Normalization (60% , 5%)
 - σ_γ : Uncorrelated "Shape" (10% , 0%)
 - σ_i^{MC} : MC Statistics

Parameter Estimation: High Statistics (Model 1)

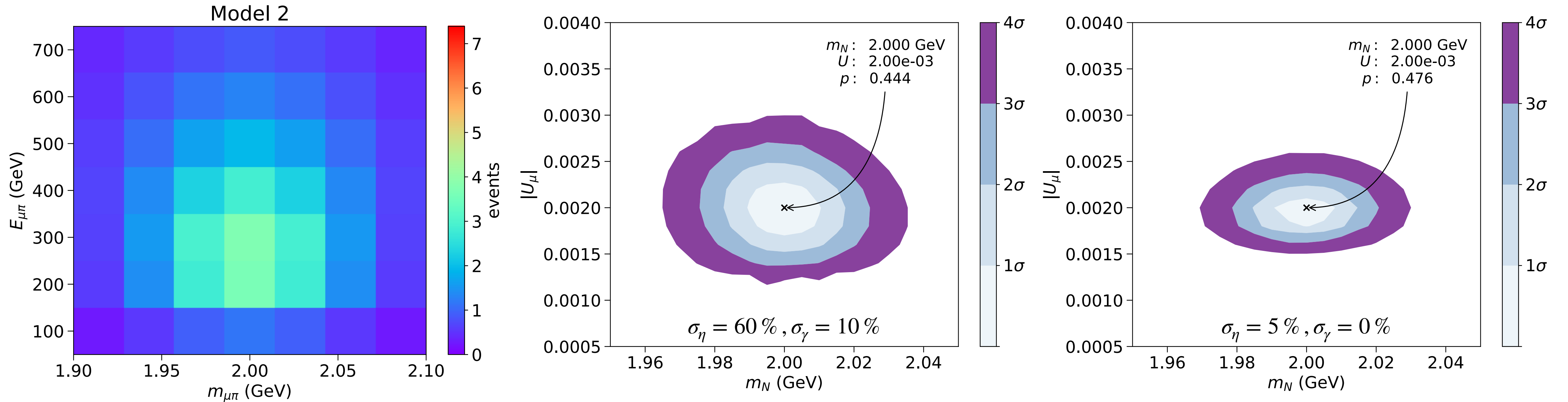
$$L(\vec{d}, m_N; \eta, \vec{\gamma}) = \prod_i f_P(d_i | \eta \gamma_i \mu_i) \underbrace{f_G(1 | \eta, \sigma_\eta)}_{\text{Normalization Uncertainty}} \underbrace{f_G(1 | \gamma, \sigma_\gamma \oplus \sigma_i^{MC})}_{\text{Bin-by-bin Uncertainty (Shape)}}$$



- Assuming current flux uncertainties, m_N & U_μ can be constrained to 0.1 % and 3 % respectively.
- Assuming improved flux uncertainties, m_N & U_μ can be constrained to 0.1 % and 2 % respectively.
- The mass measurement is limited by detector resolution and the coupling measurement is limited by flux uncertainties.

Parameter Estimation: Low Statistics (Model 2)

$$L(\vec{d}, m_N; \eta, \vec{\gamma}) = \prod_i f_P(d_i | \eta \gamma_i \mu_i) \underbrace{f_G(1 | \eta, \sigma_\eta)}_{\text{Normalization Uncertainty}} \underbrace{f_G(1 | \gamma, \sigma_\gamma \oplus \sigma_i^{MC})}_{\text{Bin-by-bin Uncertainty (Shape)}}$$

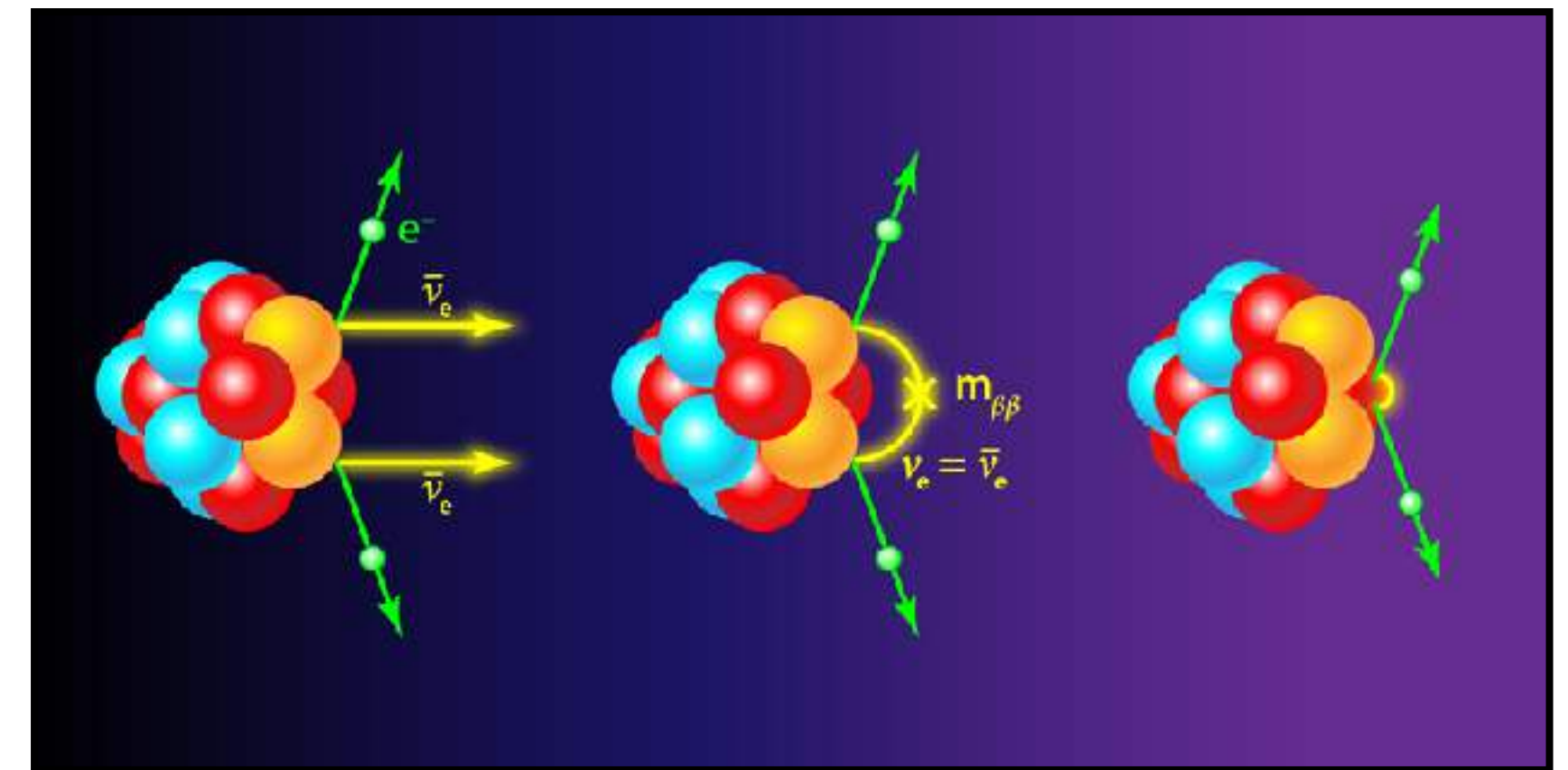


- With ~ 80 events, the procedure is statistics-limited:
 - Assuming current flux uncertainties, m_N & U_μ can be constrained to 1 % and 25 % respectively.
 - Assuming improved flux uncertainties, m_N & U_μ can be constrained to 1 % and 15 % respectively.

Majorana or Dirac?

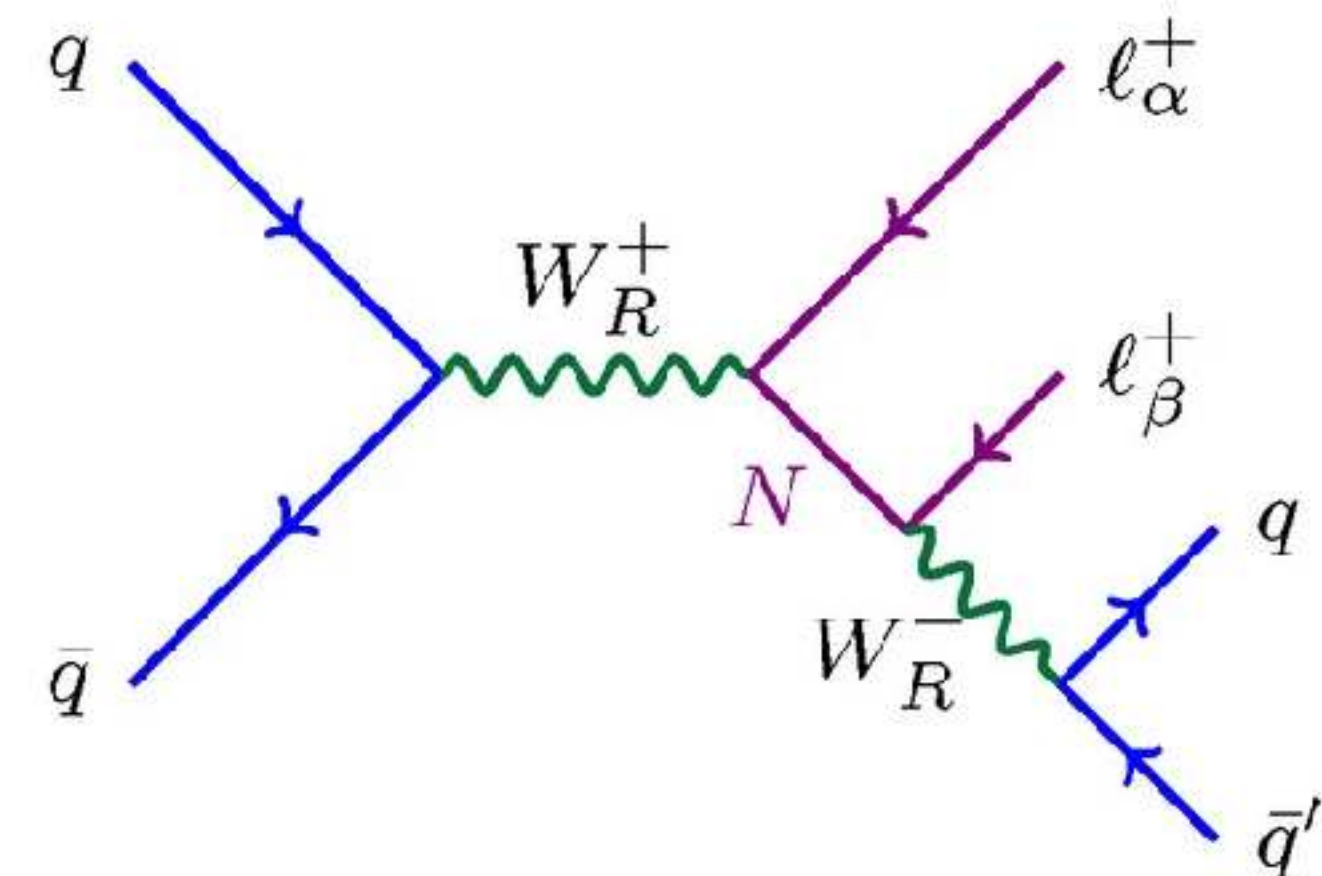
- Open question: is the SM neutrino Dirac or Majorana?
 - Carries strong implications for cosmology, neutrino oscillations, and dark matter
- Direct tests of LNV have been carried out for decades in lab and collider based experiments:
 - $0\nu\beta\beta$ (Furry 1939)
 - $pp \rightarrow l^\pm l^\pm jj$ (Keung and Senjanović 1983)

Neutrinoless double beta decay



[<https://physics.aps.org/articles/v11/30>]

$$pp \rightarrow l^\pm l^\pm jj$$



[<https://www.mdpi.com/2218-1997/8/3/164>]

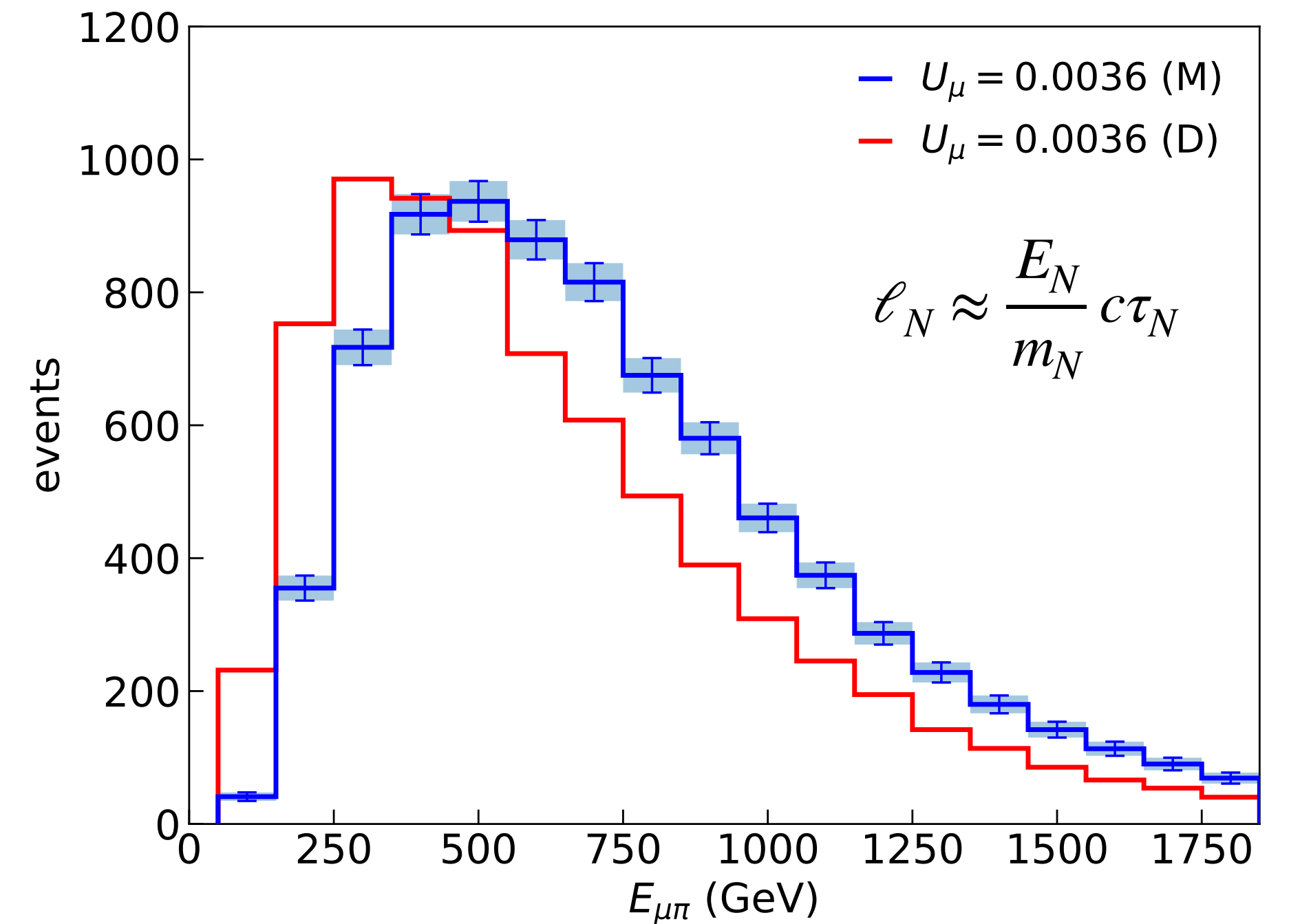
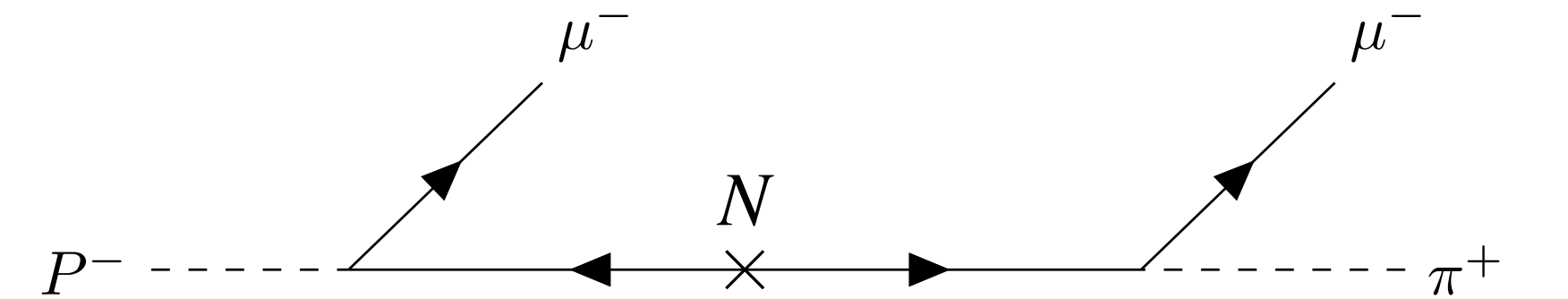
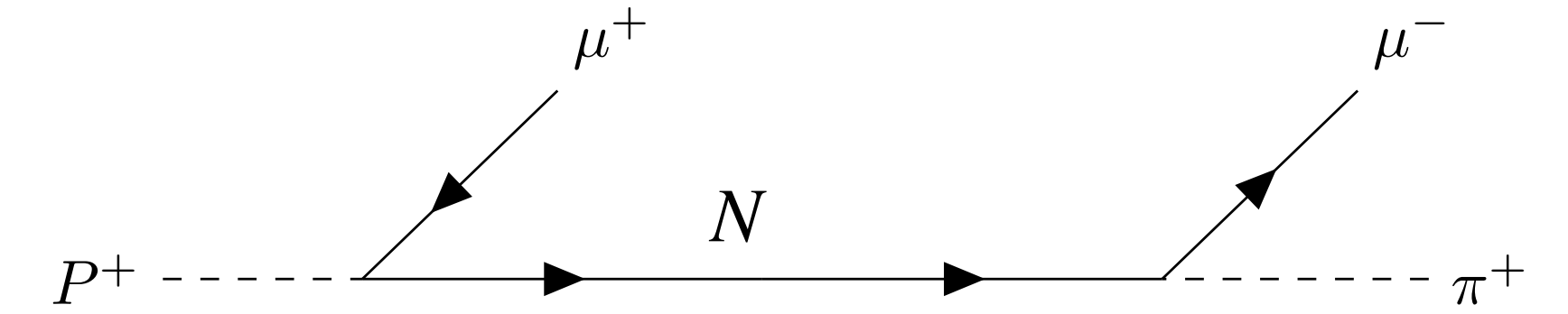
Majorana vs. Dirac HNLs: The Kinematic Lever

- Since SM neutrinos are massless, helicity and chirality eigenstates coincide, causing LNV processes involving SM neutrino to be suppressed by

$$|M|^2 \sim \left(\frac{m_\nu}{E_\nu} \right)^2$$

- However, for on-shell HNL decay, the mass dependence of the amplitude exactly cancels, resulting in LNC and LNV processes which are **equal and unsuppressed**
- Therefore, the lifetimes between Majorana and Dirac HNLs are related by

$$\tau_M = \frac{1}{\Gamma_{LNC} + \Gamma_{LNV}} = \frac{\tau_D}{2}$$

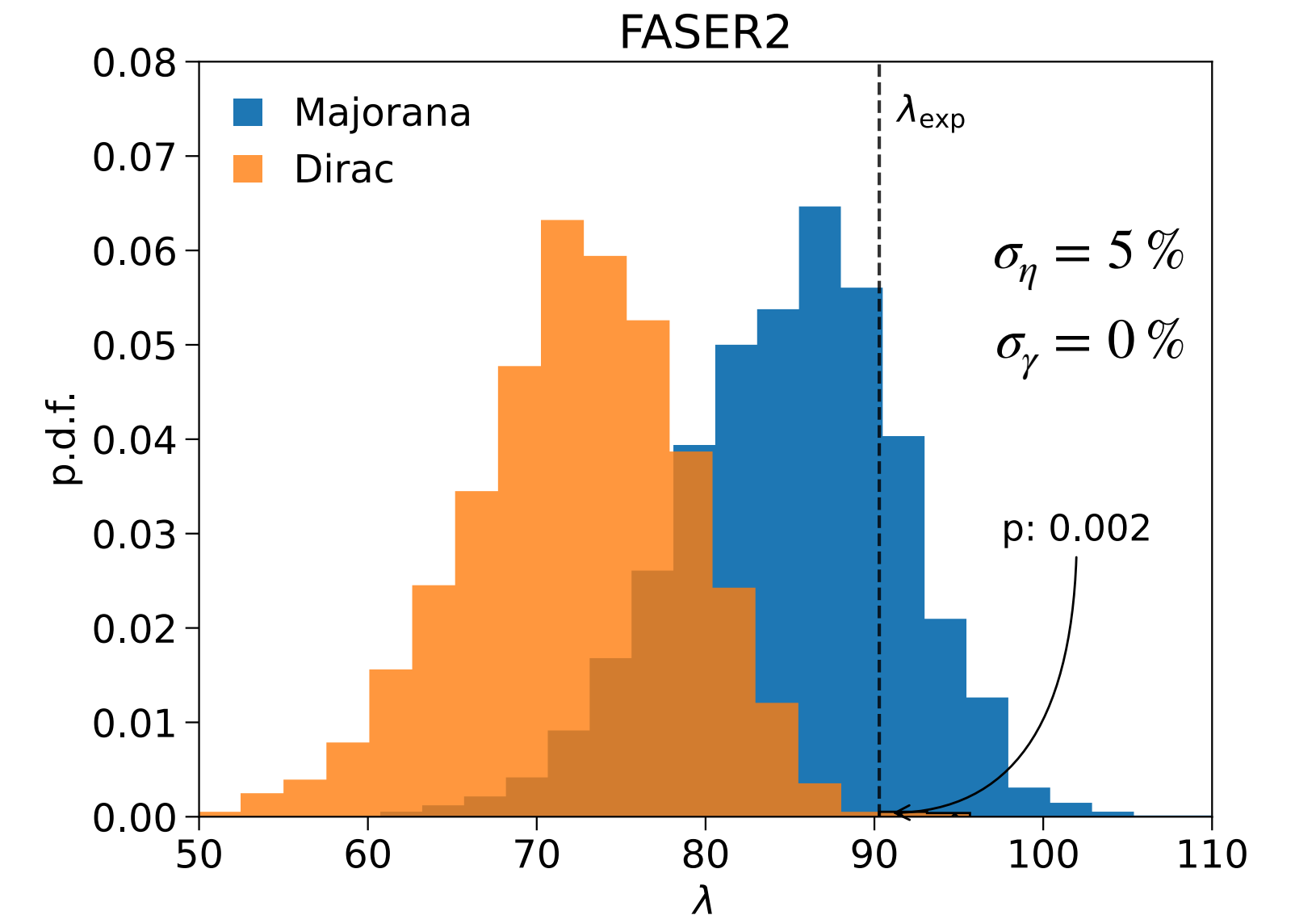
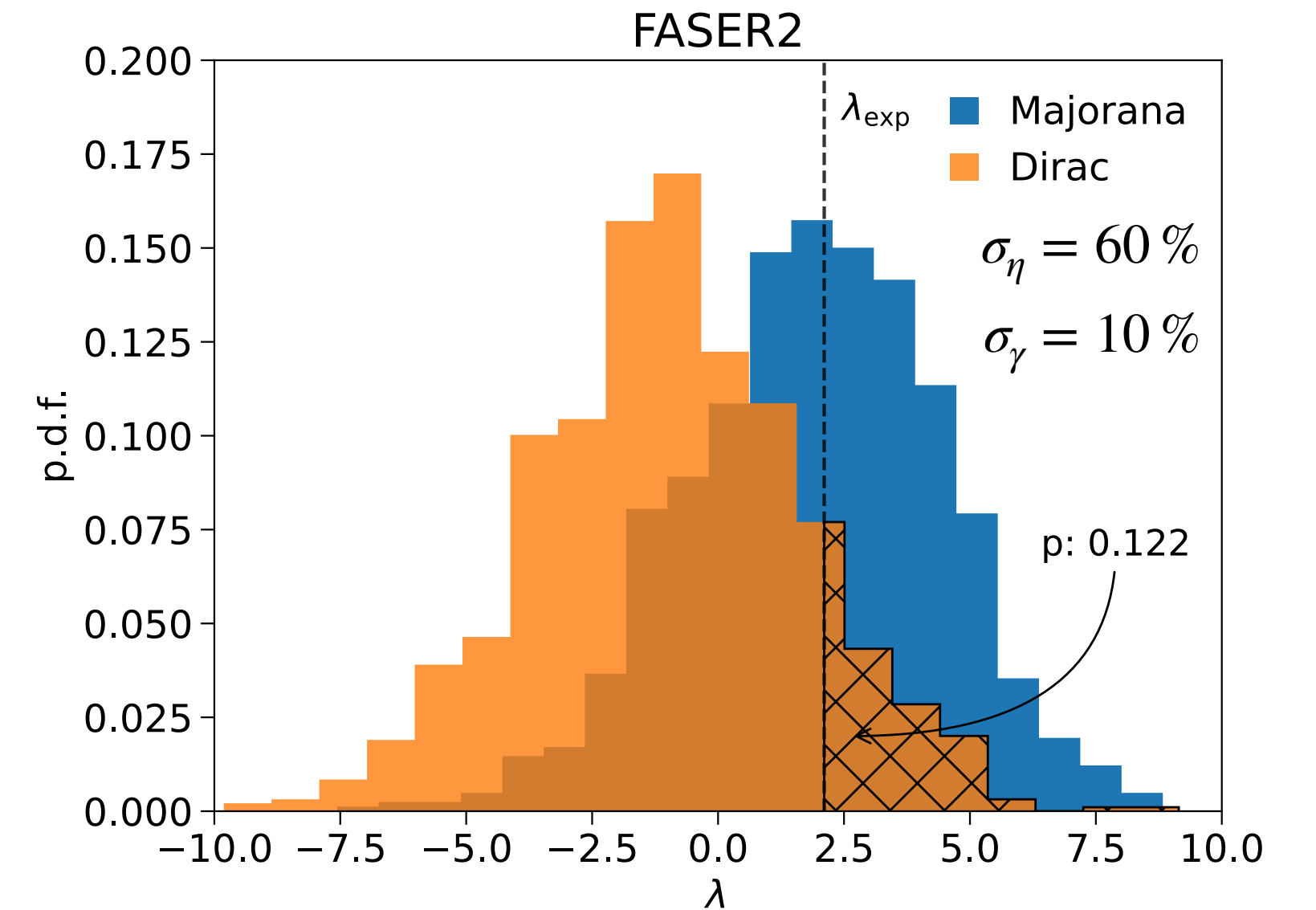


Discrimination via Kinematics: High Statistics (Model 1)

Profiled Likelihood Ratio Test

$$\lambda(\vec{d}) := -2 \ln \frac{L_D(\vec{d} | \hat{m}_N^D, \hat{U}_\mu^D; \hat{\eta}^D, \hat{\vec{\gamma}}^D)}{L_M(\vec{d} | \hat{m}_N^M, \hat{U}_\mu^M; \hat{\eta}^M, \hat{\vec{\gamma}}^M)}$$

- Using λ , we compute the probability that a Dirac HNL would fluctuate to produce a signal that looks just as (or more) “Majorana-like” than the expected observation
- Generated via $\mathcal{O}(10^3)$ pseudo-experiments, where the nuisance parameters for both models have been fixed to their best-fit values.
- Assuming flux uncertainties are improved, we find that FASER2 can expect to reject the Dirac hypothesis with 99.8 % CL (2.9σ)

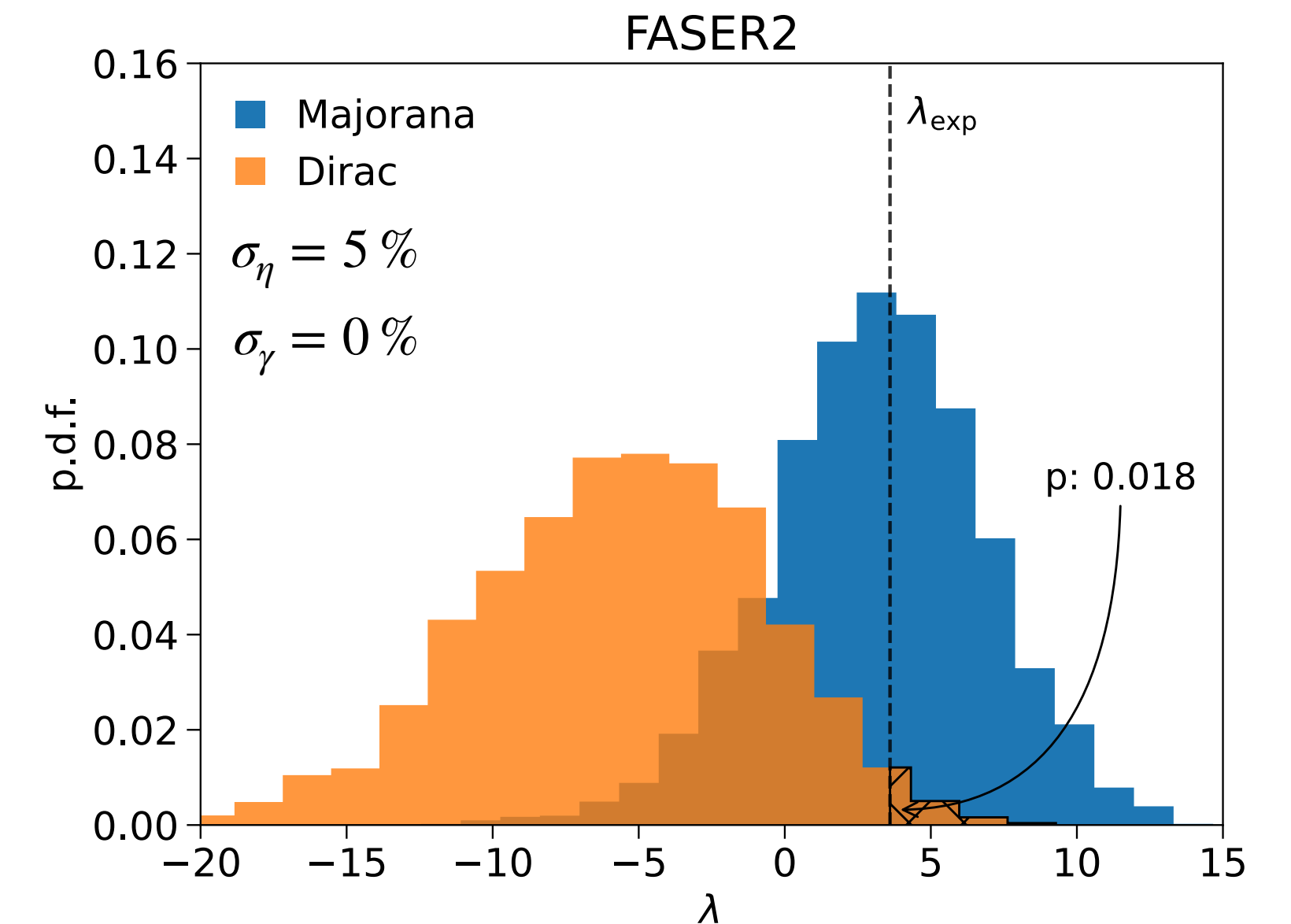
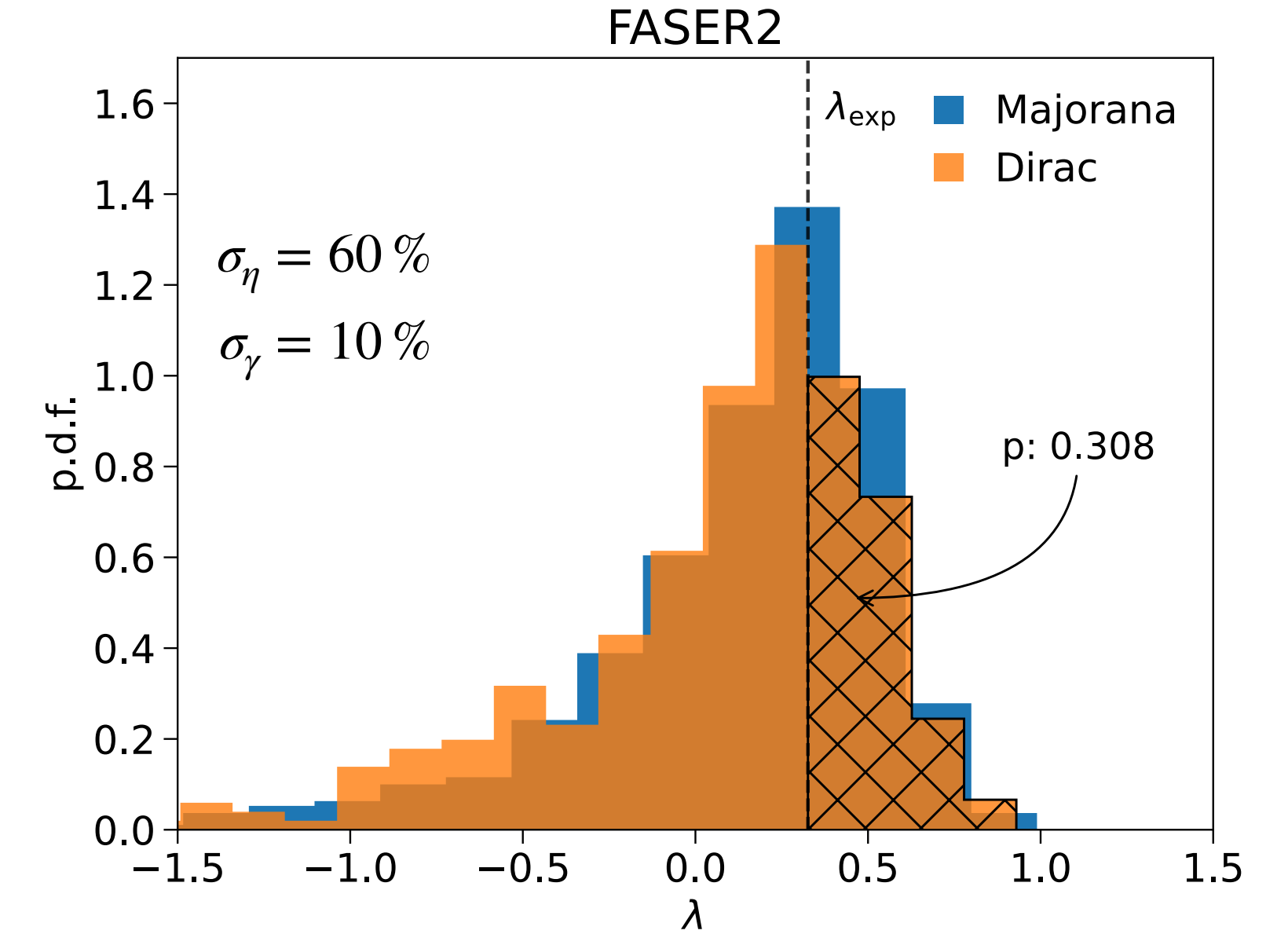


Discrimination via Kinematics: Low Statistics (Model 2)

Profiled Likelihood Ratio Test

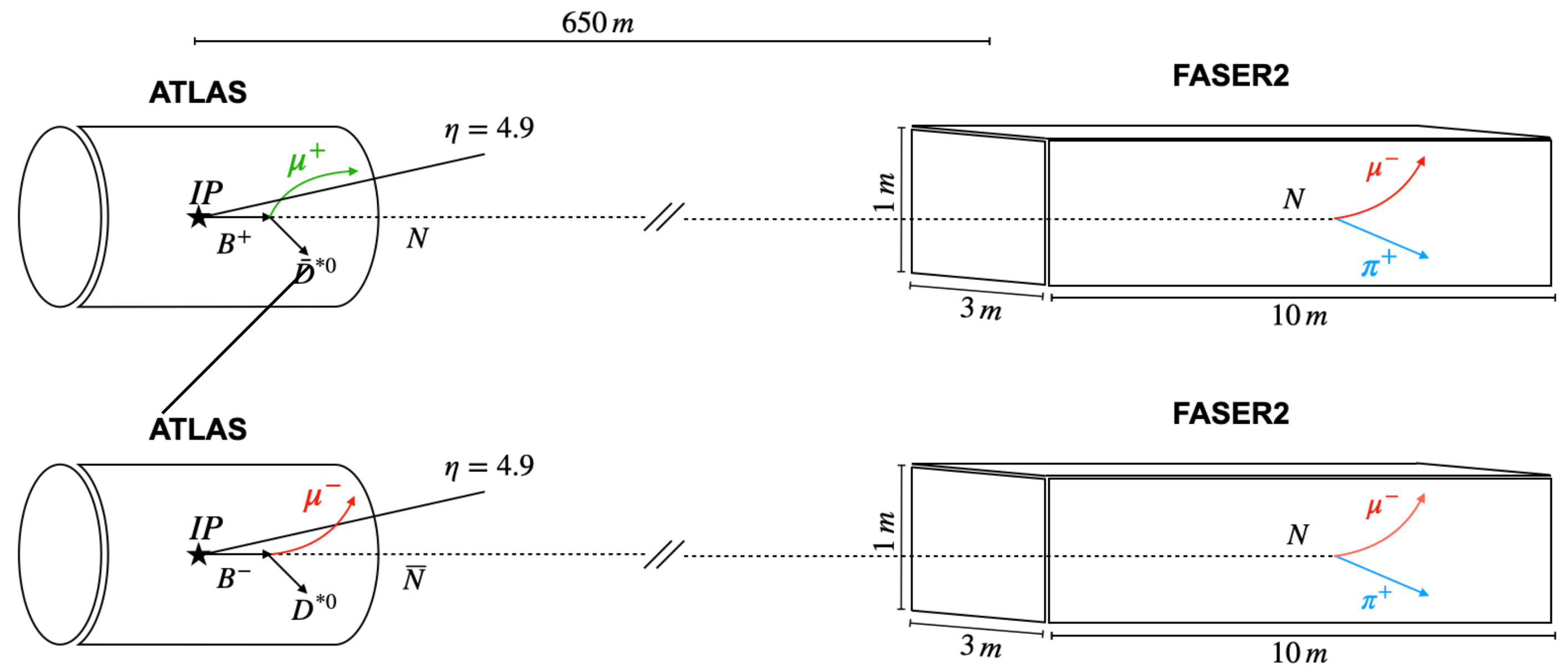
$$\lambda(\vec{d}) := -2 \ln \frac{L_D(\vec{d} | \hat{m}_N^D, \hat{U}_\mu^D; \hat{\eta}^D, \hat{\gamma}^D)}{L_M(\vec{d} | \hat{m}_N^M, \hat{U}_\mu^M; \hat{\eta}^M, \hat{\gamma}^M)}$$

- With significantly lower event rates, the procedure is statistics limited.
- Assuming flux uncertainties are improved, we find that FASER2 can only expect to reject the Dirac hypothesis with 98.2 % CL (2.1σ)
- Therefore, in order to obtain comparable results to the high-statistics scenario, additional information is necessary.



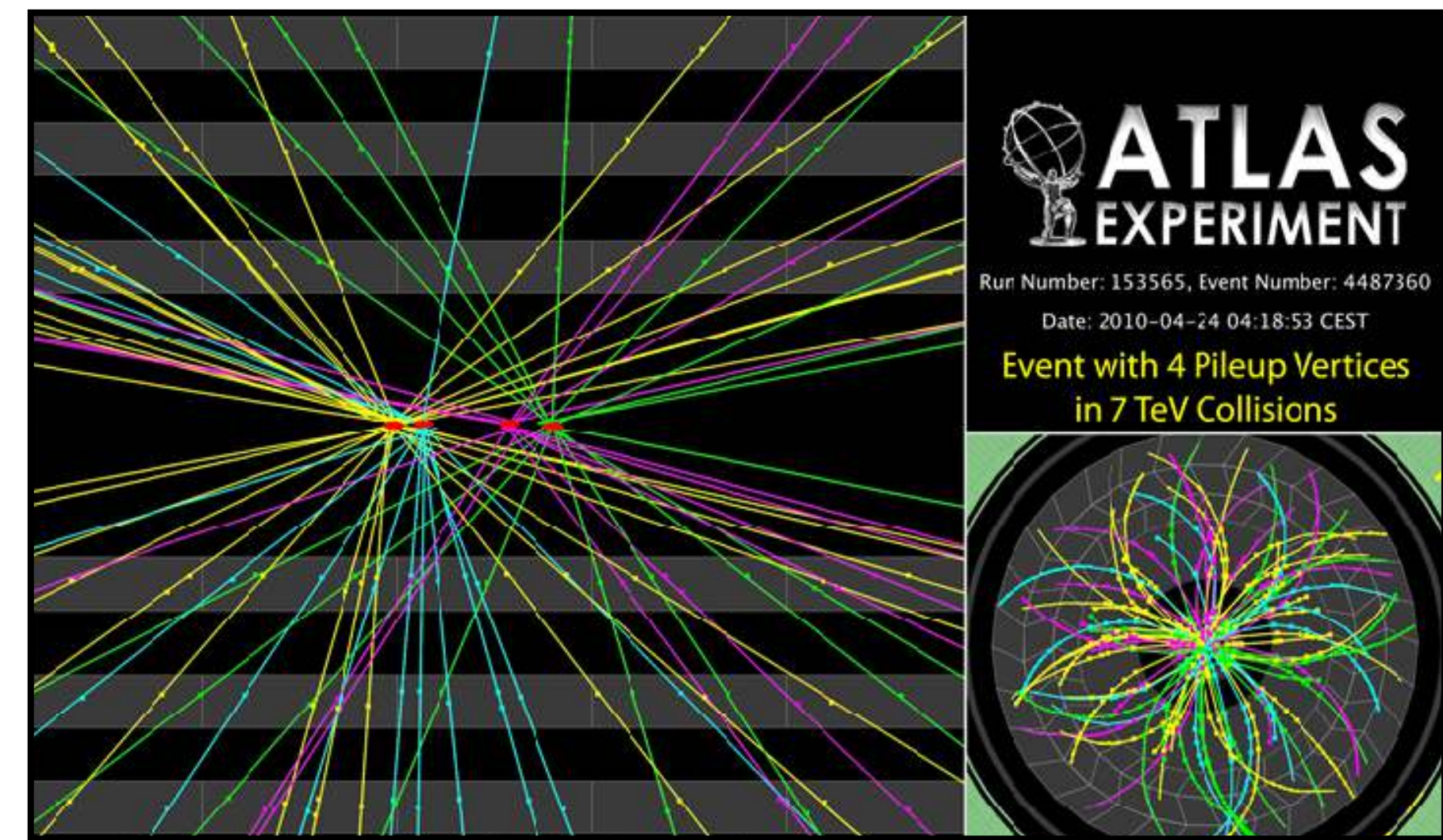
A Novel Detector Concept: FASER2 → ATLAS

- In our work, we explore the idea of using FASER2 as a trigger for ATLAS to measure muons produced in association with HNLs as a probe of LNV.
 - ATLAS L0 Latency: $6\ \mu\text{s}$
 - Triggering Time: $2.2\ \mu\text{s} + 2.9\ \mu\text{s}$
- In the absence of background, one could determine LNV on an event-by-event basis.
However this is not in the absence of background!

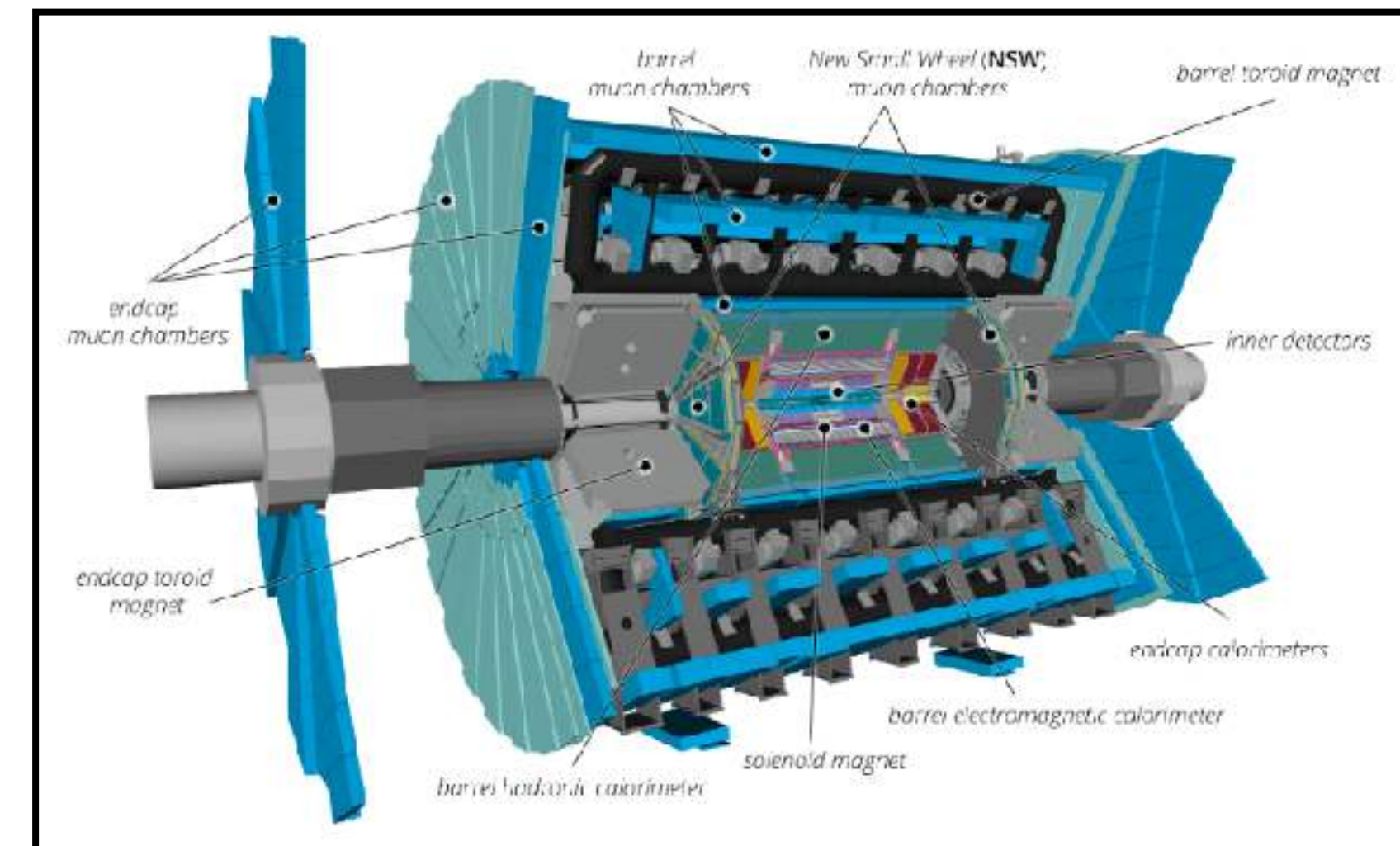


Pileup Background at the HL-LHC

- At the HL-LHC, there are $\sim 10^{11}$ protons per bunch, resulting in $\langle \mu \rangle = 200$ pile-up events per crossing.
- We use Pythia8 to overlay 200 pp collisions to generate an estimate for the number of background tracks per bunch crossing
- A high- η muon tagger is required to reduce background from charged tracks
 - Proposed in ATLAS Phase-II upgrade (ATLAS Collaboration 2015)



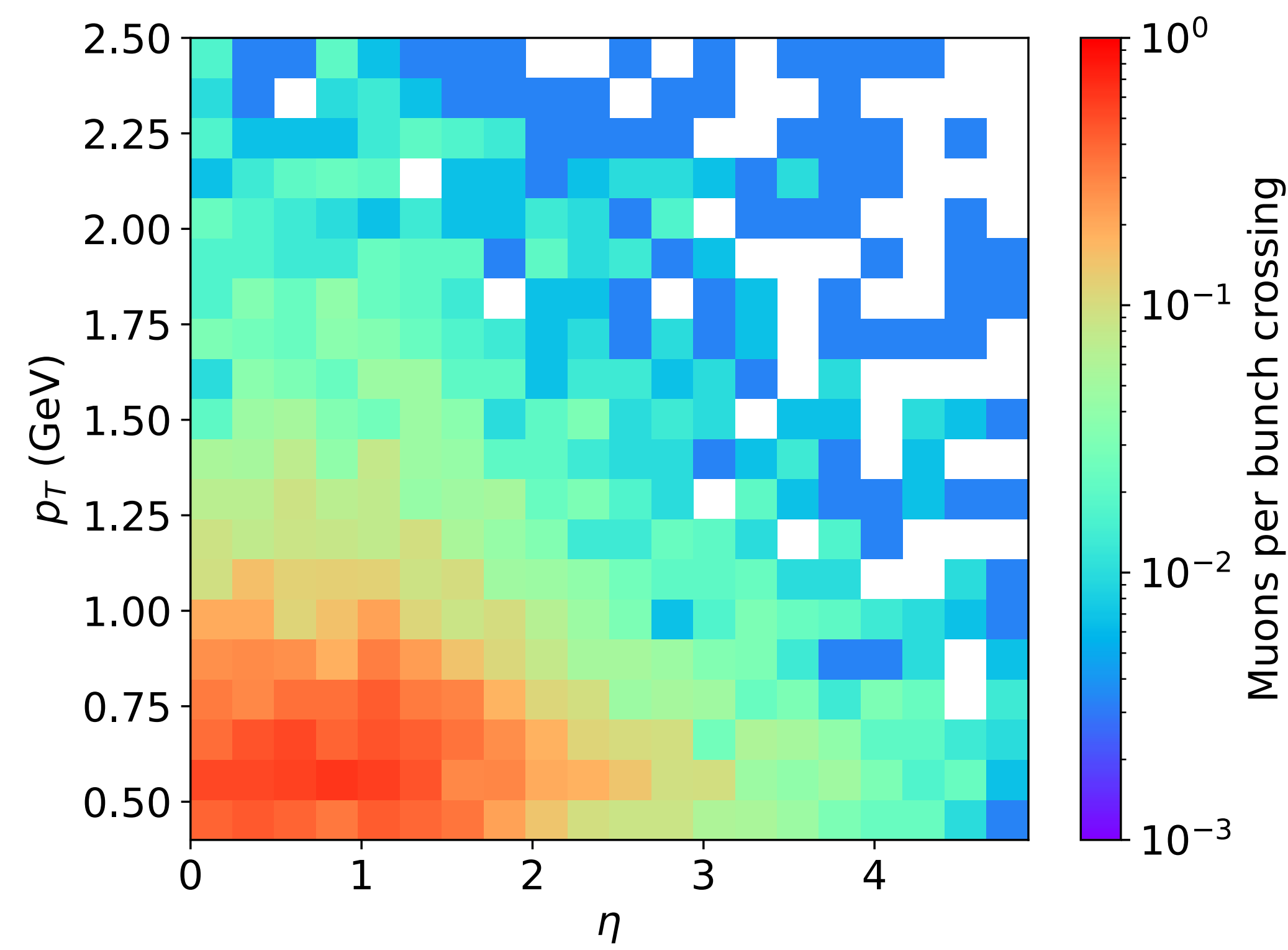
https://lhc-closer.es/taking_a_closer_look_at_lhc/0.pile_up



ATLAS Collaboration 2023

Pileup Background at the HL-LHC

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- We use Pythia8 to overlay 200 pp collisions to generate an estimate for the number of background tracks per bunch crossing
- A high- η muon tagger is required to reduce background from charged tracks
 - Proposed in ATLAS Phase-II upgrade (ATLAS Collaboration 2015)
- We find that there is **roughly 1 uncorrelated background muon per bunch crossing**.



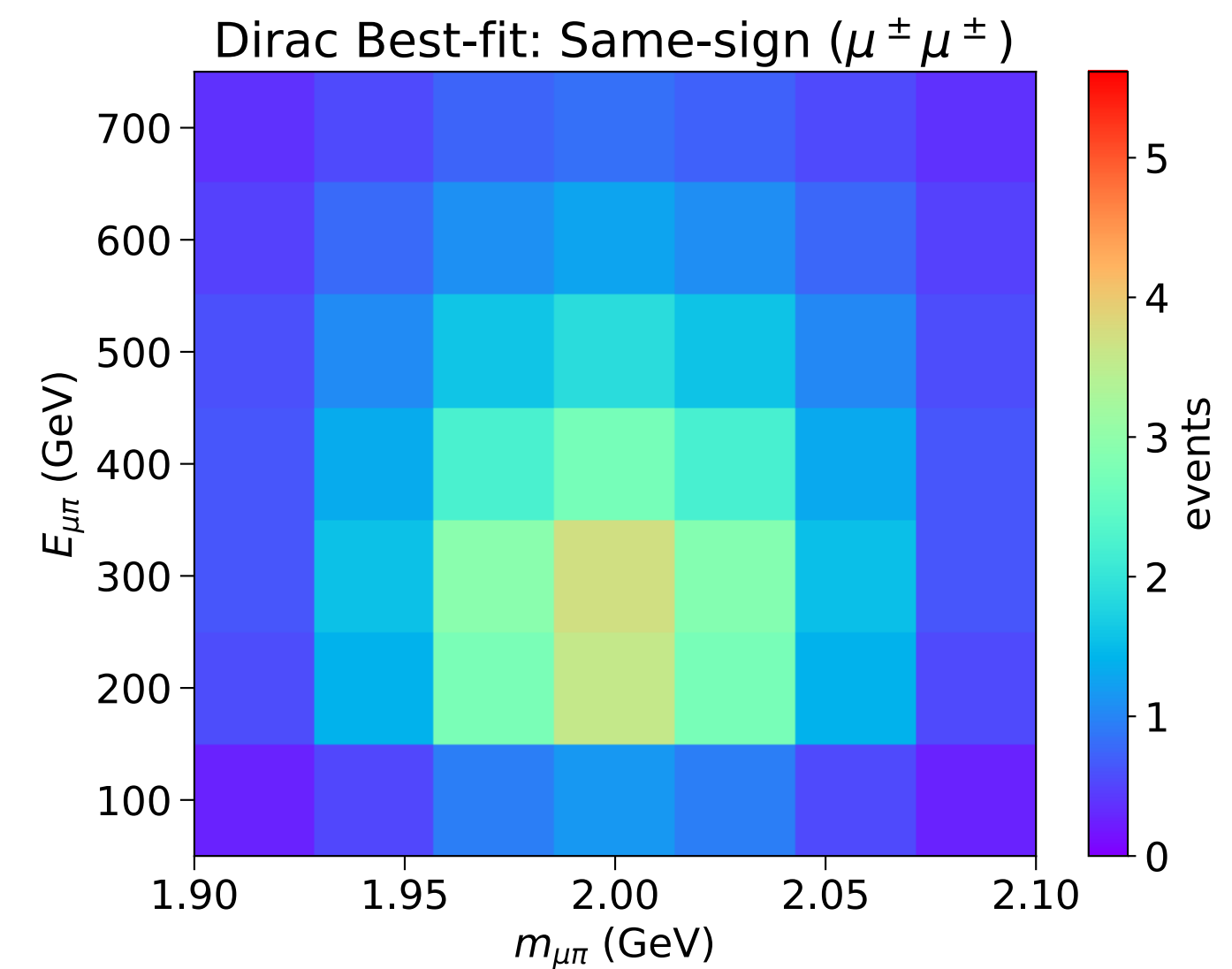
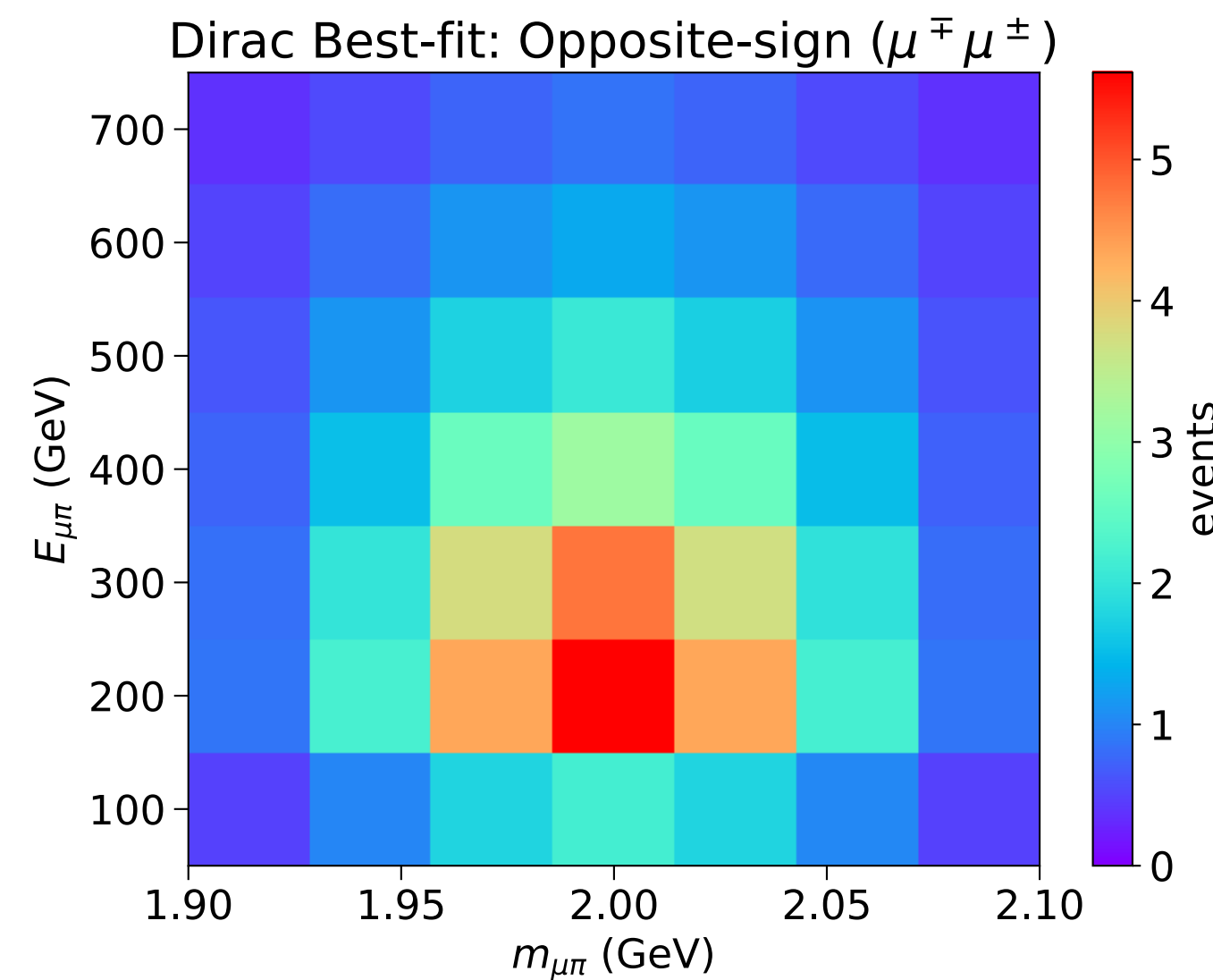
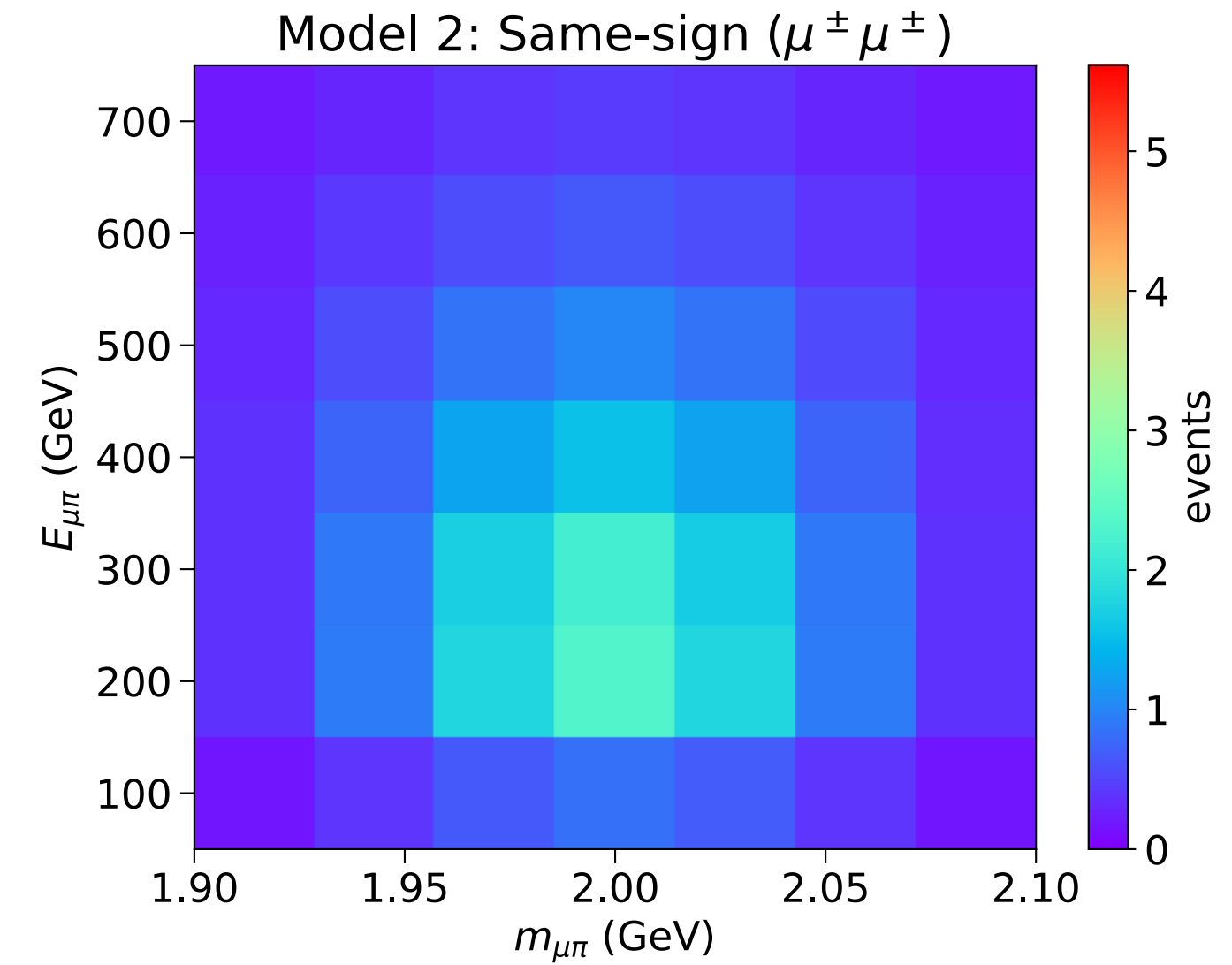
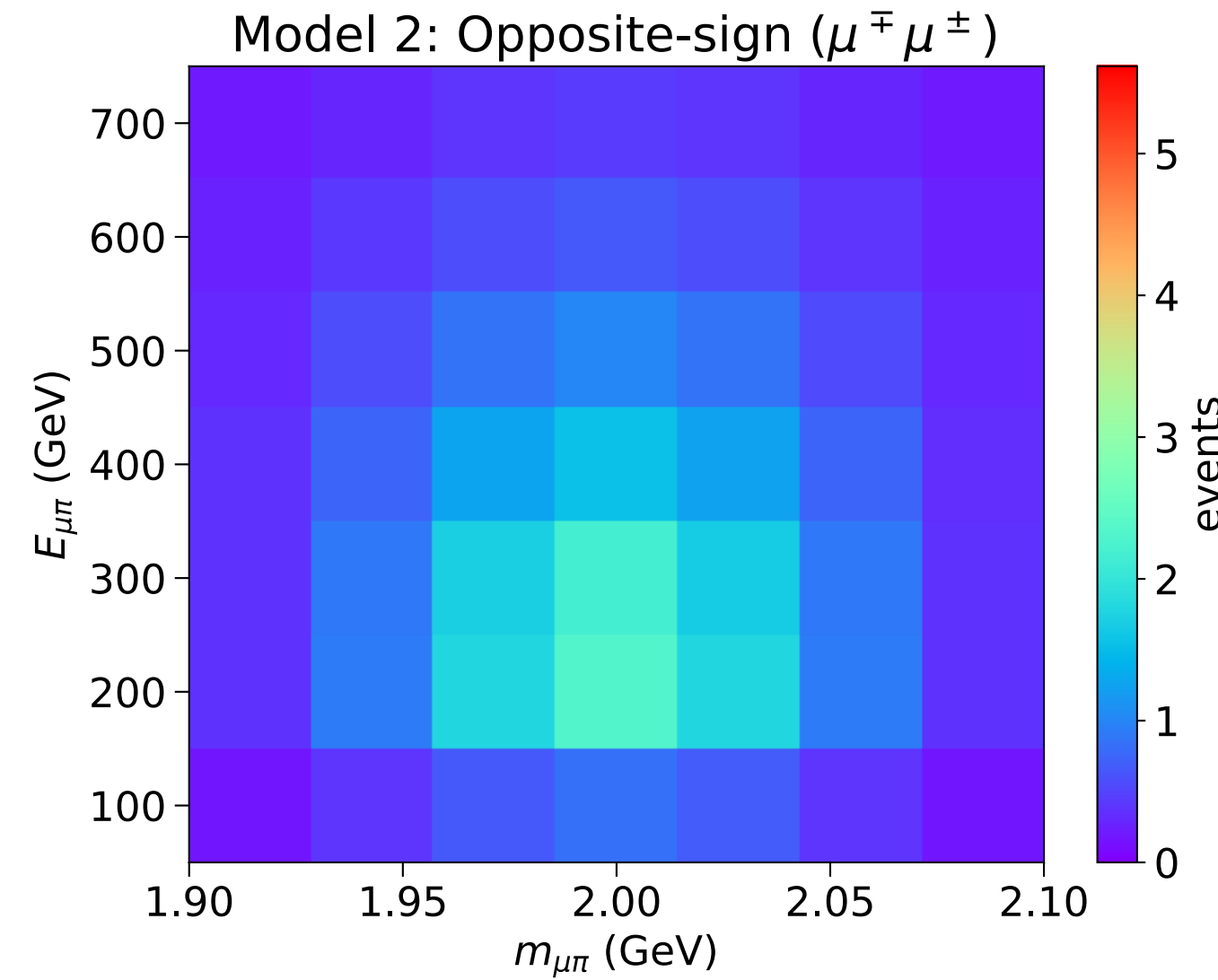
Parent	Muons ($ \eta < 4.9$)	Muons ($3.5 < \eta < 4.9$)
$\pi^\pm$	31.41	0.43
$K^\pm$	16.09	0.26
K_L^0	0.37	0.0033
Heavy Mesons	3.34	0.33

Correlated Signal: Same-Sign vs. Opposite-Sign

$$\mu_i^{SS} = \frac{b}{2} \mu_i^F + \frac{\eta_M}{2} \mu_i^A \quad \eta_M = \begin{cases} 1 & \text{Majorana} \\ 0 & \text{Dirac} \end{cases}$$

$$\mu_i^{OS} = \frac{b}{2} \mu_i^F + \left(1 - \frac{\eta_M}{2}\right) \mu_i^A$$

- We separate each triggered HNL event into the number of muons detected in ATLAS that have either the **same sign (SS)** or **opposite sign (OS)** as the muon detected in FASER2.
- Muons produced in association with **Majorana HNL's** can be found with OS or SS muons in FASER2 with equal rates, resulting in an **symmetric distribution**.
- In contrast, **Dirac HNL's** can only be found with OS muons in FASER2, resulting in an **asymmetric distribution**.

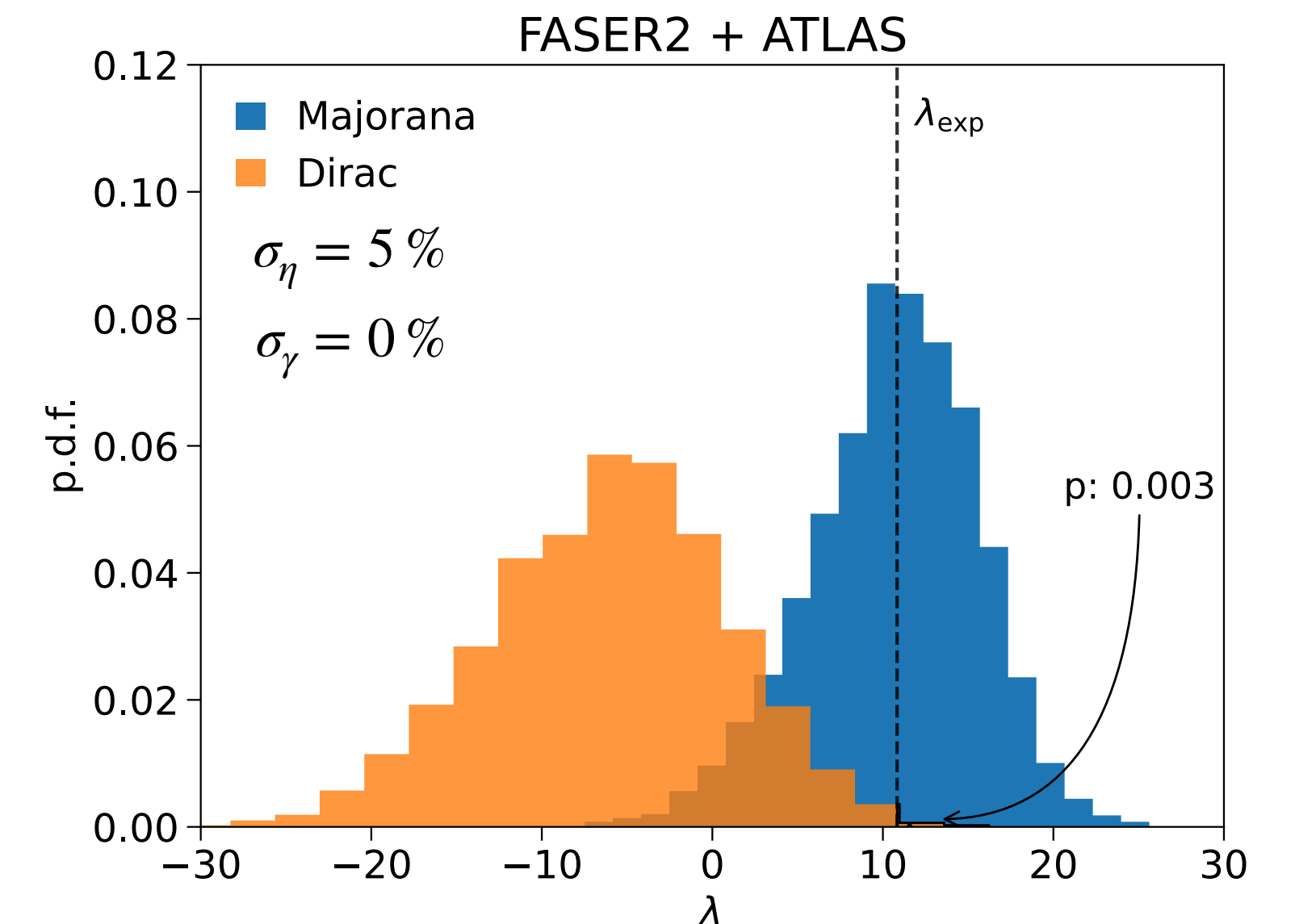
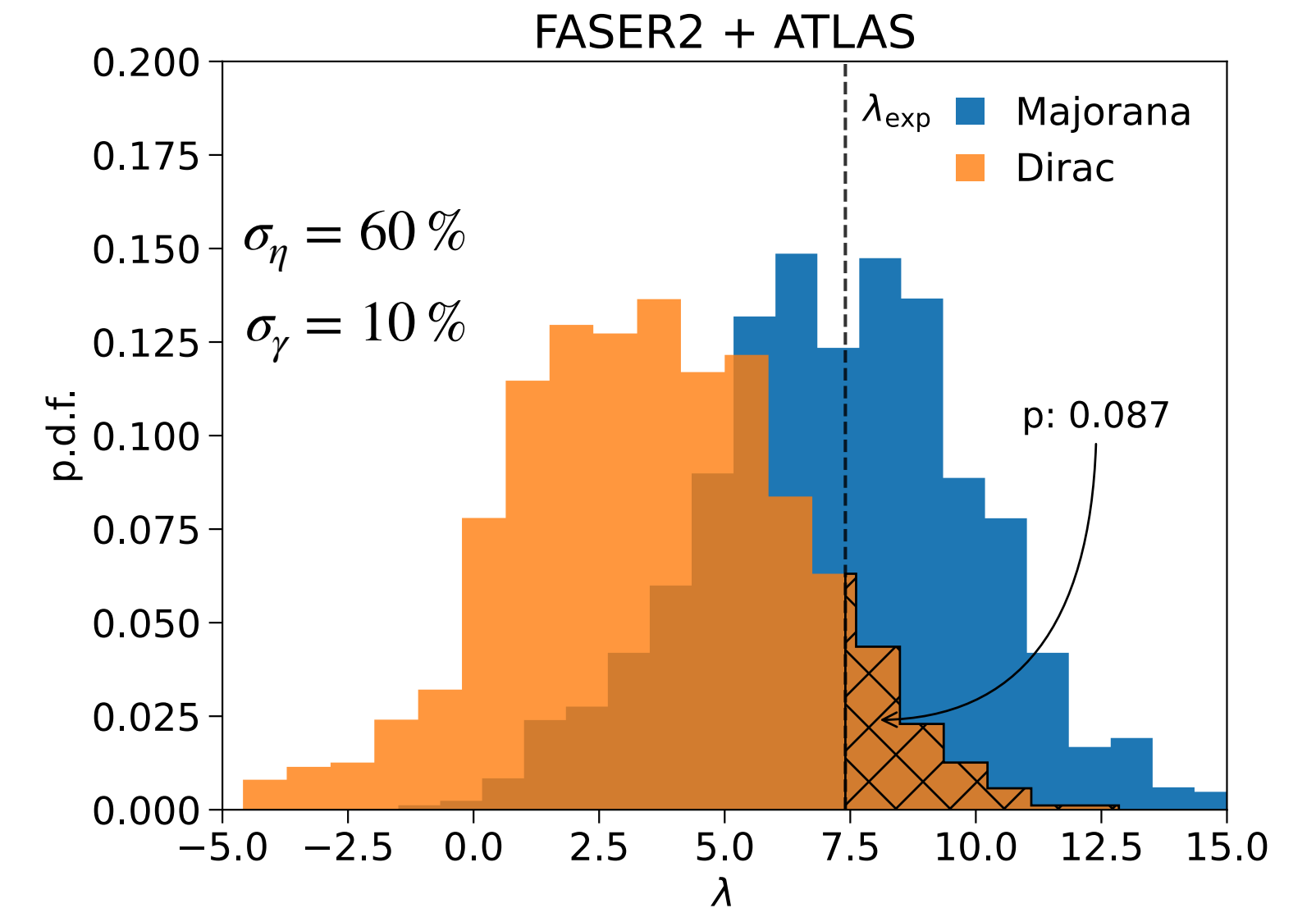


Discrimination Results with the FASER2 + ATLAS Trigger (Model 2)

Profiled Likelihood Ratio Test

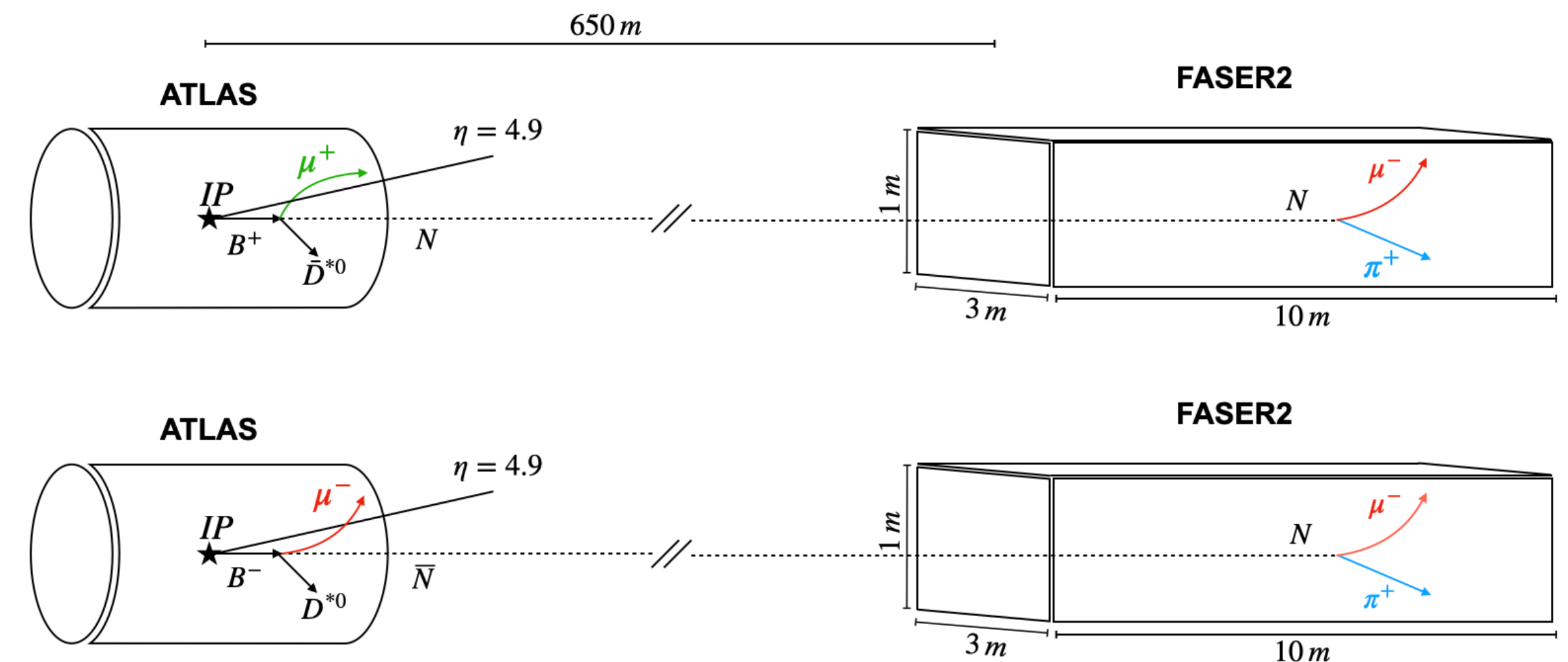
$$\lambda(\vec{d}) := -2 \ln \frac{L_D(\vec{d} | \hat{m}_N^D, \hat{U}_\mu^D; \hat{\eta}^D, \hat{\gamma}^D)}{L_M(\vec{d} | \hat{m}_N^M, \hat{U}_\mu^M; \hat{\eta}^M, \hat{\gamma}^M)}$$

- Assuming improved flux uncertainties, we find that FASER2 can expect to reject the Dirac hypothesis with 99.7 % CL (2.7σ), **despite having significantly lower statistics!**
- This illustrates that the **main LHC detectors** and proposed **auxiliary detectors** can have a **synergistic relationship** to improve signal characterization.



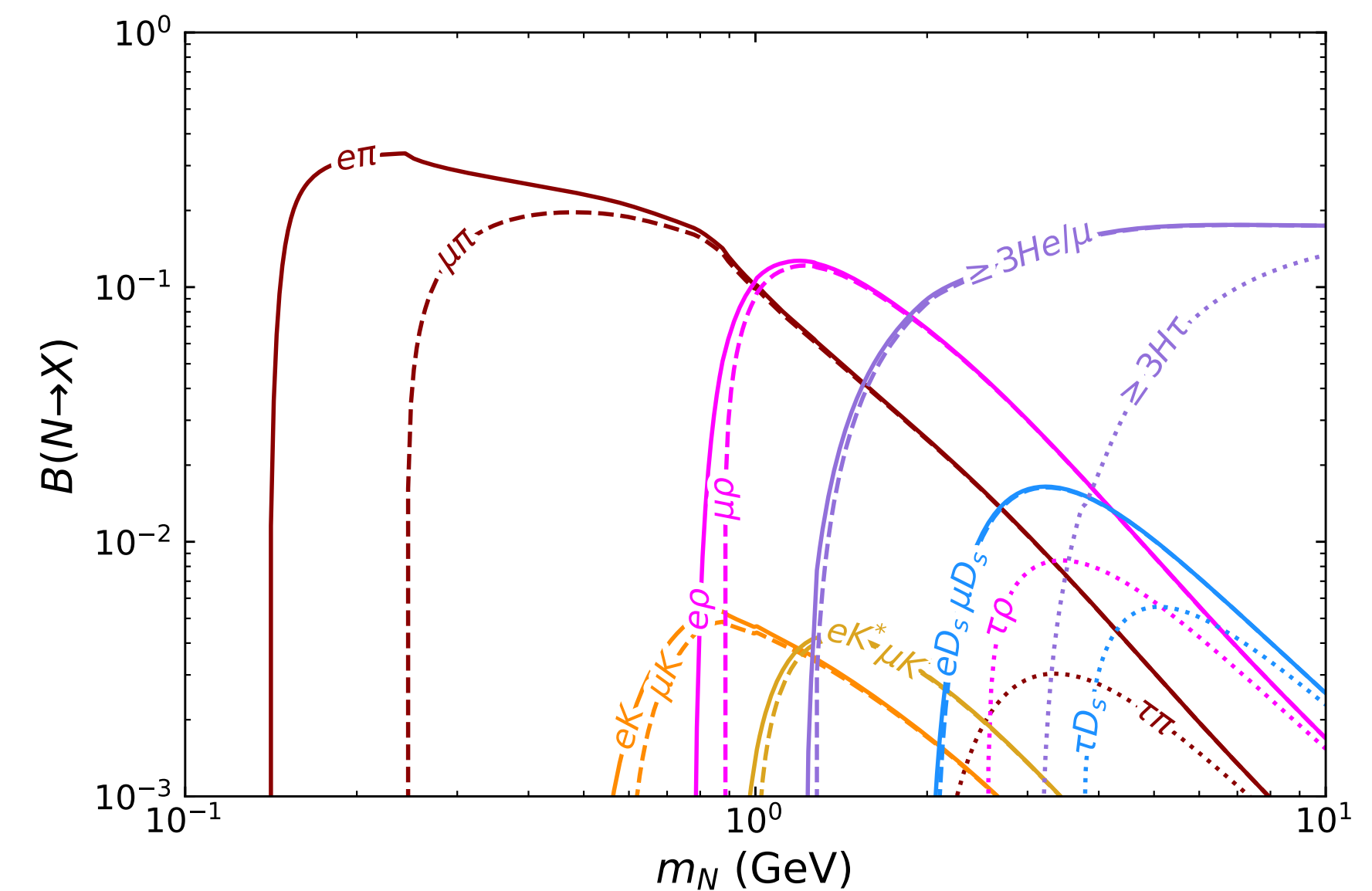
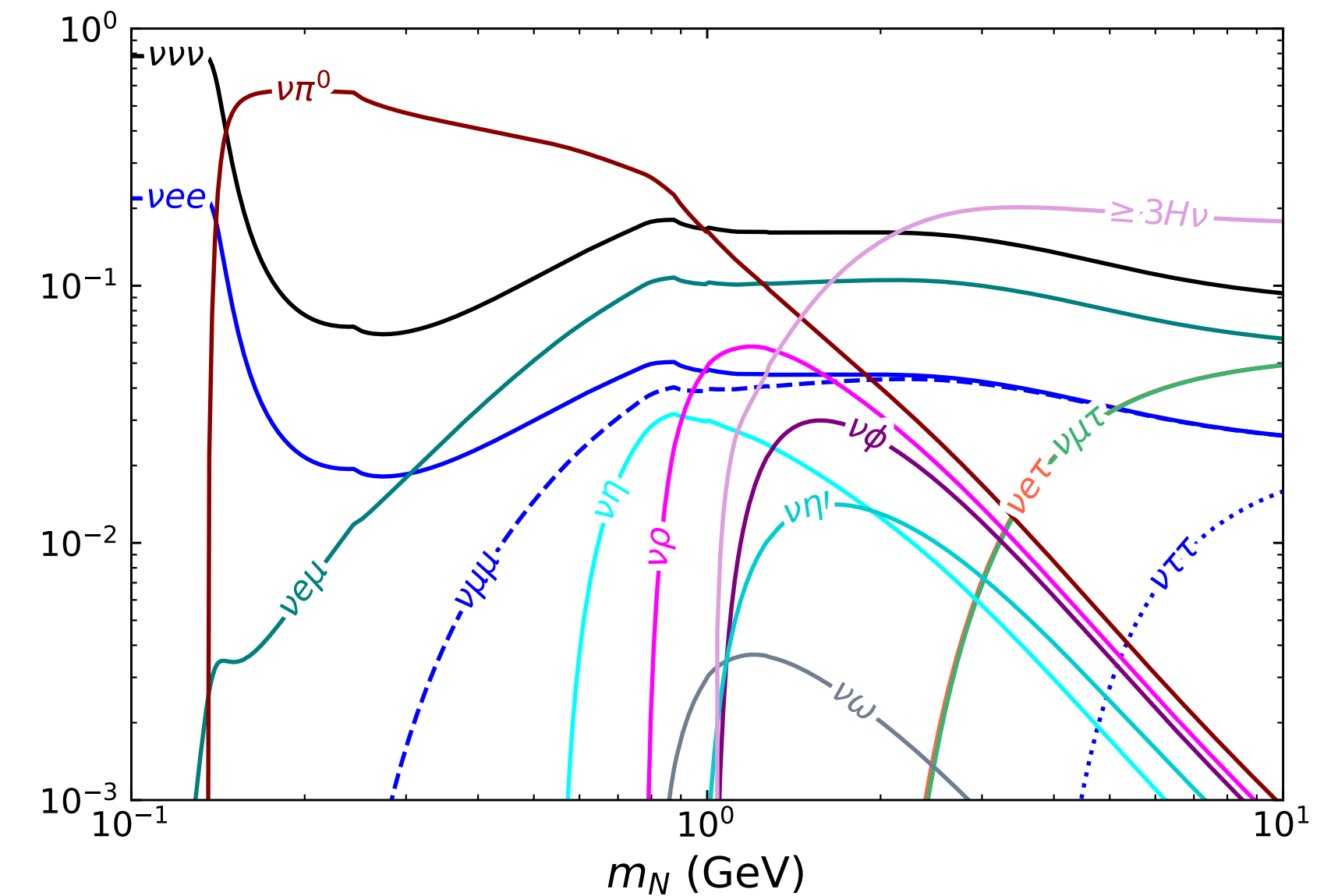
Modeling & Discovery Summary (Chapter 4)

- In the event of a Majorana HNL discovery at FASER2 we found:
 - The HNL mass and coupling would be measurable at 0.1% and 3% (high statistics) / 1% and 15% (low statistics) levels given current and future theory uncertainties respectively.
 - The Dirac hypothesis could be rejected at 2.9σ in the high statistics scenario via kinematics alone, assuming future reduced theory uncertainties.
 - Using FASER2 as a trigger for ATLAS recovers comparable model discrimination (2.7σ) even in the low-statistics regime.



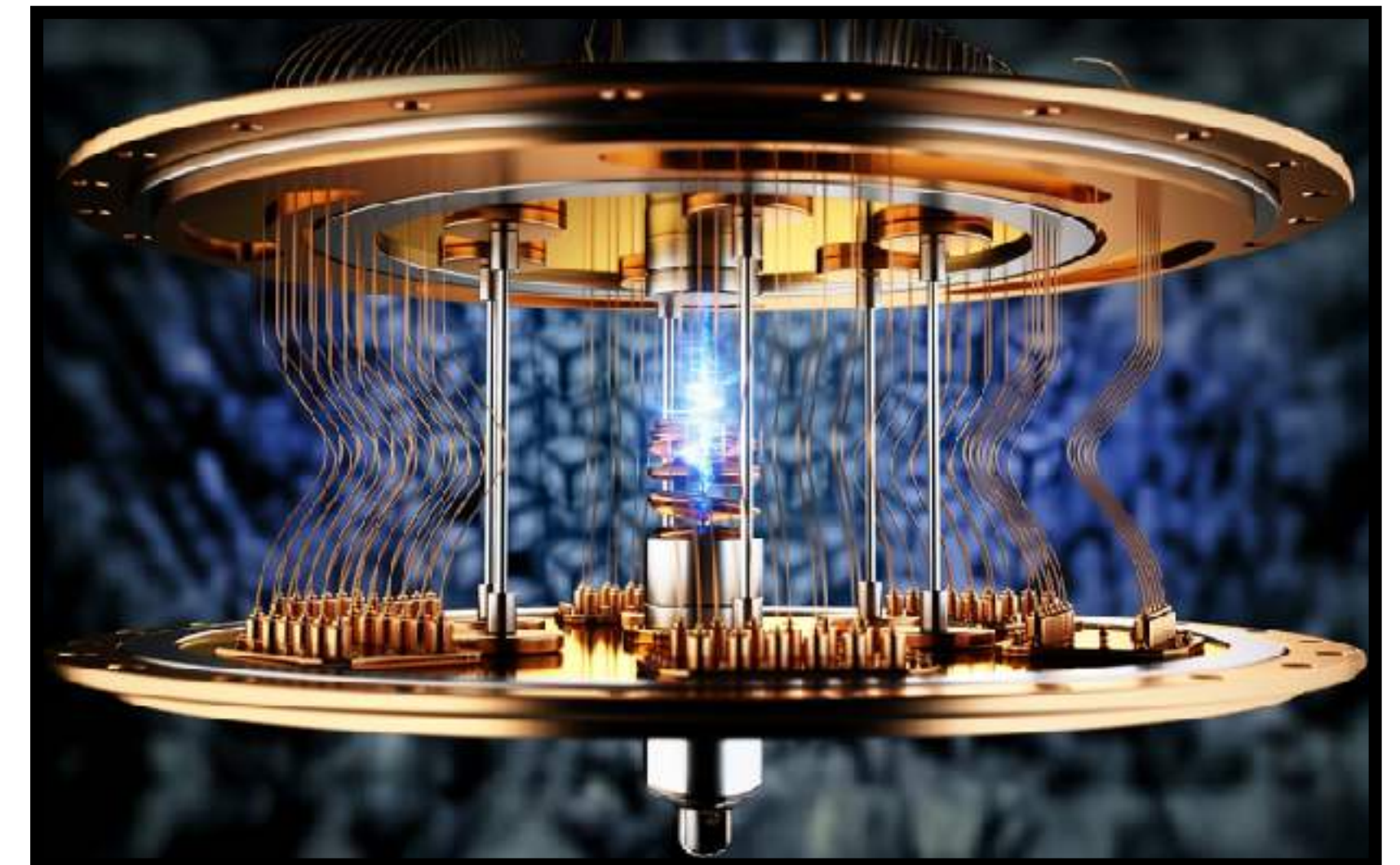
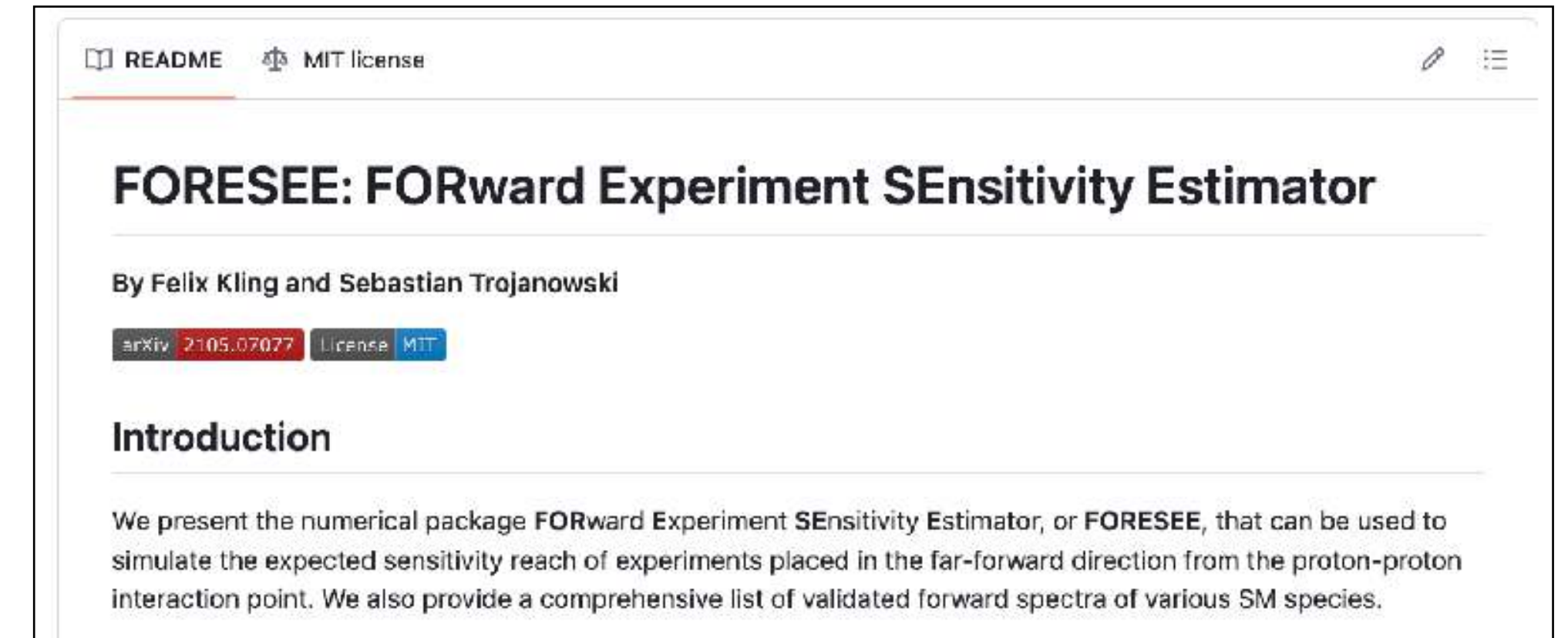
What This Dissertation Contributed

- **Modeling:** HNLCalc + FORESEE HNL module – a fast and comprehensive simulation pipeline for HNLs with arbitrary couplings
- **Discovery:** Forward-LHC sensitivity estimates for FASER and FASER2 across the single and mixed coupling (011, 111) PBC benchmarks
- **Characterization:** Demonstrated FASER2's ability to measure HNL mass and coupling and to discriminate Majorana from Dirac, and proposed a FASER2-as-trigger concept for ATLAS that recovers discrimination power in the low-statistics regime



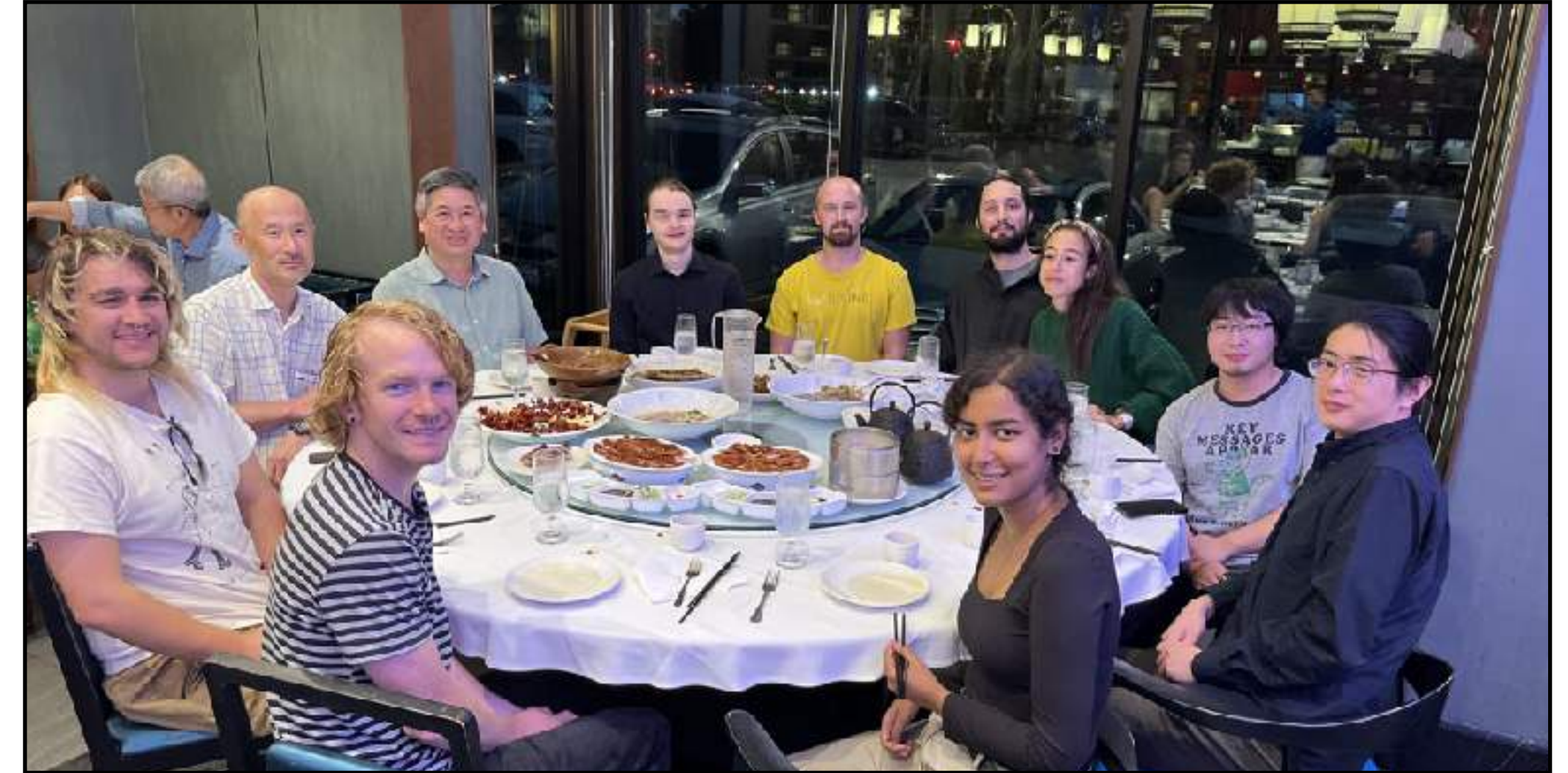
Outlook and Future Directions

- Future Directions:
 - Extend characterization analysis to **realistic see-saw models** with mixed-couplings.
 - Consider **additional decay signatures**: $N \rightarrow \mu\rho$ or $N \rightarrow \nu\mu^+\mu^-$.
 - Explore whether FASER2-Atlas correlated data could be of benefit for **different LLP models**, like a dark photon or a dark higgs.
- Whats next:
 - **FORESEEv2 Update Paper** (Collaboration: Felix Kling, Toni Makela, Sebastian Trojanowski)
 - Extended model library, uncertainty estimates, updated hadron and Bremsstrahlung production, package improvements...
 - Applying for jobs in **industry** (scientific computing / simulation engineering) and **university education**.

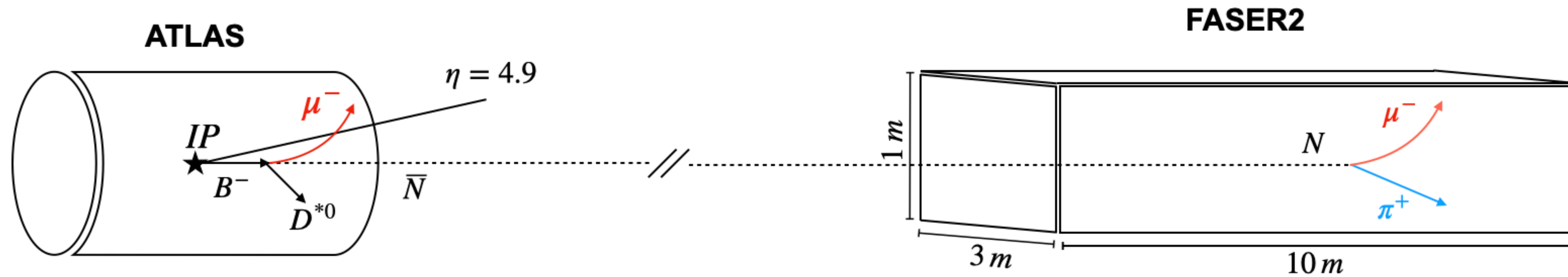
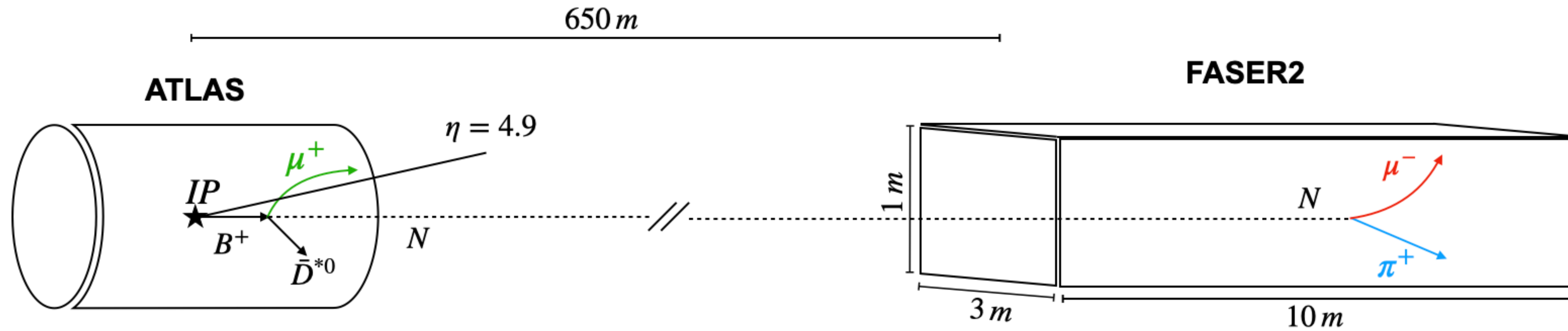


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 - Committee: Shirley Li, Tim Tait, Daniel Whiteson
 - Collaborators: Alec Hewitt, Felix Kling, Toni Makela, and Sebastian Trojanowski
 - Particle Theory Group



Thank You / Questions?



Production/Decay Inputs

Hadron Decay Constants

$$\langle 0 | \bar{q}_1 \gamma^\mu \gamma_5 q_2 | P(k) \rangle = i f_P k^\mu$$

$$\langle 0 | \bar{q}_1 \gamma^\mu q_2 | V(k, \epsilon) \rangle = i f_V M_V \epsilon^\mu$$

P	f_P (MeV)	V	f_V (MeV)
π^0 [55]	130.3	ρ^0 [56]	220
π^+ [55]	130.3	ρ^+ [56]	220
K^+ [55]	156.4	ω [56]	195
η [57]	78.4	K^{*+} [55]	204
η' [57]	-95.7	ϕ [55]	229
D^+ [58]	222.6		
D_s^+ [59]	280.1		
B^+ [21]	190		
B_c^+ [21]	480		

Hadron Transition Form Factors

$$\langle P_2 | \bar{q}_1 \gamma^\mu q_2 | P_1 \rangle = f_+(q^2)(p_1^\mu + p_2^\mu) + f_-(q^2)q^\mu$$

$$\langle V | \bar{q}_1 \gamma_\mu \gamma^5 q_2 | P \rangle = i \epsilon^{*\nu} \left[f(q^2) g_{\mu\nu} + a_+(q^2) p_{1\nu} (p_{1\mu} + p_{2\mu}) + a_-(q^2) p_{1\nu} q_\mu \right]$$

	A_0			A_1			A_2			V			
Decay Channel	$f_0(0)$	σ_1	σ_2	$f_0(0)$	σ_1	σ_2	$f_0(0)$	σ_1	σ_2	$f(0)$	σ_1	σ_2	c_V
$D^0 \rightarrow \rho^-$ [64]	0.66	0.36	0	0.59	0.50	0	0.49	0.89	0	0.90	0.46	0	1
$D^0 \rightarrow K^{*-}$ [64]	0.76	0.17	0	0.66	0.3	0	0.49	0.67	0	1.03	0.27	0	1
$D^+ \rightarrow \rho^0$ [64]	0.66	0.36	0	0.59	0.50	0	0.49	0.89	0	0.90	0.46	0	1/2
$D^+ \rightarrow \omega$ [64]	0.66	0.36	0	0.59	0.50	0	0.49	0.89	0	0.90	0.46	0	1/2
$D^+ \rightarrow \bar{K}^{*0}$ [64]	0.76	0.17	0	0.66	0.3	0	0.49	0.67	0	1.03	0.27	0	1
$D_s^+ \rightarrow K^{*0}$ [64]	0.67	0.2	0	0.57	0.29	0.42	0.42	0.58	0	1.04	0.24	0	1
$D_s^+ \rightarrow \phi$ [64]	0.73	0.10	0	0.64	0.29	0	0.47	0.63	0	1.10	0.26	0	1
$B^+ \rightarrow \rho^0$ [64]	0.30	0.54	0	0.26	0.73	0.1	0.29	1.4	0.5	0.31	0.59	0	1/2
$B^+ \rightarrow \omega$ [64]	0.30	0.54	0	0.26	0.54	0.1	0.24	1.40	0.50	0.31	0.59	0	1/2
$B^+ \rightarrow \bar{D}^{*0}$ [64]	0.69	0.58	0	0.66	0.78	0	0.62	1.04	0	0.76	0.57	0	1
$B^0 \rightarrow \rho^-$ [64]	0.30	0.54	0	0.26	0.54	0.1	0.24	1.40	0.50	0.31	0.59	0	1
$B^0 \rightarrow D^{*-}$ [64]	0.69	0.58	0	0.66	0.78	0	0.62	1.04	0	0.76	0.57	0	1
$B_s^0 \rightarrow K^{*-}$ [64]	0.37	0.60	0.16	0.29	0.86	0.6	0.26	1.32	0.54	0.38	0.66	0.30	1
$B_s^0 \rightarrow D_s^{*-}$ [71]	0.67	0.35	0	0.70	0.463	0	0.75	1.04	0	0.95	0.372	0	1
$B_c^+ \rightarrow D^{*0}$ [69]	0.56	0	0	0.64	0	0	-1.17	0	0	0.98	0	0	1

Majorana vs. Dirac HNLs: The Kinematic Lever

- Since SM neutrinos are massless, helicity and chirality eigenstates coincide, causing LNV processes involving SM neutrino to be suppressed by

$$|M|^2 \sim \left(\frac{m_\nu}{E_\nu} \right)^2$$

- However, for on-shell HNL decay, the mass dependence of the amplitude exactly cancels, resulting in LNC and LNV processes which are **equal and unsuppressed**
- Therefore, the lifetimes between Majorana and Dirac HNLs are related by

$$\tau_M = \frac{1}{\Gamma_{LNC} + \Gamma_{LNV}} = \frac{\tau_D}{2}$$

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